

ANCHORAGE NOTES

ELECTRICAL ANCHORAGE NOTES:

ALL ELECTRICAL COMPONENTS SHALL BE ANCHORED AND INSTALLED PER THE DETAILS ON THE APPROVED CONSTRUCTION DOCUMENTS. THE FOLLOWING COMPONENTS SHALL BE ANCHORED OR BRACED TO MEET THE FORCE AND DISPLACEMENT REQUIREMENTS PRESCRIBED IN THE 2022 CBC SECTIONS 1617A.1.18 THROUGH 1617A.1.26 AND ASCE 7-16, CHAPTER 13, 26, AND 30.

- ALL PERMANENT EQUIPMENT AND COMPONENTS.
- TEMPORARY, MOVABLE OR MOBILE EQUIPMENT THAT IS PERMANENTLY ATTACHED (e.g. HARD WIRED) TO THE BUILDING UTILITY SERVICES SUCH AS ELECTRICITY, GAS OR WATER. "PERMANENTLY ATTACHED" SHALL INCLUDE ALL ELECTRICAL CONNECTIONS EXCEPT PLUGS FOR 110/220 VOLT RECEPTACLES HAVING A FLEXIBLE CABLE.
- TEMPORARY, MOVABLE OR MOBILE EQUIPMENT WHICH IS HEAVIER THAN 400 POUNDS OR HAS A CENTER OF MASS LOCATED 4 FEET OR MORE ABOVE THE ADJACENT FLOOR OR ROOF LEVEL THAT DIRECTLY SUPPORT THE COMPONENT IS REQUIRED TO BE RESTRAINED IN A MANNER APPROVED BY DSA.

THE FOLLOWING ELECTRICAL COMPONENTS SHALL BE BE POSITIVELY ATTACHED TO THE STRUCTURE, BUT NEED NOT DEMONSTRATE DESIGN COMPLIANCE WITH THE REFERENCES NOTED ABOVE. THESE COMPONENTS SHALL HAVE FLEXIBLE CONNECTIONS PROVIDED BETWEEN THE COMPONENT AND ASSOCIATED CONDUIT. FLEXIBLE CONNECTIONS MUST ALLOW MOVEMENT IN BOTH TRANSVERSE AND LONGITUDINAL DIRECTIONS.

- COMPONENT WEIGHING LESS THAN 400 POUNDS AND HAVING A CENTER OF MASS LOCATED 4 FEET OR LESS ABOVE THE ADJACENT FLOOR OR ROOF LEVEL THAT DIRECTLY SUPPORT THE COMPONENT.
- COMPONENTS WEIGHING LESS THAN 20 POUNDS, OR IN THE CASE OF DISTRIBUTED SYSTEMS, LESS THAN 5 POUNDS PER FOOT, WHICH ARE SUSPENDED FROM A ROOF OR FLOOR OR HUNG FROM WALL.

THE ANCHORAGE OF ALL ELECTRICAL COMPONENTS SHALL BE SUBJECT TO THE APPROVAL OF THE DESIGN PROFESSIONAL IN GENERAL RESPONSIBLE CHARGE OR STRUCTURAL ENGINEER DELEGATED RESPONSIBILITY.

ELECTRICAL DISTRIBUTION SYSTEM BRACING NOTE:

ELECTRICAL DISTRIBUTION SYSTEMS SHALL BE BRACED TO COMPLY WITH THE FORCES AND DISPLACEMENTS PRESCRIBED IN ASCE 7-16 SECTION 13.3 AS DEFINED IN ASCE 7-16 SECTION 13.6.5, 13.6.6, 13.6.7, 13.6.8, AND 2019 CBC, SECTIONS 1617A.1.24, 1617A.1.25, AND 1617A.1.26.

THE METHOD OF SHOWING BRACING AND ATTACHMENTS TO THE STRUCTURE FOR THE IDENTIFIED DISTRIBUTION SYSTEM ARE AS NOTED BELOW. WHEN BRACING AND ATTACHMENTS ARE BASED ON A PREAPPROVED INSTALLATION GUIDE, COPIES OF THE BRACING SYSTEM INSTALLATION GUIDE OR MANUAL SHALL BE AVAILABLE ON THE JOBSITE PRIOR TO THE START OF AND DURING THE HANGING AND BRACING OF THE DISTRIBUTION SYSTEMS. THE STRUCTURAL ENGINEER OF RECORD SHALL VERIFY THE ADEQUACY OF THE STRUCTURE TO SUPPORT THE HANGER AND BRACE LOADS.

LIGHT FIXTURES:

ALL LIGHT FIXTURES SHALL BE POSITIVELY ATTACHED TO THE CEILING SUSPENSION SYSTEMS BY MECHANICAL MEANS TO RESIST A HORIZONTAL FORCE EQUAL TO THE WEIGHT OF THE FIXTURE. A MINIMUM OF TWO SCREWS OR APPROVED FASTENERS ARE REQUIRED AT EACH LIGHT FIXTURE, PER ASTM E580, SECTION 5.3.1.

SURFACE-MOUNTED LIGHT FIXTURES SHALL BE ATTACHED TO THE MAIN RUNNER WITH AT LEAST TWO POSITIVE CLAMPING DEVICES. THE CLAMPING DEVICE SHALL COMPLETELY SURROUND THE SUPPORTING CEILING RUNNER AND BE MADE OF STEEL WITH A MINIMUM THICKNESS OF #14 GAGE. ROTATIONAL SPRING CATCHES DO NOT COMPLY. A #12 GAGE SLACK SAFETY WIRE SHALL BE CONNECTED FROM EACH CLAMPING DEVICE TO THE STRUCTURE ABOVE. PROVIDE ADDITIONAL SUPPORTS WHEN LIGHT FIXTURES ARE EIGHT (8) FEET OR LONGER OR EXCEED 56 LB. MAXIMUM SPACING BETWEEN SUPPORTS SHALL NOT EXCEED EIGHT (8) FEET.

LIGHT FIXTURES WEIGHING LESS THAN OR EQUAL TO 10 LB. SHALL HAVE A MINIMUM OF ONE (1) #12 GAGE SLACK SAFETY WIRE CONNECTED FROM THE FIXTURE HOUSING TO THE STRUCTURE ABOVE.

LIGHT FIXTURES WEIGHING GREATER THAN 10 LB. BUT LESS THAN OR EQUAL TO 56 LBS. MAY BE SUPPORTED DIRECTLY ON THE CEILING RUNNERS, BUT THEY SHALL HAVE A MINIMUM OF TWO (2) #12 GAGE SLACK SAFETY WIRES CONNECTED FROM THE FIXTURE HOUSING AT DIAGONAL CORNERS TO THE STRUCTURE ABOVE. EXCEPTION: ALL LIGHT FIXTURES GREATER THAN TWO BY FOUR FEET WEIGHING LESS THAN 56 LBS. SHALL HAVE A #12 GAGE SLACK SAFETY WIRE AT EACH CORNER.

ALL LIGHT FIXTURES WEIGHING GREATER THAN 56 LB. SHALL BE INDEPENDENTLY SUPPORTED BY NOT LESS THAN FOUR (4) TAUT #12 GAGE HANGER WIRES (ONE AT EACH CORNER) ATTACHED FROM THE FIXTURE HOUSING TO THE STRUCTURE ABOVE OR OTHER APPROVED HANGERS. THE FOUR (4) TAUT #12 GAGE WIRES OR OTHER APPROVED HANGERS, INCLUDING THEIR ATTACHMENT TO THE STRUCTURE ABOVE, SHALL BE CAPABLE OF SUPPORTING FOUR (4) TIMES THE WEIGHT OF THE FIXTURE.

ABBREVIATIONS

AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
C	CONDUIT
CATV	CABLE TV
CB	CIRCUIT BREAKER
CO	CONDUIT ONLY
CU	COPPER
DP	DISTRIBUTION PANEL
E.C.	ELECTRICAL CONTRACTOR
E.G.C.	EQUIPMENT GROUNDING CONDUCTOR
EM	EMERGENCY
EMS	ENERGY MANAGEMENT SYSTEM
EQPT	EQUIPMENT
EXT	EXTERIOR
(E)	EXISTING
(ER)	EXISTING EQUIPMENT TO BE RELOCATED
(EX)	EXISTING EQUIPMENT TO BE DEMOLISHED
FA	FIRE ALARM
FMC	FLEXIBLE METALLIC CONDUIT
FO	FIBER OPTIC
FTL	FEED THROUGH LUGS
G.E.C.	GROUNDING ELECTRODE CONDUCTOR
GFI	GROUND FAULT CIRCUIT INTERRUPTING TYPE RECEPTACLE
INV	INVERTER, EM LIGHTING OR PHOTOVOLTAIC.
IDF	INTERMEDIATE DISTRIBUTION FRAME
L	LOCKABLE
LTG	LIGHTING
LV	LOW VOLTAGE
MC	METAL CLAD CABLING
MCB	MAIN CIRCUIT BREAKER
MDF	MAIN DISTRIBUTION FRAME
MFR	MANUFACTURER
MLO	MAIN LUGS ONLY
MTD	MOUNTED
(N)	NEW
N.E.C.	NATIONAL ELECTRICAL CODE
NEU	NEUTRAL
NIEC	NOT IN ELECTRICAL CONTRACT
OAH	OVERALL HEIGHT
OFCI	OWNER FURNISHED, CONTRACTOR INSTALLED
P	INDICATES FIXTURES ON PHOTOCCELL CONTROL
PA	PUBLIC ADDRESS
PNL	PANEL
S.A.D.	SEE ARCHITECTURAL DRAWINGS
SIG	SIGNAL SYSTEM
SPD	SURGE PROTECTION DEVICE
STC	SIGNAL TERMINAL CABINET
SWBD	SWITCHBOARD
TELE	TELEPHONE
UFER	CONCRETE ENCASED CU G.E.C.
UON	UNLESS OTHERWISE NOTED
UG	UNDERGROUND
VAV	VAV BOX. SEE MECHANICAL DIVISION DRAWINGS FOR LOCATIONS. PROVIDE TOGGLE TYPE DISCONNECT SWITCH.
WP	WEATHER PROOF. NEMA 3R. EQUALS "WHILE IN USE" TYPE WHEN APPLIED TO EXTERIOR POWER RECEPTACLES
XFMR	TRANSFORMER

GENERAL ELECTRICAL NOTES

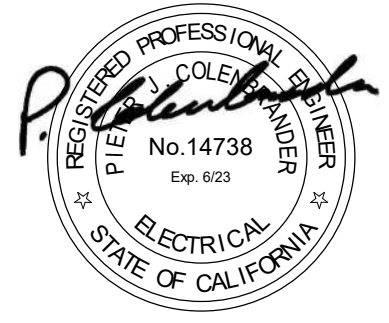
- THE CONTRACTOR SHALL PAY FOR ALL REQUIRED PERMITS AND INSPECTION FEES.
- THE CONTRACTOR SHALL VERIFY ALL CRITICAL DIMENSIONS WITH THE ARCHITECTURAL DRAWINGS PRIOR TO ROUGH-IN.
- ALL EXIT SIGNS SHALL COMPLY WITH THE RELEVANT PORTIONS OF SECTIONS 1008 AND 1013 OF THE CBC.
- ALL MECHANICAL DIVISION EQUIPMENT LOW VOLTAGE CONTROL WIRING AND RACEWAY SHALL BE PROVIDED AND INSTALLED AS SPECIFIED IN MECHANICAL DIVISION U.O.N.
- COORDINATE INSTALLATION OF ALL RECESSED LUMINAIRE(S) WITH MECHANICAL DIVISION PRIOR TO INSTALLATION OF HVAC DUCTS AND SPRINKLER HEADS. ENSURE AFTER INSTALLATION OF LUMINAIRE(S) THAT THERE IS NO CONTACT BETWEEN DUCTS AND LUMINAIRE(S) TO AVOID VIBRATION IN LUMINAIRE(S).
- USE FLEXIBLE CONDUIT FOR ALL MOTOR, TRANSFORMER, RECESSED LUMINAIRE CONNECTIONS, AND CONNECTIONS BETWEEN TWO SEPARATE STRUCTURES AND FOR ALL FINAL CONNECTIONS TO "CRITICAL EQUIPMENT" AS DEFINED IN SPECIFICATIONS. MINIMUM 1/2" DIAMETER, LIQUID TIGHT TYPE USED OUTDOORS AND IN ALL WET LOCATIONS; PROVIDE WITH CODE-SIZE (MINIMUM #12) BARE GROUND WIRE IN ALL FLEXIBLE CONDUIT.
- PROVIDE A DEDICATED NEUTRAL CONDUCTOR FOR ALL BRANCH CIRCUITS FEEDING OUTLETS AS NOTED ON THE DRAWINGS.
- FOR FLUSH MOUNTED PANELBOARDS THE CONTRACTOR SHALL STUB A MINIMUM OF FOUR (4) 3/4" CONDUITS FROM THE PANEL UP INTO THE ACCESSIBLE CEILING ABOVE FOR FUTURE CIRCUITS.
- ALL CONDUIT CONNECTORS TO OUTLET OR JUNCTION BOXES SHALL HAVE INSULATED THROATS (MANUFACTURED AS AN INTEGRAL PART OF THE CONNECTOR). AFTER-MARKET INSERTABLE THROATS ARE NOT ACCEPTABLE.
- ALL CIRCUITS IN ALL JUNCTION BOXES AND DEVICES SHALL BE CLEARLY IDENTIFIED BY MEANS OF "E2" NUMBERING TAGS OR EQUIVALENT, TO IDENTIFY THE CIRCUIT NUMBER OR RELAY SUPPLYING THE CONDUCTOR. ALL JUNCTION BOXES SHALL BE LABELED PER SPECIFICATIONS.
- ALL SURFACE MOUNTED POWER AND SIGNAL BOXES IN FINISHED AREAS SHALL BE "WIREMOLD" TYPE, WITH MATCHING RACEWAYS. SURFACE MOUNTED STEEL JUNCTION BOXES AND/OR EMT ARE NOT ACCEPTABLE.
- ALL LOCATIONS OF BARE METAL SURFACE MOUNTED CONDUIT, BOXES, PANEL COVERS, AND RELATED FITTINGS OR ACCESSORIES INSTALLED IN FINISHED AREAS (BOTH INTERIOR AND EXTERIOR) SHALL BE FINISH PAINTED TO MATCH THE SURFACE TO WHICH THEY ARE MOUNTED TO (AFTER INSTALLATION). PAINTING SHALL INCLUDE DIFFERENT COLORS AS REQUIRED TO MATCH EXISTING STRIPING OR OTHER BUILDING FEATURES TO WHICH THE EQUIPMENT IS ATTACHED AND VISIBLE. VERIFY EXACT JUNCTION BOX LOCATION(S) AND ROUTING OF EXPOSED RACEWAYS WITH THE ARCHITECT PRIOR TO ROUGH-IN.
- PROVIDE A BLANK COVER PLATE (COLOR TO MATCH ADJACENT DEVICES OR AS SPECIFICALLY CALLED FOR IN SPECIFICATIONS) FOR ALL JUNCTION BOXES (NEW AND EXISTING) ON THE PROJECT WHEN NO DEVICE IS INSTALLED.
- FOR OUTDOOR 15 AND 20-AMPERE, 125 AND 250-VOLT RECEPTACLES: RECEPTACLES LOCATED IN "WET" LOCATIONS SHALL HAVE "IN-USE" TYPE WEATHERPROOF COVER PLATES PROVIDED AND INSTALLED; RECEPTACLES LOCATED IN "DAMP" LOCATIONS SHALL HAVE "IN-USE" TYPE WEATHERPROOF COVER PLATES IN LOCATIONS DEEMED TO BE "IN-USE" WITH CORD AND PLUG ATTACHED.
- TWO OR THREE DIFFERENT PHASES SUPPLIED BY A 3-PHASE PANEL MAY SHARE A SINGLE NEUTRAL ONLY IF CIRCUIT POSITIONS ARE ADJACENT IN THE PANEL. PROVIDE COMMON HANDLE-TIE ON BREAKERS FOR MULTI-WIRE BRANCH CIRCUITS, WITH COMMON NEUTRAL, PER NEC REQUIREMENTS.

GENERAL ELECTRICAL NOTES

- PRIOR TO BID THE CONTRACTOR SHALL VISIT THE SITE TO ADEQUATELY DETERMINE ALL PRE-EXISTING CONDITIONS. BY THE ACT OF SUBMITTING A BID, THE CONTRACTOR WILL BE DEEMED TO HAVE COMPLIED WITH THE FOREGOING, TO HAVE ACCEPTED SUCH CONDITIONS, AND TO HAVE MADE ALLOWANCES THEREFORE IN PREPARING THE BID.
- PROVIDE PARITY SIZED GREEN GROUND WIRE IN ALL POWER CONDUITS, BRANCH CIRCUITS (LIGHTING & POWER) AND HOMERUNS. PROVIDE ADDITIONAL ISOLATED GROUND, GREEN WITH YELLOW STRIPE, TO ALL ISOLATED GROUND RECEPTACLES.
- PROVIDE PULLROPE IN ALL EMPTY CONDUITS THROUGHOUT THE PROJECT.
- REFER TO ARCHITECTURAL PLANS AND ELEVATIONS FOR EXACT LOCATION & CONNECTION REQUIREMENTS OF ALL LUMINAIRE(S) AND ALL OUTLET, SWITCH, AND ELECTRICAL-RELATED DEVICE MOUNTING HEIGHTS AND LOCATIONS. COORDINATE LOCATIONS OF ALL LUMINAIRE(S) AND JUNCTION BOXES WITH MECHANICAL DIVISION PRIOR TO ROUGH-IN. COORDINATE LOCATIONS OF ELECTRICAL DEVICES WITH FURNITURE PLANS PRIOR TO ROUGH-IN.
- REFER TO MECHANICAL PLANS FOR EXACT LOCATION(S) OF ALL MECHANICAL EQUIPMENT, AND CONFIRM EXACT CONNECTION REQUIREMENTS OF ALL MECHANICAL EQUIPMENT WITH MECHANICAL DIVISION, PRIOR TO ROUGH-IN. VERIFY EXACT REQUIREMENTS FOR VOLTAGE, PHASE, HORSE-POWER, OR KVA RATINGS, OF ALL MECHANICAL DIVISION EQUIPMENT REQUIRING ELECTRICAL CONNECTION.
- VERIFY EXACT CONNECTION REQUIREMENTS, OUTLET TYPE(S), MOUNTING HEIGHT(S) AND LOCATION(S) OF ALL OWNER-SUPPLIED EQUIPMENT, AND ALL EQUIPMENT PROVIDED UNDER OTHER SECTIONS OF THE SPECIFICATIONS, PRIOR TO ROUGH-IN. REFER TO ARCHITECTURAL DRAWINGS FOR EQUIPMENT LOCATIONS.
- COORDINATE TRENCHING WITH OWNER AND OTHER TRADES BEFORE BEGINNING WORK.
- ALL CONDUIT PENETRATIONS THROUGH FIRE-RATED WALLS AND FLOORS SHALL BE SEALED WITH LISTED FIRE RATED FIRE PENETRATION ASSEMBLIES TO MAINTAIN FIRE SEPARATION RATING.
- DO NOT INSTALL ANY OUTLETS BACK TO BACK IN STUD WALLS OR DE-MOUNTABLE PARTITIONS.
- THE CONTRACTOR SHALL VERIFY ALL CEILING TYPES BEFORE ORDERING OF LUMINAIRE(S). ALSO VERIFY THAT ALL FEATURES CALLED FOR IN LUMINAIRE DESCRIPTIONS ON THE LUMINAIRE SCHEDULE ARE INCLUDED WITH CATALOG NUMBERS LISTED ON THE LUMINAIRE SCHEDULE WHEN LUMINAIRE ORDERS ARE PLACED, AND ARE INCLUDED AS PART OF THE LIGHTING SUBMITTALS FOR THIS PROJECT. IF A DISCREPANCY EXISTS, CONTACT THE ARCHITECT AND ELECTRICAL ENGINEER FOR CLARIFICATION PRIOR TO BID.
- CIRCUITRY AND CONDUIT ROUTING SHOWN ON THE PLANS IS DIAGRAMMATIC ONLY. THIS CONTRACTOR IS RESPONSIBLE FOR BECOMING COMPLETELY FAMILIAR WITH THE ARCHITECTURAL AND STRUCTURAL CONDITIONS AND LIMITATIONS IN THE BUILDING AND TO PROVIDE ALL LABOR, TOOLS AND MATERIALS REQUIRED TO PRODUCE A COMPLETELY CONCEALED INSTALLATION WHEREVER INDICATED ON THE PLANS.
- MAINTAIN "AS-BUILT" RECORDS AT ALL TIMES, SHOWING EXACT LOCATION OF ALL UNDERGROUND AND/OR CONCEALED CONDUITS AND SERVICES INSTALLED UNDER THIS CONTRACT, INCLUDING CIRCUIT IDENTIFICATION WHERE APPLICABLE. PROVIDE OWNER WITH "AS-BUILT" DOCUMENTS AS INDICATED IN THE SPECIFICATIONS, AND/OR CALLED FOR IN THE SPECIFICATIONS.
- DRAWINGS INDICATE THE LOCATION(S) OF DEVICES, LUMINAIRE(S) AND EQUIPMENT, AND THE CIRCUIT NUMBER AND PANEL DESIGNATED TO SUPPLY THEM. THE CONTRACTOR SHALL BE RESPONSIBLE FOR COMPLETELY CONNECTING ALL ELECTRICAL DEVICES TO CIRCUITS INDICATED ON THE DRAWINGS.
- UNLESS OTHERWISE NOTED, ALL WORK SHOWN ON DRAWINGS IS NEW AND TO BE PROVIDED AND INSTALLED COMPLETE UNDER THIS CONTRACT.
- ALL EQUIPMENT GROUNDING SHALL CONFORM TO ARTICLE 250 OF THE NATIONAL ELECTRICAL CODE, LATEST EDITION.
- ALL EXTERIOR CONDUIT ABOVE GRADE, INCLUDING ALL ROOF MOUNTED CONDUIT, SHALL BE GALVANIZED RIGID STEEL. COAT ALL EXPOSED THREADS WITH GALVANIZING PAINT. PAINT ALL SURFACE MOUNTED RACEWAYS AND PULLBOXES TO MATCH SURROUNDING CONDITIONS, AS DIRECTED BY THE ARCHITECT.
- ALL ELECTRICAL WORK SHALL BE CARRIED OUT IN ACCORDANCE WITH THE LATEST EDITION OF THE N.E.C., AS WELL AS STATE, AND LOCAL CODES AND REQUIREMENTS.
- ALL CONDUIT SHALL BE CONCEALED, UNLESS OTHERWISE NOTED.
- THE CONTRACTOR SHALL BE RESPONSIBLE TO VERIFY THE AVAILABLE SHORT CIRCUIT CURRENT AT THE MAIN SWITCHBOARD INCOMING TERMINALS WITH THE UTILITY COMPANY, AND TO VERIFY THAT ALL POWER AND SIGNAL SERVICE PROVISIONS, INCLUDING CONCRETE EQUIPMENT PADS, CONDUITS, PULLBOXES AND CLEARANCES, MEET THE UTILITY COMPANY'S REQUIREMENTS, PRIOR TO INSTALLATION.
- EQUIPMENT OVERLOADS AND FUSES SHALL BE PROVIDED AND INSTALLED AS PER NAME PLATE ON THE EQUIPMENT ACTUALLY PROVIDED.

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PROJECT:

CITY OF MORGAN HILL

BUTTERFIELD FIRE
STATION

17285 Butterfield Blvd.
Morgan Hill, CA 95037



DESCRIPTION:

DATE:

95% CONSTR. 01/16/23
DRAWINGS

100% CONSTR. 03/23/23
DRAWINGS

NOT FOR
CONSTRUCTION

PROJECT NUMBER:

211101

SHEET TITLE:

GENERAL
NOTES, LIST OF
DRAWINGS

SHEET NUMBER:

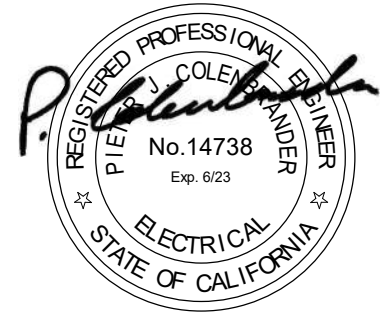
E0.1

ELECTRICAL SYMBOLS LIST	
	SECURITY SYSTEM DOOR CONTACT. PROVIDE 1/2" C.O. ROUGH-IN FROM NEAREST ACCESSIBLE CEILING AREA, TO DOOR FRAMING. PROVIDE 1" C.O. WITHIN ALL NON-ACCESSIBLE AREAS FOR WIRING AND DEVICE BY OTHERS.
	SECURITY SYSTEM CEILING MOUNT 360° MOTION DETECTOR. PROVIDE 1" C.O. WITHIN ALL NON-ACCESSIBLE AREAS FOR WIRING, DEVICE AND MOUNT, BY OTHERS
	SECURITY SYSTEM CEILING MOUNT GLASS BREAK DETECTOR. PROVIDE 1" C.O. WITHIN ALL NON-ACCESSIBLE AREAS FOR WIRING, DEVICE AND MOUNT, BY OTHERS
	WALL MOUNTED SECURITY SYSTEM KEYPAD. UP 48" TO TOP OF BOX. PROVIDE 3/4" STUB-UP FROM WALL MOUNT LOCATION TO ACCESSIBLE CEILING SPACES, AND PROVIDE 1" C.O. WITHIN ALL NON-ACCESSIBLE AREAS FOR WIRING AND DEVICE BY OTHERS.
	FIRE ALARM SYSTEM SMOKE DETECTOR, CEILING MOUNTED
	FIRE ALARM SYSTEM MONITOR MODULE
	FIRE ALARM SYSTEM STROBE, WALL MOUNTED, UP 80" U.O.N. ADJACENT NUMBER INDICATES STROBE CANDELLA RATING
	FIRE ALARM SYSTEM HORN/STROBE, WALL MOUNTED, UP 80" U.O.N. ADJACENT NUMBER INDICATES STROBE CANDELLA RATING
	FIRE ALARM SYSTEM WEATHERPROOF HORN, WALL MOUNTED, UP 90" U.O.N.
	PROJECT NUMBERED NOTE, OR SHEET NUMBERED, AS NOTED ON PLAN.
	COPPER GROUND ROD, 10' L x 3/4"Ø, <u>SEE</u> SPECS
	CIRCUIT BREAKER, INDICATES 100 AMP, 3 POLE
	UTILITY METER
	CONDUCTOR LANDING LUGS
	CONDUIT TURN DOWN
	CONDUIT TURN UP
	CONTINUATION
	CONDUIT STUB
	ELECTRICAL FEEDER TAG, PER COPPER FEEDER SCHEDULE
	MECHANICAL EQUIPMENT DESIGNATION, REFER TO MECHANICAL PLANS.
	DETAIL OR SHEET REFERENCE CALLOUT. INDICATES DETAIL 1, SHEET E-0.1. WHEN ADJACENT EQUIPMENT, APPLIES TO EQUIPMENT IDENTIFIED ONLY.
	DETAIL OR SHEET REFERENCE CALLOUT. INDICATES DETAIL 1, SHEET E-0.1. WHEN ADJACENT EQUIPMENT, APPLIES TO TYPICAL EQUIPMENT SERIES.
BRANCH CIRCUIT NOMENCLATURE	
EXAMPLES:	
LA1-3	1-POLE BRANCH CIRCUIT TO CB
LA1-1,3,5	1-POLE BRANCH CIRCUIT FOR MULTI CIRCUIT HOMERUNS TO SEPARATE CB'S.
LA1-[1,3]	2-POLE BRANCH CIRCUIT TO COMMON CB
LA1-[1,3,5]	3-POLE BRANCH CIRCUIT TO COMMON CB

ELECTRICAL SYMBOLS LIST	
	MAIN SWITCHBOARD, DISTRIBUTION PANEL, OR MOTOR CONTROL CENTER
	SURFACE MOUNTED PANELBOARD OR EQUIPMENT AS NOTED ON DRAWINGS. 6" - 6" TO TOP
	FLUSH MOUNTED PANELBOARD OR EQUIPMENT AS NOTED ON DRAWINGS. 6" - 6" TO TOP.
	PAD MOUNTED UTILITY TRANSFORMER, PER UTILITY CO. REQUIREMENTS.
	CONDUIT AND WIRE CONCEALED IN CEILING OR WALL
	CONDUIT AND WIRE UNDERGROUND, OR CONCEALED UNDER SLAB
	CONDUIT AND WIRE RUN EXPOSED, PAINTED TO MATCH ALL ADJACENT FINISHES WITHIN FINISHED SPACES
	HOMERUN TO PANELBOARD OR TERMINAL BOARD, AS NOTED ON PLANS
	CROSSMARKS INDICATE QUANTITY OF #12 CONDUCTORS PLUS PARITY SIZED GROUND CONDUCTOR. NO HASHMARKS INDICATES (2) #12 PLUS PARITY SIZED GROUND CONDUCTOR.
	GROUND WIRE
	SHORT CROSSMARK(S) INDICATE PHASE CONDUCTOR(S); LONG CROSSMARK(S) INDICATE NEUTRAL CONDUCTOR(S)
	(#10) WIRE SIZE 10 AWG FOR ALL CONDUCTORS, INCLUDING GROUND WIRE THROUGHOUT THE COMPLETE CIRCUIT
	20A 3PG 125V DUPLEX RECEPTACLE, UP 18" U.O.N. - "WP" INDICATES WEATHERPROOF.
	20A 3PG 125V DOUBLE DUPLEX RECEPTACLE, UP 18" U.O.N.
	20A 3PG 125V DUPLEX RECEPTACLE, MOUNTED ABOVE COUNTER.
	20A 3PG 125V DUPLEX RECEPTACLE, WITH INTEGRAL TYPE A USB OUTLETS, QUANTITY INDICATED BY "x" _
	20A 3PG 125V DUPLEX RECEPTACLE, FLUSH CEILING MOUNT.
	SURFACE MOUNTED WIREMOLD RACEWAY WITH RECEPTACLES AS INDICATED ON PLANS
	SURFACE MOUNTED WIREMOLD RACEWAY RISER
	DATA OUTLET, WALL MOUNTED, UP 18" U.O.N.
	DATA OUTLET, WALL MOUNTED, MOUNTED ABOVE COUNTER.
	DATA OUTLET, FLUSH CEILING MOUNT.
	"AP" - ACCESS POINT
	"P" - PROJECTOR
	FLUSH WALL MOUNTED INDOOR PUBLIC ADDRESS SPEAKER & SIGNAL SYSTEM CLOCK, UP 96" U.O.N.
	FLUSH WALL MOUNTED INDOOR PUBLIC ADDRESS SPEAKER, UP 96" U.O.N. - "WP" INDICATES WEATHERPROOF
	FLUSH WALL MOUNTED SIGNAL SYSTEM CLOCK, UP +96" U.O.N.
	FLOOR BOX ASSEMBLY, IN-SLAB, DOUBLE DUPLEX RECEPTACLE, AND DATA OUTLET.
	FIRE RATED POKE THRU ASSEMBLY FLOOR BOX
	LINE VOLTAGE MOTOR RATED SWITCH INSTALLED AT EQUIPMENT SHOWN
	MOTOR DISCONNECT SWITCH, HORSEPOWER RATED, FUSED
	PLAN SPECIFIC DIMENSIONED SYMBOL, BASED ON INDUSTRY STANDARD FRAME SIZES
	DIAGRAMAMTIC SYMBOL
	MOTOR DISCONNECT SWITCH, HORSEPOWER RATED, NON FUSED
	PLAN SPECIFIC DIMENSIONED SYMBOL, BASED ON INDUSTRY STANDARD FRAME SIZES
	DIAGRAMAMTIC SYMBOL
	VARIABLE FREQUENCY DRIVE, FURNISHED BY MECHANICAL. INSTALLED AND CONNECTED COMPLETE BY ELECTRICAL.

ELECTRICAL SYMBOLS LIST	
	LINE VOLTAGE SINGLE POLE TOGGLE SWITCH. LETTER ADJACENT INDICATES RESPECTIVE ZONE CONTROLLED, UP 48" U.O.N.
	LINE VOLTAGE TWO POLE TOGGLE SWITCH, UP 48" U.O.N.
	LINE VOLTAGE THREE-WAY TOGGLE SWITCH, UP 48" U.O.N.
	LINE VOLTAGE KEY OPERATED TOGGLE SWITCH
	LINE VOLTAGE MOTOR RATED TOGGLE SWITCH INSTALLED AT EQPT SHOWN
	LINE VOLTAGE TOGGLE SWITCH WITH PILOT LIGHT, LIGHT IS ON WHEN CIRCUIT IS CLOSED. UP 48" U.O.N.
	LOW VOLTAGE MOMENTARY CONTACT SWITCH - SEE LOW VOLTAGE RELAY SCHEDULE. LOWER CASE LETTER ADJACENT INDICATES RESPECTIVE ZONE CONTROLLED, UP 48" U.O.N.
	LOW VOLTAGE KEYED MOMENTARY CONTACT SWITCH - SEE LOW VOLTAGE RELAY SCHEDULE. LOWER CASE LETTER ADJACENT INDICATES RESPECTIVE ZONE CONTROLLED, UP 48" U.O.N.
	WALL MOUNTED SWITCH TYPE INFRARED OCCUPANCY SENSOR; UP 48" U.O.N.; SINGLE OR DUAL AS NOTED BY LETTERS ADJACENT. SET TO FIXED 20 MINUTE TIME DELAY AND MAX SENSITIVITY
	WALL MOUNTED DUAL TECHNOLOGY OCCUPANCY SENSOR; UP 48" U.O.N.; SINGLE OR DUAL AS NOTED BY LETTERS ADJACENT. SET TO FIXED 20 MINUTE TIME DELAY AND MAX SENSITIVITY
	WALL MOUNTED DIGITAL DUAL TECHNOLOGY DIMMING OCCUPANCY SENSOR SWITCH; UP 48" U.O.N.
	WALL MOUNTED DIGITAL SWITCH, UP 48" U.O.N.; LOWER CASE LETTER ADJACENT INDICATES RESPECTIVE ZONE CONTROLLED
	WALL MOUNTED SINGLE OR MULTI-ZONE DIGITAL DIMMER SWITCH, UP 48" U.O.N.; LOWER CASE LETTERS ADJACENT INDICATE RESPECTIVE ZONES TO BE SIMULTANEOUSLY MANUALLY CONTROLLED; NUMERAL DESIGNATES NUMBER OF ZONES ASSIGNED TO THE DEVICE
	CEILING MOUNTED DUAL TECHNOLOGY DIGITAL OCCUPANCY SENSOR
	WALL MOUNTED DUAL TECHNOLOGY DIGITAL OCCUPANCY SENSOR
	LOW VOLTAGE COLD TEMPERATURE PIR OCCUPANCY SENSOR
	CEILING MOUNTED LINE VOLTAGE DUAL TECHNOLOGY OCCUPANCY SENSOR
	SINGLE OR MULTI-ZONE SWITCHING OR DIMMING OPEN LOOP DIGITAL DAYLIGHTING SENSOR; NOTATIONS ADJACENT IDENTIFY DAYLIGHT ZONES ASSIGNED TO THE DEVICE. VERIFY EXACT LOCATION PRIOR TO ROUGH-IN
	SINGLE ZONE SWITCHING OR DIMMING CLOSED LOOP DIGITAL DAYLIGHTING SENSOR; NOTATIONS ADJACENT IDENTIFY DAYLIGHT ZONES ASSIGNED TO THE DEVICE. VERIFY EXACT LOCATION PRIOR TO ROUGH-IN
	DAYLIGHT CONTROL PHOTOCELL - BRACKET MOUNTED; NOTATIONS ADJACENT IDENTIFY DAYLIGHT ZONES ASSIGNED TO THE DEVICE. VERIFY EXACT LOCATION PRIOR TO ROUGH-IN
	INDICATES DAYLIGHT ZONE CONTROLLED VIA PHOTOCELL
	ROOM CONTROLLER ADJACENT NUMERAL REFERS TO THE NUMBER OF ZONES TO BE CONTROLLED. VENDOR OR CONTRACTOR TO PROVIDE QUANTITY OF ROOM CONTROLLERS REQUIRED FOR THE NUMBER OF CONTROLLED ZONES.
	PLUG LOAD ROOM CONTROLLER
	NETWORK BRIDGE
	MASTER WIRELESS BORDER ROUTER & NB - SWITCH IN NETWORK CABINET
	SECONDARY WIRELESS BORDER ROUTER
	ISOLATED RELAY INTERFACE
	EMERGENCY LIGHTING CONTROL MODULE
	OCCUPANCY SENSOR POWER PACK MOUNTED IN CONCEALED ACCESSIBLE LOCATION
CALIFORNIA GREEN BUILDING STANDARDS COMPLIANCE ALL EXTERIOR LUMINAIRES SPECIFIED IN THESE CONTRACT DOCUMENTS COMPLY WITH THE REQUIREMENTS OF THE CALIFORNIA ENERGY CODE AND THE CALIFORNIA GREEN BUILDING STANDARDS CODE, SECTION A5.106.8 LIGHT POLLUTION REDUCTION. EXTERIOR LUMINAIRES COMPLY WITH BACKLIGHT, UPLIGHT, AND GLARE (BUG) RATINGS AS DEFINED IN IESNA TM-15-11 AND BUG RATINGS DO NOT EXCEED THE MAXIMUM ALLOWABLE RATINGS FOR THIS PROJECT.	

ELECTRICAL SYMBOLS LIST	
ALL SWITCH AND CONTROL MOUNTING HEIGHTS OF 48" SHALL BE TO TOP OF THE DEVICE BOX. ALL RECEPTACLES WITH MOUNTING HEIGHT OF UP TO 18" SHALL BE NO LOWER THAN 15" TO BOTTOM OF THE DEVICE BOX, TYPICAL, U.O.N.	
	AREA BRANCH CIRCUIT TAG. INDICATES BRANCH CIRCUIT FOR ALL LIGHTING SHOWN IN AREA (OR ADJACENT CONNECTED AREA), U.O.N.
	INDICATES LUMINAIRE TYPE, SEE LUMINAIRE SCHEDULE
	RECESSED 2'x2', 2'x4' OR 1'x4' LUMINAIRE, FULLY LENSED
	INDICATES EMERGENCY LUMINAIRE. SEE ABBREVIATIONS FOR TYPE OF EMERGENCY SOURCE
	SUSPENDED LINEAR LUMINAIRE
	INDICATES AIRCRAFT CABLE SUPPORT POINT (VERIFY WITH MFG'R)
	INDICATES COMBINATION AIRCRAFT CABLE/ELECTRICAL FEED POINT (VERIFY WITH MFG'R)
	SURFACE CEILING, WALL OR COVE MOUNTED LUMINAIRE
	UNDER CABINET LUMINAIRE
	SURFACE CEILING MOUNTED LUMINAIRE
	PENDANT MOUNTED LUMINAIRE
	WALL MOUNTED LUMINAIRE
	RECESSED DOWNLIGHT LUMINAIRE
	RECESSED WALLWASH LUMINAIRE
	POLE ARM-MOUNTED AREA LUMINAIRE; ARROW INDICATES DIRECTION OF LIGHT DISTRIBUTION WHEN NOT PARALLEL TO ARM ORIENTATION
	POST-TOP PEDESTRIAN-SCALE AREA LUMINAIRE; ARROW INDICATES DIRECTION OF LIGHT DISTRIBUTION
	WALL MOUNTED EXIT SIGN. ARROWS AS NOTED ON PLANS. SHADED AREA INDICATES NUMBER OF FACES



PROJECT:

CITY OF MORGAN HILL

BUTTERFIELD FIRE STATION

17285 Butterfield Blvd.
Morgan Hill, CA 95037



DESCRIPTION: DATE:

95% CONSTR. 01/16/23
DRAWINGS

100% CONSTR. 03/23/23
DRAWINGS

NOT FOR CONSTRUCTION

PROJECT NUMBER:

211101

SHEET TITLE:

ELECTRICAL SYMBOLS LIST

SHEET NUMBER:

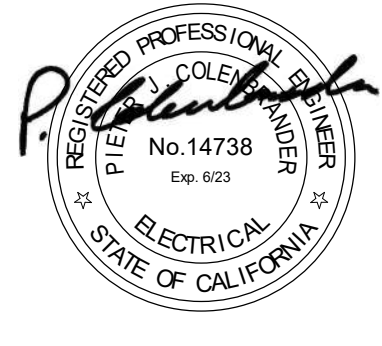
E0.2

LUMINAIRE SCHEDULE

TYPE	MOUNTING	DESCRIPTION	MANUFACTURER CATALOG NUMBER	LIGHT SOURCE	POWER SUPPLY	VOLT	INPUT WATTS	REMARKS
AM1	PENDANT	DECORATIVE PENDANT AT ISLAND WITH STEPPED SPUN AND EXTRUDED ALUMINUM SHADE AND HOUSING WITH FIELD ADJUSTABLE SINGLE CABLE SUSPENSION WITH DIRECT/INDIRECT DISTRIBUTION. 9" DIAMETER X 12" HEIGHT. WHITE INTERNAL DIFFUSER. BLACK OR WHITE CANOPY AND CORD FINISH AND BLACK, WHITE, OR CUSTOM PAINTED SHADE FINISH TO BE DETERMINED BY THE ARCHITECT.	EUREKA - STELLA 4272DI-9-LED-35-80-120V-DV-C-36-CANOPY FINISH-POWER CORD FINISH-SHADE FINISH-WH	3500K LED 80 CRI 2253 LM	0-10V DIMMING (100%-1%)	120V	29W	
AN1	RECESSED	RECESSED 2X2 LUMINAIRE WITH EXTRUDED ALUMINUM FRAME HOUSING, INJECTION MOLDED COMPOSITE END PLATES, END PLATE GRID LOCK HARDWARE, OPTICAL GRADE ACRYLIC LENS WITH WAVESTREAM REFRACTIVE OPTICS. 3 1/16" DEEP.	COOPER METALUX 22EN-LD2-34-UNV-L835-CD	3500K LED 80 CRI 3400 LM	0-10V DIMMING (100%-1%)	120V	29W	
AN2	RECESSED	SIMILAR TO TYPE AN1 EXCEPT WITH 1X4 CONFIGURATION.	COOPER METALUX 14EN-LD2-33-UNV-L835-CD	3500K LED 80 CRI 3300 LM	0-10V DIMMING (100%-1%)	120V	29W	
BA1	SURFACE	CANOPY DOWNLIGHT WITH MARINE GRADE ALUMINUM BASE PLATE, INTEGRAL HEAT SINKS WITH INTERLOCKING LENS BASE AND FOUR POINT MOUNTING HOLES AND ONE WIREWAY HOLE. FULL REFLECTOR COVER. UV STABILIZED, HIGH IMPACT RESISTANT, INJECTION MOLDED POLYCARBONATE LENS WITH CAPTIVE 'TORX' FASTENERS. HIGH IMPACT RESISTANT, INJECTION MOLDED POLYCARBONATE LENS BASE. DIE CUT CLOSED CELL EPDM GASKET AND STAINLESS FASTENERS. 15" SQUARE X 6" DEEP. STANDARD FINISH TO BE DETERMINED BY THE ARCHITECT.	KENALL MS15FL-PP-FINISH-25L35K-120	3500K LED 80 CRI 3259 LM	0-10V DIMMING (100%-1%)	120V	30W	
BB1	SURFACE	SURFACE MOUNTED SCONCE WITH CAST AND EXTRUDED ALUMINUM HOUSING CONSTRUCTION WITH FROSTED 1/8" HIGH IMPACT WHITE ACRYLIC DIFFUSER WITH CUSTOM RED LEDS. 19" H X 3" W X 4" D. POWDER COATED FINISH TO BE DETERMINED BY THE ARCHITECT.	VISA OW5524-MOD(RE D LEDS)- MVOLT-FINISH	RED LED	NON-DIMMED	UNV	11W	
BC1	SURFACE	WALL MOUNTED DOWNLIGHT SCONCE WITH PRESSURE DIE CAST MARINE GRADE ALUMINUM HOUSING, SILICONE GASKETING, STAINLESS STEEL FASTENERS, AND LOW IRON GLASS LENS. IES TYPE IV OPTICAL DISTRIBUTION. PROVIDE 4" OCTAGONAL J-BOX MOUNTING PLATE. 10.5" W X 8" H X 5" D. PRE PAINT DE-GREASE AND PHOSPHATE PROCESS. POWDER COATED FINISH TO BE DETERMINED BY THE ARCHITECT.	LIGMAN ULEW-30021-28W-T4-W35- FINISH-120/277V-4MP	3500K LED 80 CRI 3200 LM	NON-DIMMED	UNV	28W	
BC2	SURFACE	SIMILAR TO TYPE BC1 EXCEPT WITH SMALLER HOUSING SIZE, TYPE III OPTICAL DISTRIBUTION, AND LOWER LUMEN OUTPUT. 8.11" W X 8" H X 5" D.	LIGMAN ULEW-30011-14W-T3-W35- FINISH-120/277V-4MP	3500K LED 80 CRI 1660 LM	NON-DIMMED	UNV	14W	
BD1	SURFACE	WALL MOUNTED UPLIGHT SCONCE WITH PRESSURE DIE CAST MARINE GRADE ALUMINUM HOUSING, SILICONE GASKETING, STAINLESS STEEL FASTENERS, AND LOW IRON FROSTED GLASS LENS. IES TYPE IV OPTICAL DISTRIBUTION. PROVIDE 4" OCTAGONAL J-BOX MOUNTING PLATE. 10.5" W X 8" H X 5" D. PRE PAINT DE-GREASE AND PHOSPHATE PROCESS. POWDER COATED FINISH TO BE DETERMINED BY THE ARCHITECT.	LIGMAN ULEW-30021-20W-T4-W35- FINISH-120/277V-4MP-F	3500K LED 80 CRI 2422 LM	NON-DIMMED	UNV	20W	
BE1	POLE MTD	POLE MOUNTED AREA LIGHT WITH DIE CAST ALUMINUM DRIVER HOUSING AND STRAIGHT ANGLED SPUN ALUMINUM ROUND SHADE. INDIVIDUAL REFRACTOR OPTICS WITH IES TYPE IV OPTICAL DISTRIBUTION. CLEAR LENS WITH BACKLIGHT SHIELD. 120MA OPERATION. 18" DIA. X 13.5" H. 16" TAPERED POLE WITH STRAIGHT BRACKET ARM AND HANG STRAIGHT ASSEMBLY MOUNTED ON 36" RAISED CONCRETE ANCHOR BASE FOR 16' OVERALL HEIGHT FROM GROUND TO BOTTOM OF LUMINAIRE. POWDER COATED FINISH TO BE DETERMINED BY THE ARCHITECT.	STERNBERG LIGHTING LUMINAIRE: 1A-GL1960-S-BFR-12L-35-T4- MDL012-CA-EZ-BLOC-TAHPM- FINISH POLE: 550T5-316-.125-RCC-FINISH	3500K LED 80 CRI 3599 LM	NON-DIMMED	UNV	39W	
BE2	POLE MTD	SIMILAR TO TYPE BE1 EXCEPT WITH IES TYPE V OPTICAL DISTRIBUTION AND WITHOUT BACKLIGHT SHIELD.	STERNBERG LIGHTING LUMINAIRE: 1A-GL1960-S-BFR-12L-35-T5- MDL012-CA-EZ-TAHPM-FINISH POLE: 550T5-316-.125-RCC-FINISH	3500K LED 80 CRI 3599 LM	NON-DIMMED	UNV	39W	
BE3	POLE MTD	SIMILAR TO TYPE BE1 EXCEPT WITH IES TYPE II OPTICAL DISTRIBUTION, LOWER LUMEN OUTPUT, 60MA OPERATION, 15' POLE HEIGHT WITH 12' OVERALL HEIGHT FROM GROUND TO BOTTOM OF LUMINAIRE, AND LIGHTLY DIFFUSED ACRYLIC LENS.	STERNBERG LIGHTING LUMINAIRE: 1A-GL1960-S-BFR-12L-35-T2- MDL006-SV1-EZ-BLOC-TAHPM- FINISH POLE: 550T5-315-.125-RCC-FINISH	3500K LED 80 CRI 2057 LM	NON-DIMMED	UNV	20W	
BF1	GROUND	WALKWAY BOLLARD WITH EXTRUDED AND DIE CAST MARINE GRADE ALUMINUM WITH IES TYPE III OPTICAL DISTRIBUTION. 2" DEEP X 4" WIDE X 39.3" OVERALL HEIGHT SHAFT WITH 2.5" X 5.7" TOP. PAINTED POLYESTER POWDER COATED FINISH TO BE DETERMINED BY THE ARCHITECT.	LIGMAN LIGHTING: UPRA-10002-8W-T3-W35-FINISH- 120/277V	3500K LED 80 CRI 855 LM	NON-DIMMING	UNV	8W	
BG1	GROUND	FLUSH IN GRADE UPLIGHT WITH COMPRESSION MOLDED POLYESTER HOUSING WITH BRUSHED STAINLESS STEEL FACEPLATE, TEMPERED CLEAR BOROSILICATE GLASS LENS WITH ANTI SLIP SURFACE. TWO CONDUIT ENTRIES FOR THROUGH WIRING, INTEGRAL POWER MODULE, NARROW SPOT DISTRIBUTION WITH INTERNAL HONEYCOMB LOUVER, 360 DEGREE ROTATION, AND 15 DEGREE TILT ADJUSTMENT. 10" APERTURE. 18" L X 12" W X 12" DEEP HOUSING.	HYDREL PDX10-BSS-18LED-WHT30K- MVOLT-NSP-FLCAS-34S-IHL	3000K LED 80 CRI 1000 LM	NON-DIMMING	UNV	19W	
BH1	GROUND	LOW PROFILE LINEAR WALLWASH LUMINAIRE; ANODIZED ALUMINUM HOUSING WITH DIE CAST ENDCAPS; DOUBLE SEALED POLYCARBONATE LENS; IP66 RATED SUITABLE FOR WET LOCATIONS; STANDARD OUTPUT WITH 45° BEAM SPREAD, GLARE SHIELD, AND BLADE LOUVER. SWIVEL MOUNT WITH PERMAPOST MOUNTING OPTION. FINISH AS SELECTED BY ARCHITECT.	ORGATECH 1400-4-LS-35-U-ND-45-SW- FINISH-B-G-P	3500K LED 80 CRI 1092 LM	NON-DIMMING	UNV	14W	

LUMINAIRE SCHEDULE

TYPE	MOUNTING	DESCRIPTION	MANUFACTURER CATALOG NUMBER	LIGHT SOURCE	POWER SUPPLY	VOLT	INPUT WATTS	REMARKS
AA1	SURFACE	SURFACE MOUNTED LINEAR VANITY LUMINAIRE, 20-GAUGE STEEL HOUSING WITH SPRING FASTENED ALUMINUM END CAPS, EXTRUDED ROUND PROFILE ACRYLIC LENS, MEDIUM OUTPUT. NOM. 4' LENGTH X 2" W X 3" H. SINGLE OR DUAL GANG ELECTRICAL BOX COVER PLATE. GLOSS WHITE POLYESTER POWDER COATED FINISH.	PRUDENTIAL SNAP-PRO R1-PRO-LED35-MO-4'-WA-YGW- UNV-SUR-ND-EBCP1G/2G	3500K LED 80 CRI 2960 LM/4'	NON-DIMMED	120V	26W	
AA2	SURFACE	SIMILAR TO TYPE AA1 EXCEPT 6' LENGTH.	PRUDENTIAL SNAP-PRO R1-PRO-LED35-MO-6'-WA-YGW- UNV-SUR-ND-EBCP1G/2G	3500K LED 80 CRI 2960 LM/4'	NON-DIMMED	120V	39W	
AB1	SURFACE	4FT LED 'STRIP' LUMINAIRE WITH DIE-FORMED COLD ROLLED STEEL HOUSING, 'PAINT-AFTER-FABRICATION' WHITE FINISH, WITH CURVED, FROSTED ACRYLIC DIFFUSER, HEAVY-GAGE WIRE GUARD W/WHITE FINISH. 3.5" H X 2.75" W.	H.E. WILLIAMS 75R-4-L50-8-35-WG-75-DRV-UNV	3500K LED 80 CRI 4867 LM/4'	NON-DIMMED	120V	33W	
AB2	SURFACE	SIMILAR TO TYPE AB1 EXCEPT 2' LENGTH AND LOWER LUMEN OUTPUT.	H.E. WILLIAMS 75R-2-L15-8-35-WG-75-DRV-UNV	3500K LED 80 CRI 1458 LM	NON-DIMMED	120V	11W	
AB3	SURFACE	SIMILAR TO TYPE AB1 EXCEPT 8' LENGTH AND LOWER LUMEN OUTPUT.	H.E. WILLIAMS 75R-4-L60-8-35-WG-75-DRV-UNV	3500K LED 80 CRI 2916 LM/4'	NON-DIMMED	120V	40W	
AB4	PENDANT	SIMILAR TO TYPE AB1 EXCEPT PENDANT MOUNTED WITH AIRCRAFT CABLE SUSPENSION KIT.	H.E. WILLIAMS 75R-4-L50-8-35-WG-75-DRV-UNV- ACF-D-96-ACJ-D-96	3500K LED 80 CRI 4867 LM/4'	NON-DIMMED	120V	33W	
AC1	PENDANT	PENDANT MOUNTED DIRECT/INDIRECT LINEAR LUMINAIRE WITH EXTRUDED ALUMINUM CONSTRUCTION HOUSING, WIDESPREAD UPLIGHT OPTIC, FLUSH DOWNLIGHT DIFFUSER, AND CAST ALUMINUM ENDCAPS. 50" FULLY ADJUSTABLE AIRCRAFT CABLE SUSPENSION KIT WITH HARD CEILING HARDWARE. 2.5" X 2.5" X 8' LENGTH COMPRISED OF (1) 8' MODULE. WHITE, BLACK, OR SATIN ALUMINUM POLYESTER PAINTED FINISH TO BE DETERMINED BY THE ARCHITECT.	FINELITE HP-X-P-ID-8'-B-B-835-WSO-F-120- SC-FC-1%-FA50-C4-FE-FINISH	3500K LED 80 CRI 4234 LM/4'	0-10V DIMMING (100%-1%)	120V	70W	
AD1	SURFACE	CEILING MTD DOWNLIGHT WITH SOLID ALUMINUM OUTER RING WITH INTERNAL LIGHT GUIDE AND DIFFUSING LENS. 0.75" D X 7" DIAMETER. WHITE POLYESTER POWDER COATED FINISH.	JUNO JSF-7IN-35K-90CRI-MVOLT-ZT- WH	3500K LED 80 CRI 1000 LM	NON-DIMMED	120V	13W	
AE1	SURFACE	SURFACE MOUNTED LED UNDERCABINET. 22" IN LENGTH WITH EXTRUDED ALUMINUM CONSTRUCTION HOUSING WITH OPAL WHITE ACRYLIC DIFFUSER. 1" D X 3.75" W. WHITE POLYESTER PAINTED FINISH.	JUNO UPS-22IN-30K-90CRI-WH	3000K LED 90 CRI 731 LM	NON-DIMMING	120V	11W	COORDINATE VALANCE TO CONCEAL LUMINAIRE FROM VIEW WITH CASEWORK CONTRACTOR.
AF1	SURFACE	BEDSIDE TASK LIGHT WITH ARTICULATING ALUMINUM ARMS WITH 8" CORD FOR HARDWIRED APPLICATIONS AND INTEGRAL LED DRIVER. FINGERTIP TOUCH DIMMING ON HEAD. TITANIUM, BRONZE, OR BLACK PAINTED FINISH TO BE DETERMINED BY THE ARCHITECT.	WAC LIGHTING BL-1021L/R-3000K-FINISH	3000K LED 80 CRI 315 LM	INTEGRAL DIMMER	120V	9W	COORDINATE RIGHT OR LEFT HAND ARTICULATION DEPENDING UPON WHICH SIDE OF THE BED THE LUMINAIRE IS MOUNTED.
AG1	SURFACE	36" DIA DRUM WITH ROLLED AND WELDED EXTRUDED ALUMINUM CONSTRUCTION HOUSING WITH EDGE LIT WAVE GUIDE AND SECONDARY ACRYLIC DIFFUSER. 2.5" DEEP. ROUNDED OR SOFT EDGE PROFILE TO BE DETERMINED BY THE ARCHITECT. TEXTURED MATTE, GLOSS WHITE, OR CUSTOM COLOR TO BE DETERMINED BY THE ARCHITECT.	PRUDENTIAL GAZERDS/H-36-LED35-LO- FINISH-D1-SC-UNV-SURF-X3- DM01	3500K LED 80 CRI 4162 LM	0-10V DIMMING (100%-1%)	120V	35W	
AH1	SURFACE	24" SQUARE SURFACE MOUNTED LUMINAIRE WITH ROLLED AND WELDED EXTRUDED ALUMINUM CONSTRUCTION HOUSING WITH EDGE LIT WAVE GUIDE AND SECONDARY ACRYLIC DIFFUSER. 2.5" DEEP. ROUNDED OR SOFT EDGE PROFILE TO BE DETERMINED BY THE ARCHITECT. TEXTURED MATTE, GLOSS WHITE, OR CUSTOM COLOR TO BE DETERMINED BY THE ARCHITECT.	PRUDENTIAL GAZE-SQS/H-24-LED35-MO- FINISH-D1-SC-UNV-SURF-X3- DM01	3500K LED 80 CRI 4059 LM	0-10V DIMMING (100%-1%)	120V	35W	
AH1E	SURFACE	SIMILAR TO TYPE AH1 EXCEPT WITH INTEGRAL 800 LUMEN EMERGENCY BATTERY PACK.	PRUDENTIAL GAZE-SQS/H-24-LED35-MO- FINISH-D1-SC-UNV-SURF-X3- DM01-EMHE	3500K LED 80 CRI 4059 LM	0-10V DIMMING (100%-1%) + EM EMERGENCY BATTERY	120V	35W	
AJ1	SURFACE	4' WET LOCATION LUMINAIRE WITH MARINE GRADE EXTRUDED ALUMINUM CONSTRUCTION HOUSING WITH CAST ALUMINUM END CAPS. LINEAR RIBBED WHITE POLYCARBONATE LENS WITH INTERNAL SPECULAR ALUMINUM REFLECTOR. WHITE POWDER COATED FINISH AND EPDM GASKETING. 6.5" W X 4" D.	NEWSTAR VIC-4-N-L1-35-1C-RW-12-WH	3500K LED 80 CRI 2700 LM/4'	NON-DIMMED	120V	25W	
AJ2	SURFACE	SIMILAR TO TYPE AJ1 EXCEPT 8' LENGTH.	NEWSTAR VIC-4-N-L1-35-1C-RW-12-WH-TM	3500K LED 80 CRI 2700 LM/4'	NON-DIMMED	120V	50W	
AJ3	SURFACE	SIMILAR TO TYPE AJ1 EXCEPT 2' LENGTH.	NEWSTAR VIC-2-N-L1-35-1C-RW-12-WH	3500K LED 80 CRI 2700 LM/4'	NON-DIMMED	120V	13W	
AK1	SURFACE	8FT LED 'STRIP' LUMINAIRE WITH DIE-FORMED COLD ROLLED STEEL HOUSING, 'PAINT-AFTER-FABRICATION' WHITE FINISH, WITH CURVED, FROSTED ACRYLIC DIFFUSER, HEAVY-GAGE WIRE GUARD W/WHITE FINISH. 3.5" H X 2.75" W.	H.E. WILLIAMS 75R-8-L100-8-35-WG-75-DRV- UNV	3500K LED 80 CRI 4867 LM/4'	NON-DIMMED	120V	66W	CONTINUOUS ROW MOUNTED TO UNISTRUT CHANNEL IN LENGTHS AND CONFIGURATIONS INDICATED ON THE PLANS. WALL MOUNTED ON CUSTOM BRACKET ARMS.
AK2	PENDANT	8FT LED 'STRIP' LUMINAIRE WITH DIE-FORMED COLD ROLLED STEEL HOUSING, 'PAINT-AFTER-FABRICATION' WHITE FINISH, WITH CURVED, FROSTED ACRYLIC DIFFUSER, HEAVY-GAGE WIRE GUARD W/WHITE FINISH. 3.5" H X 2.75" W. PROVIDE AIRCRAFT CABLE SUSPENSION KITS AT 48" ON CENTER.	H.E. WILLIAMS 75R-8-L100-8-35-WG-75-DRV- UNV-ACF/ACJ-D-48-5-8D-W	3500K LED 80 CRI 4867 LM/4'	NON-DIMMED	120V	66W	CONTINUOUS ROW MOUNTED TO UNISTRUT CHANNEL IN LENGTHS AND CONFIGURATIONS INDICATED ON THE PLANS. SUSPENDED WITH STAINLESS STEEL AIRCRAFT CABLE.
AL1	RECESSED	RECESSED DOWNLIGHT WITH FORMED STEEL CONSTRUCTION HOUSING, 4" DIAMETER SELF FLANGED 'COMFORT CLEAR' ALUMINUM REFLECTOR. WIDE 'BATWING' DISTRIBUTION. 2" MAX. CEILING THICKNESS.	LIGHTOLIER FRAME: 4R-N ENGINE: C4L-20-8-35-W-Z10-U TRIM: C4-R-DL-CC-P	3500K LED 80 CRI 2000 LM	0-10V DIMMING (100%-1%)	120V	19W	
AL2	RECESSED	SIMILAR TO TYPE AL1 EXCEPT WITH WALLWASH DISTRIBUTION AND LOWER LUMEN OUTPUT.	LIGHTOLIER FRAME: 4R-N ENGINE: C4L-15-8-35-W-Z10-U TRIM: C4-R-WW-CC-P	3500K LED 80 CRI 1500 LM	0-10V DIMMING (100%-1%)	120V	14W	
AL3	RECESSED	SIMILAR TO TYPE AL1 EXCEPT WITH DECORATIVE GLASS CYLINDRICAL TRIM. 6" TRIM COLLAR WITH 4" DIAMETER X 2" HIGH GLASS CYLINDER.	LIGHTOLIER FRAME: 4R-N ENGINE: C4L-20-8-35-W-Z10-U TRIM: C4-R-DL-CC-D4A04	3500K LED 80 CRI 2000 LM	0-10V DIMMING (100%-1%)	120V	19W	



PROJECT:

CITY OF MORGAN HILL

BUTTERFIELD FIRE
STATION

17285 Butterfield Blvd.
Morgan Hill, CA 95037



DESCRIPTION: DATE:

95% CONSTR. 01/16/23
DRAWINGS

100% CONSTR. 03/23/23
DRAWINGS

NOT FOR
CONSTRUCTION

PROJECT NUMBER:

211101

SHEET TITLE:

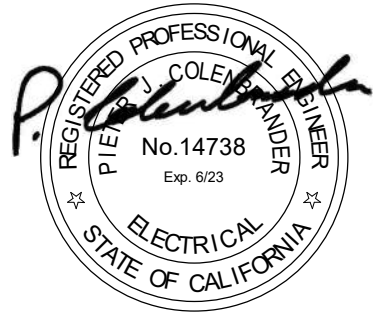
LUMINAIRE
SCHEDULE

SHEET NUMBER:

E0.3

LUMINAIRE SCHEDULE								
TYPE	MOUNTING	DESCRIPTION	MANUFACTURER CATALOG NUMBER	LIGHT SOURCE	POWER SUPPLY	VOLT	INPUT WATTS	REMARKS
EX1	SURFACE	SURFACE WALL MOUNTED. 5/8" PRESSURE DIE CAST ALUMINUM SINGLE FACE EXIT SIGN. AVAILABLE IN FINE GRAIN BRUSHED ALUMINUM. 3MM REFRACTIVE LIGHT GUIDE, LIGHT ENGINE, PHOTOLUMINESCENT PANEL, HOUSED IN A SINGLE INJECTION-MOLDED FRAME ASSEMBLY. NO EXPOSED FASTENERS. REMOVEABLE FACE WITH FIELD SELECTABLE CHEVRON DIRECTIONAL INDICATORS. 25-YEAR SERVICE LIFE. PROVIDE ALL MOUNTING ACCESSORIES, PARTS, STENCIL FACES, KNOCK-OUT DIRECTIONAL ARROW INDICATORS, ETC., AS REQUIRED FOR EXIT SIGNS THAT FULLY CONFORM TO THE CONFIGURATIONS AS INDICATED ON THE LIGHTING PLANS. STANDARD FACTORY FRAME COLOR TO BE DETERMINED BY THE ARCHITECT.	EVENLITE AUR-1-BODY COLOR-1B	N/A	N/A	120V	2W	
EX2	SURFACE	SIMILAR TO TYPE EX1 EXCEPT CEILING MOUNTED.	EVENLITE AUR-1-BODY COLOR-CN	N/A	N/A	120V	2W	

END OF LUMINAIRE SCHEDULE



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PROJECT:

CITY OF MORGAN HILL

BUTTERFIELD FIRE STATION

17285 Butterfield Blvd.
Morgan Hill, CA 95037



DESCRIPTION: **DATE:**

95% CONSTR. 01/16/23
DRAWINGS

100% CONSTR. 03/23/23
DRAWINGS

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CONSTRUCTION

PROJECT NUMBER:

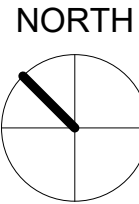
211101

SHEET TITLE:

LUMINAIRE
SCHEDULE

SHEET NUMBER:

E0.4



1

1. PROVIDE FLUSH IN-GRADE BROADBAND UTILITY PULLBOX WITH STEEL LID, LABELED PER UTILITY PROVIDER REQUIREMENTS. VERIFY PULLBOX SPEC WITH UTILITY PRIOR TO ROUGH-IN. COORDINATE EXACT LOCATION WITH UTILITY COMPANY AND OTHER TRADES PRIOR TO ROUGH-IN. ORIENT PARALLEL AND PERPENDICULAR TO SITE LINES.
2. PROVIDE (2) 1" UNDERGROUND SCHEDULE 40 PVC RACEWAYS FOR BROADBAND UTILITY AND FIBER CABLEING. STUB UP IN COMM 115, ADJACENT TO TELECOM BACKBOARD.
3. PROVIDE POWER CONNECTION AND WEATHERPROOF MOTOR RATED DISCONNECT SWITCH FOR MOTORIZED GATE. COORDINATE EXACT SWITCH LOCATION WITH GATE INSTALLER. HOMERUN CIRCUIT INDICATED TO ELEC 127 USING 1" UNDERGROUND SCHEDULE 40 PVC RACEWAY.
4. PROVIDE 3/4" UNDERGROUND SCHEDULE 40 PVC RACEWAY FOR GATE CONTROL WIRING. CONTROL WIRING TO BE PROVIDED AND INSTALLED BY GATE INSTALLER. COORDINATE LOCATION AND CONDUIT TERMINATIONS WITH GATE INSTALLER.
5. PROVIDE FLUSH IN-GRADE PULLBOX, CHRISTY N9 OR EQUAL, WITH CONCRETE LID LABELED "POWER", FOR FUTURE EV CHARGER WIRING. COORDINATE EXACT LOCATION WITH ARCHITECT. SEE 2/E7.1.
6. PROVIDE UNDERGROUND SCHEDULE 40 PVC RACEWAY OF INDICATED SIZE FOR FUTURE EV CHARGER WIRING. SEE 1/E7.1.
7. PROVIDE (2) 1" UNDERGROUND SCHEDULE 40 PVC RACEWAYS FOR FUTURE EV CHARGER WIRING. STUB, CAP AND MARK ADJACENT TO PARKING LOT, PER 5/E7.3.
8. STUB UP UNDERGROUND RACEWAY AND CAP IN ELEC 127 ADJACENT TO MAIN SWITCHBOARD. SEE 6/E-7.3.
9. VERIFY GENERATOR FEEDER, JACKET HEATER, BATTERY CHARGER, AEROSOL CONTROL, CONDUNIT STUB-UP LOCATIONS WITH GENERATOR EQUIPMENT, PRIOR TO ROUGH-IN.
10. PROVIDE FLUSH IN-GRADE PULLBOX, CHRISTY N9 OR EQUAL, WITH CONCRETE LID LABELED "TRAFFIC CONTROL", FOR FUTURE TRAFFIC WARNING LIGHTS WIRING. COORDINATE EXACT LOCATION WITH ARCHITECT. SEE 2/E7.1.
11. PROVIDE 1" UNDERGROUND SCHEDULE 40 PVC RACEWAY FOR FUTURE TRAFFIC WARNING LIGHTS POWER WIRING. SEE 1/E7.1.
12. PROVIDE 1" UNDERGROUND SCHEDULE 40 PVC RACEWAY FOR FUTURE TRAFFIC WARNING LIGHTS CONTROL WIRING. STUB UP AND CONNECT TO MASTER APP BAY DOOR CONTROL PANEL. SEE E3.1 AND 1/E7.1.
13. STUB UP UNDERGROUND RACEWAY AND CAP IN ELEC 127, FOR FUTURE EXTENSION TO TRAFFIC SIGNAL POWER SOURCE. SEE 6/E-7.3.

PROJECT:

CITY OF MORGAN HILL

BUTTERFIELD FIRE
STATION

17285 Butterfield Blvd.
Morgan Hill, CA 95037



DESCRIPTION:	DATE
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95% CONSTR. 01/16/2
DRAWINGS

100% CONSTR. 03/23/2017
DRAWINGS

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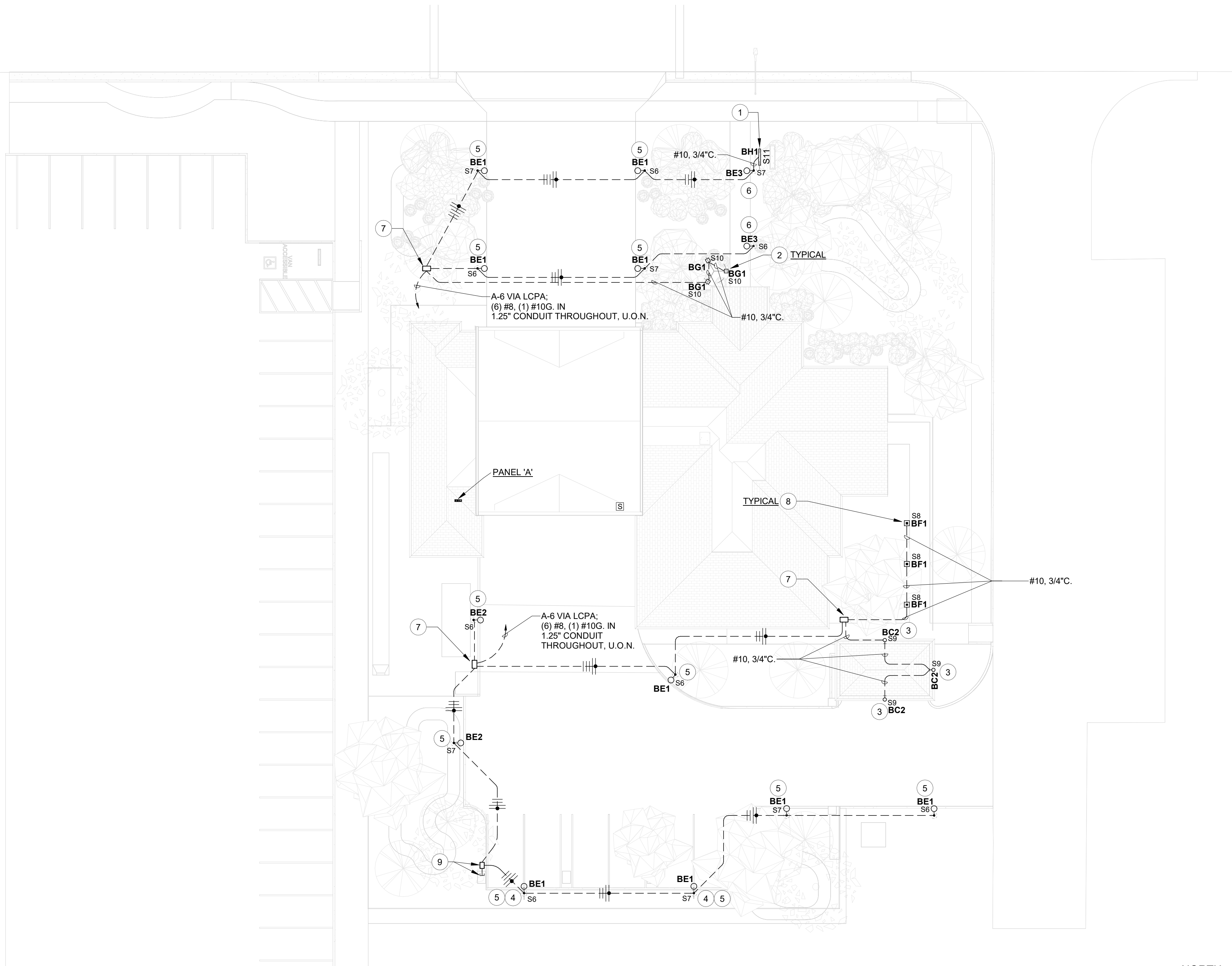
PROJECT NUMBER:
211101

SHEET TITLE:

**SITE PLAN -
ELECTRICAL**

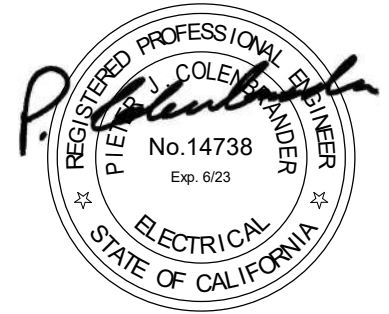
SHEET NUMBER:

E1.1



1 SITE PLAN - LIGHTING
1/16" = 1'-0"

- # SHEET NUMBERED NOTES
- 1 MONUMENT SIGN LIGHTER.
 - 2 IN GRADE FLAGPOLE UPLIGHTS. SEE DETAIL 9/E7.1.
 - 3 WALL MOUNTED AT 6'-9" A.F.F. TO THE BOTTOM OF THE LUMINAIRE.
 - 4 PROVIDE AND INSTALL LUMINAIRE IF ADDITIVE ALTERNATE FOR PV CANOPY IS NOT ACCEPTED.
 - 5 SEE DETAIL 6/E7.2.
 - 6 SEE DETAIL 7/E7.2.
 - 7 PROVIDE FLUSH IN-GRADE PULLBOX WITH CONCRETE LID LABELED "LIGHTING". SIZE PULLBOX PER NEC.
 - 8 SEE DETAIL 1/E7.3.
 - 9 STUB OUT 1" CONDUIT WITH PULL ROPE FOR ADD-ALTERNATE CANOPY LIGHTING.



PROJECT:

CITY OF MORGAN HILL

BUTTERFIELD FIRE STATION

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DESCRIPTION: DATE:

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PROJECT NUMBER:

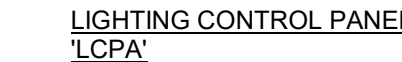
211101

SHEET TITLE:

SITE PLAN - LIGHTING

SHEET NUMBER:

E1.2



① LIGHTING
1/8" = 1'-0"



③

- VERIFY EXACT PROGRAMMING WITH OWNER.



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MAHONY & MYER
 ELECTRICAL ENGINEERING & LIGHTING DESIGN
 140 REDWOOD HWY., SUITE 245
 SAN RAFAEL, CALIFORNIA 94903
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OBJECT:

CITY OF MORGAN HILL

UTTERFIELD FIRE
STATION

35 Butterfield Blvd.
Gan Hill, CA 95037



DESCRIPTION:

DATE:

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OBJECT NUMBER:

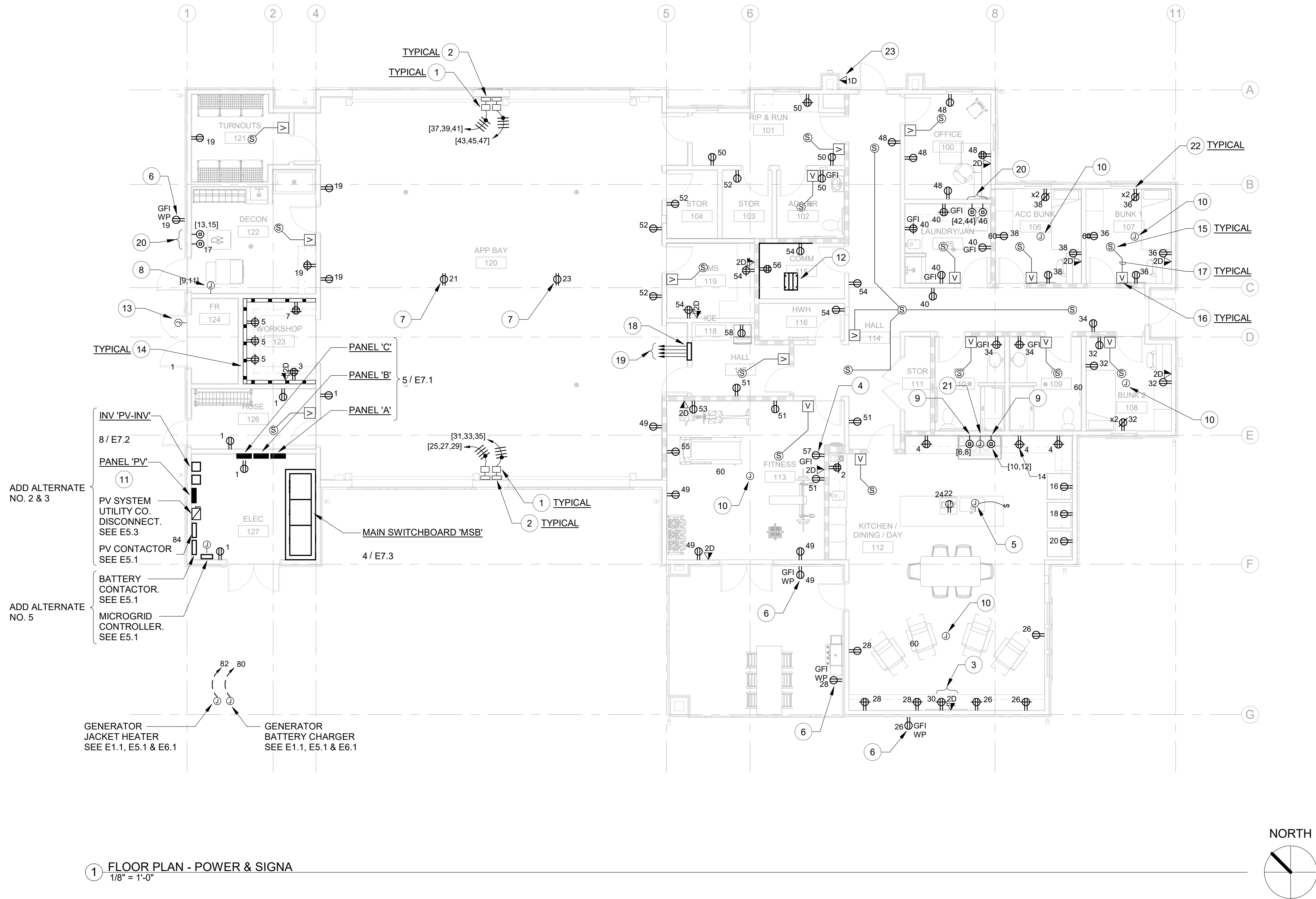
211101

EET TITLE:

DOOR PLAN - LIGHTING

EET NUMBER:

E2.1



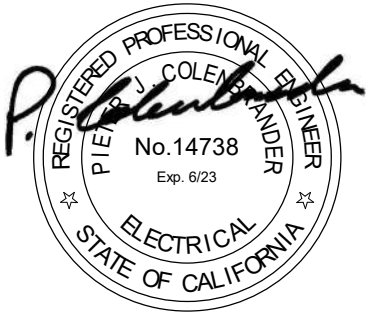
1 FLOOR PLAN - POWER & SIGNA
1/8" = 1'-0"

SHEET NUMBERED NOTES

- 1 COILING DOOR CONTROL PANEL/DISCONNECT SWITCH, FLUSH MOUNTED HIGH ON WALL. ROUTE CIRCUITRY SHOWN THROUGH PANEL/SWITCH AND CONNECT COMPLETE TO DOOR MOTOR. SEE 2/E5.1.
- 2 PROVIDE 3-PHASE, 208V-240V BUCK-BOOST XFMR FOR COILED APP BAY DOOR MOTOR. MOTOR REQUIRES 240V MINIMUM. WALL MOUNT ADJACENT TO MOTOR. SEE 2/E5.1.
- 3 FLUSH WALL MOUNTED AV/POWER BOX ASSEMBLY (BEHIND FLAT PANEL DISPLAY). HUBBELL NET SELECT FPTV 2-GANG BOX #NSAV62M, OR EQUAL, WITH DUPLEX RECEPTACLE AND (2) DATA JACKS. VERIFY MOUNTING HEIGHT AND LOCATION WITH ARCHITECT PRIOR TO ROUGH-IN. SEE ARCHITECTURAL ELEVATION DRAWINGS.
- 4 PROVIDE GFI RECEPTACLE FOR BOTTLE FILLER. COORDINATE EXACT MOUNTING HEIGHT WITH BOTTLE FILLER INSTALLER.
- 5 PROVIDE POWER CONNECTION FOR UNDER COUNTER GARBAGE DISPOSAL. RUN CIRCUIT THROUGH TOGGLE SWITCH FLUSH MOUNTED TO SIDE OF CABINETRY, AS SHOWN. COORDINATE SWITCH LOCATION WITH CABINETRY MANUFACTURER AND INSTALLER, TO AVOID CONFLICTS WITHIN CABINETRY.
- 6 PROVIDE FLUSH WALL MOUNTED WEATHERPROOF COVER WITH LOCKING LATCH, PASS & SEYMOUR 4600 SERIES, OR EQUAL. REQUIRES SPECIAL ROUGH-IN.
- 7 CEILING MOUNTED RECEPTACLE FOR CORD DROP POWER REEL. PROVIDE MOTORIZED CORD REEL ATTACHED TO STRUCTURE ABOVE USING FACTORY CEILING MOUNT, WITHIN 12" OF CEILING RECEPTACLE. CORD REEL SHALL BE RATED 15 AMPS, 120 VOLTS WITH 40 FOOT LONG 12/3 CORD AND THREE SINGLE NEMA 5-15R RECEPTACLES AT END OF CORD. PROVIDE CEILING MOUNTED HUBBELL MODEL HBLCA40123TT. VERIFY EXACT LOCATION WITH ARCHITECT PRIOR TO ROUGH-IN. SEE 7/E7.1.
- 8 PROVIDE POWER CONNECTION FOR WASHER-EXTRACTOR. COORDINATE EXACT LOCATION WITH ARCHITECT.
- 9 PROVIDE NEMA 14-50R RECEPTACLE FOR COMBINATION OVEN/RANGE. COORDINATE MOUNTING HEIGHT WITH ARCHITECT.
- 10 PROVIDE POWER CONNECTION IN CEILING FOR SURFACE MOUNTED CEILING FAN. COORDINATE EXACT LOCATION WITH ARCHITECT. CONNECT COMPLETE.
- 11 PV AC COLLECTOR PANEL. SEE 1/E-5.4.
- 12 PROVIDE FLOOR MOUNTED 2-POST RACK, CHATSWORTH MODEL #55053-703, OR EQUAL. COORDINATE EXACT LOCATION WITH CITY IT STAFF.
- 13 PROVIDE POWER CONNECTION FOR FIRE SPRINKLER ALARM BELL AND WATER FLOW ALARM SWITCH. COORDINATE EXACT LOCATION WITH DIVISION 21.
- 14 PROVIDE 4000 SERIES WIREMOLD RACEWAY, MOUNTED ABOVE COUNTER, WITH SEPARATE CHANNELS FOR LINE AND LOW VOLTAGE WIRING AND DEVICES. INCLUDE ALL REQUIRED FITTINGS, COVERS AND BLANK PLATES.
- 15 PROVIDE SPEAKER, FLUSH IN CEILING, FOR RING DOWN ALERT SYSTEM. SEE 3/E5.2 FOR RING DOWN ALERT SYSTEM DIAGRAM.
- 16 PROVIDE SOUND ATTENUATOR FOR ROOM RINGDOWN ALERT SYSTEM SPEAKER. MOUNT FLUSH IN WALL ADJACENT TO ROOM LIGHTING CONTROL SWITCH. SEE 3/E5.2 FOR RING DOWN ALERT SYSTEM DIAGRAM.
- 17 PROVIDE 1/2" CONDUIT FROM BOX TO CEILING SPEAKER, AS SHOWN, FOR VOLUME CONTROL SWITCH AND CONTROL WIRING, BY RING DOWN SYSTEM VENDOR. SEE 3/E5.2 FOR RING DOWN ALERT SYSTEM DIAGRAM.
- 18 RECESSED NEMA 1 ENCLOSURE FOR MASTER APP BAY DOOR CONTROLS, PROVIDED BY DOOR CONTRACTOR, INSTALLED BY ELECTRICAL.
- 19 3/4" CONDUITS, (1) PER APP BAY DOOR, FOR MASTER DOOR CONTROL WIRING. ROUTE CONCEALED IN WALLS AND CEILING TO WALL MOUNTED DOOR CONTROLLER FOR EACH DOOR. CONTROL WIRING TO BE INSTALLED BY DOOR CONTRACTOR. SEE 2/E5.1.
- 20 PROVIDE (2) RECEPTACLES FOR COMBO WASHER/DRYER UNIT. COORDINATE NEMA TYPE REQUIRED WITH WASHER/DRYER MANUFACTURER. MOUNT AT +48" AFF ON CENTER. LOCATION AND MOUNTING HEIGHT WITH ARCHITECT.
- 21 PROVIDE POWER CONNECTION FOR RANGE HOOD. COORDINATE LOCATION AND MOUNTING HEIGHT WITH ARCHITECT.
- 22 PROVIDE COMBINATION DUPLEX POWER-USB OUTLET RECEPTACLE IN WALL PHONE CHARGING NICHE. SEE ARCHITECTURAL INTERIOR ELEVATIONS AND A2/A7.4. TEXT INDICATES QUANTITY OF USB OUTLETS PER RECEPTACLE.
- 23 PROVIDE DATA OUTLET FOR STATION 911 TELEPHONE. COORDINATE EXACT LOCATION AND MOUNTING HEIGHT WITH TELEPHONE INSTALLER.

GENERAL SHEET NOTES

1. ALL BRANCH CIRCUITRY SHOWN ON THIS PLAN SHALL HOMERUN TO PANEL 'B', U.O.N.



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4340 REDWOOD HWY, SUITE 245
SAN RAFAEL, CALIFORNIA 94903
(415) 402-0430/FAX (415) 429-9602

PROJECT:

CITY OF MORGAN HILL
BUTTERFIELD FIRE STATION
17285 Butterfield Blvd.
Morgan Hill, CA 95037



DESCRIPTION: DATE:

95% CONSTR. 01/16/23
DRAWINGS
100% CONSTR. 03/23/23
DRAWINGS

NOT FOR
CONSTRUCTION

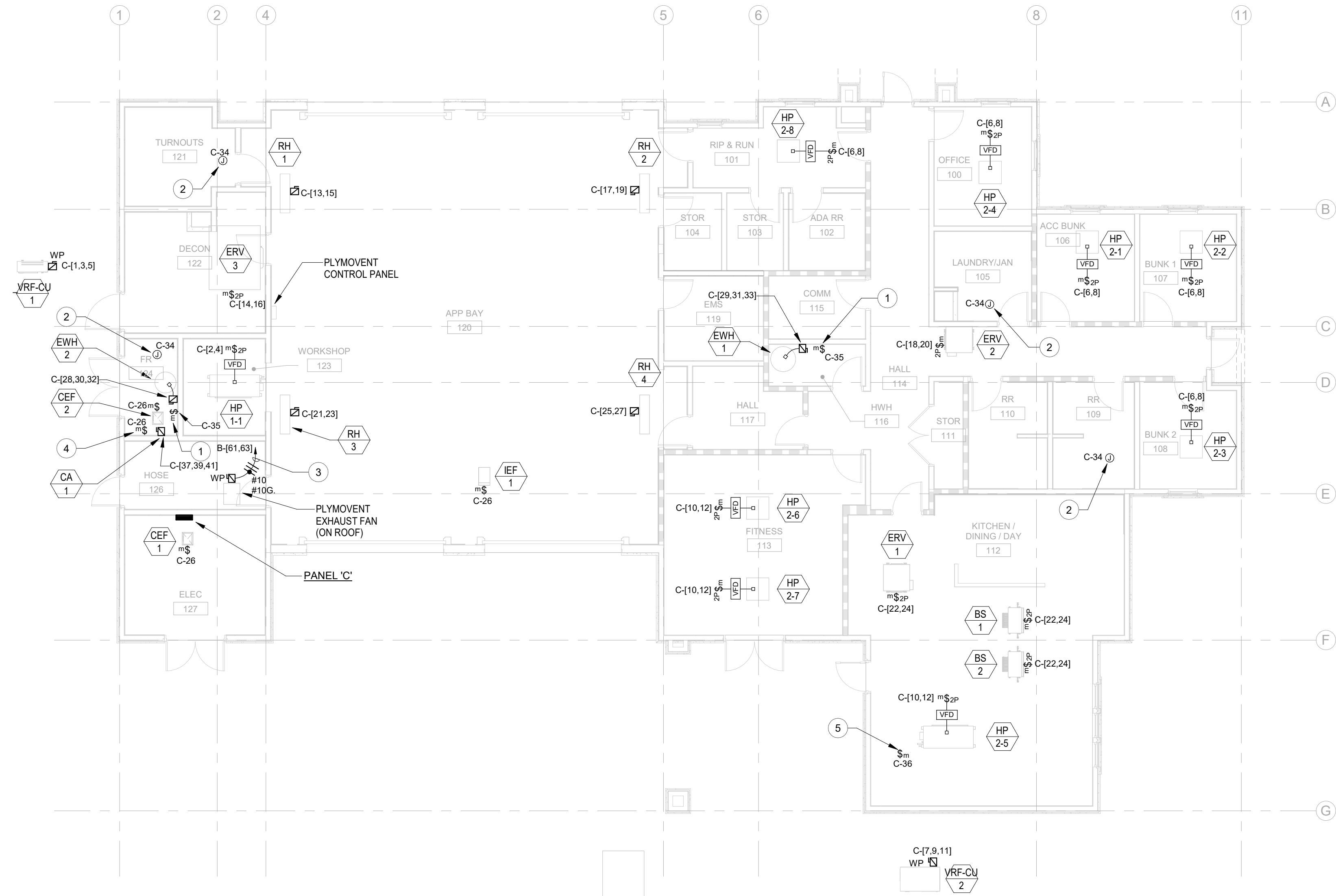
PROJECT NUMBER:
211101

SHEET TITLE:
FLOOR PLAN -
POWER &
SIGNAL

SHEET NUMBER:

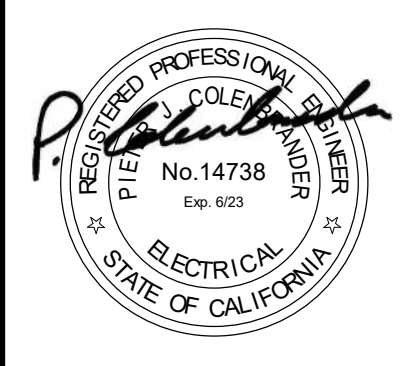
E3.1

MECHANICAL EQUIPMENT FEEDER SCHEDULE									
TYPE	NUMBER	VOLTAGE	POLES	MCA	KVA	MOCP	WIRE & CONDUIT	PANEL	CKT #
BS	1	208 V	2	0.09 A	0.02 KVA		(2) #12 + (1) #12G. IN 3/4" C.	C 22.24	
BS	2	208 V	2	0.09 A	0.02 KVA		(2) #12 + (1) #12G. IN 3/4" C.	C 22.24	
CA	1	208 V	3	26.6 A	9.58 KVA	35	(3) #10 + (1) #10G. IN 3/4" C.	C 37.39,41	
CEF	1	120 V	1	0.304 A	0.04 KVA	15	(2) #12 + (1) #12G. IN 3/4" C.	C 26	
CEF	2	120 V	1	0.304 A	0.04 KVA	15	(2) #12 + (1) #12G. IN 3/4" C.	C 26	
ERV	1	208 V	2	6.88 A	1.14 KVA	15	(2) #12 + (1) #12G. IN 3/4" C.	C 22.24	
ERV	2	208 V	2	6.88 A	1.14 KVA	15	(2) #12 + (1) #12G. IN 3/4" C.	C 18.20	
ERV	3	208 V	2	6.82 A	1.13 KVA	10	(2) #12 + (1) #12G. IN 3/4" C.	C 14.16	
EWH	1	208 V	3	50 A	18.01 KVA	60	(3) #4 + (1) #10G. IN 1" C.	C 29.31,33	
EWH	2	208 V	3	33.3 A	12.00 KVA	40	(3) #8 + (1) #10G. IN 3/4" C.	C 28.30,32	
HP	1-1	208 V	2	3.1 A	0.52 KVA		(2) #12 + (1) #12G. IN 3/4" C.	C 2.4	
HP	2-1	208 V	2	0.25 A	0.04 KVA		(2) #12 + (1) #12G. IN 3/4" C.	C 6.8	
HP	2-2	208 V	2	0.25 A	0.04 KVA		(2) #12 + (1) #12G. IN 3/4" C.	C 6.8	
HP	2-3	208 V	2	0.25 A	0.04 KVA		(2) #12 + (1) #12G. IN 3/4" C.	C 6.8	
HP	2-4	208 V	2	0.25 A	0.04 KVA		(2) #12 + (1) #12G. IN 3/4" C.	C 6.8	
HP	2-5	208 V	2	5.2 A	0.87 KVA		(2) #12 + (1) #12G. IN 3/4" C.	C 10.12	
HP	2-6	208 V	2	0.25 A	0.04 KVA		(2) #12 + (1) #12G. IN 3/4" C.	C 10.12	
HP	2-7	208 V	2	0.25 A	0.04 KVA		(2) #12 + (1) #12G. IN 3/4" C.	C 10.12	
HP	2-8	208 V	2	0.25 A	0.04 KVA		(2) #12 + (1) #12G. IN 3/4" C.	C 6.8	
IEF	1	120 V	1	0.86 A	0.10 KVA	15	(2) #12 + (1) #12G. IN 3/4" C.	C 26	
PLY	EF	208 V	2	16 A	3.33 KVA	20	(2) #10 + (1) #10G. IN 3/4" C.	B 61.63	
RH	1	208 V	2	29.5 A	6.14 KVA	35	(2) #10 + (1) #10G. IN 3/4" C.	C 13.15	
RH	2	208 V	2	29.5 A	6.14 KVA	35	(2) #10 + (1) #10G. IN 3/4" C.	C 17.19	
RH	3	208 V	2	29.5 A	6.14 KVA	35	(2) #10 + (1) #10G. IN 3/4" C.	C 21.23	
RH	4	208 V	2	29.5 A	6.14 KVA	35	(2) #10 + (1) #10G. IN 3/4" C.	C 25.27	
VRV-HP	1	208 V	3	24 A	6.92 KVA	40	(3) #10 + (1) #10G. IN 3/4" C.	C 1.3,5	
VRV-HP	2	208 V	3	30.9 A	8.91 KVA	40	(3) #10 + (1) #10G. IN 3/4" C.	C 7.9,11	



1 FLOOR PLAN - MECHANICAL POWER
1/8" = 1'-0"

- # SHEET NUMBERED NOTES
- 1 PROVIDE POWER CONNECTION AND 1-POLE MOTOR RATED SWITCH FOR WATER HEATER CIRCULATION PUMP. COORDINATE EXACT LOCATION WITH DIVISION 22.
 - 2 PROVIDE POWER CONNECTION FOR TRAP PRIMER IN FLOOR. COORDINATE EXACT LOCATION WITH DIVISION 22.
 - 3 HOMERUN TO CIRCUIT INDICATED VIA PLYMOVENT CONTROL PANEL, MOUNTED TO WEST APP BAY WALL. COORDINATE PANEL LOCATION WITH ARCHITECT.
 - 4 PROVIDE POWER CONNECTION AND MOTOR RATED DISCONNECT SWITCH FOR REFRIGERANT DRYER, ADJACENT TO AIR COMPRESSOR. COORDINATE EXACT LOCATION WITH DIVISION 22.
 - 5 PROVIDE MOTOR RATED SWITCH AND POWER CONNECTION FOR MOTORIZED DAMPER. COORDINATE EXACT LOCATION WITH DIVISION 23.



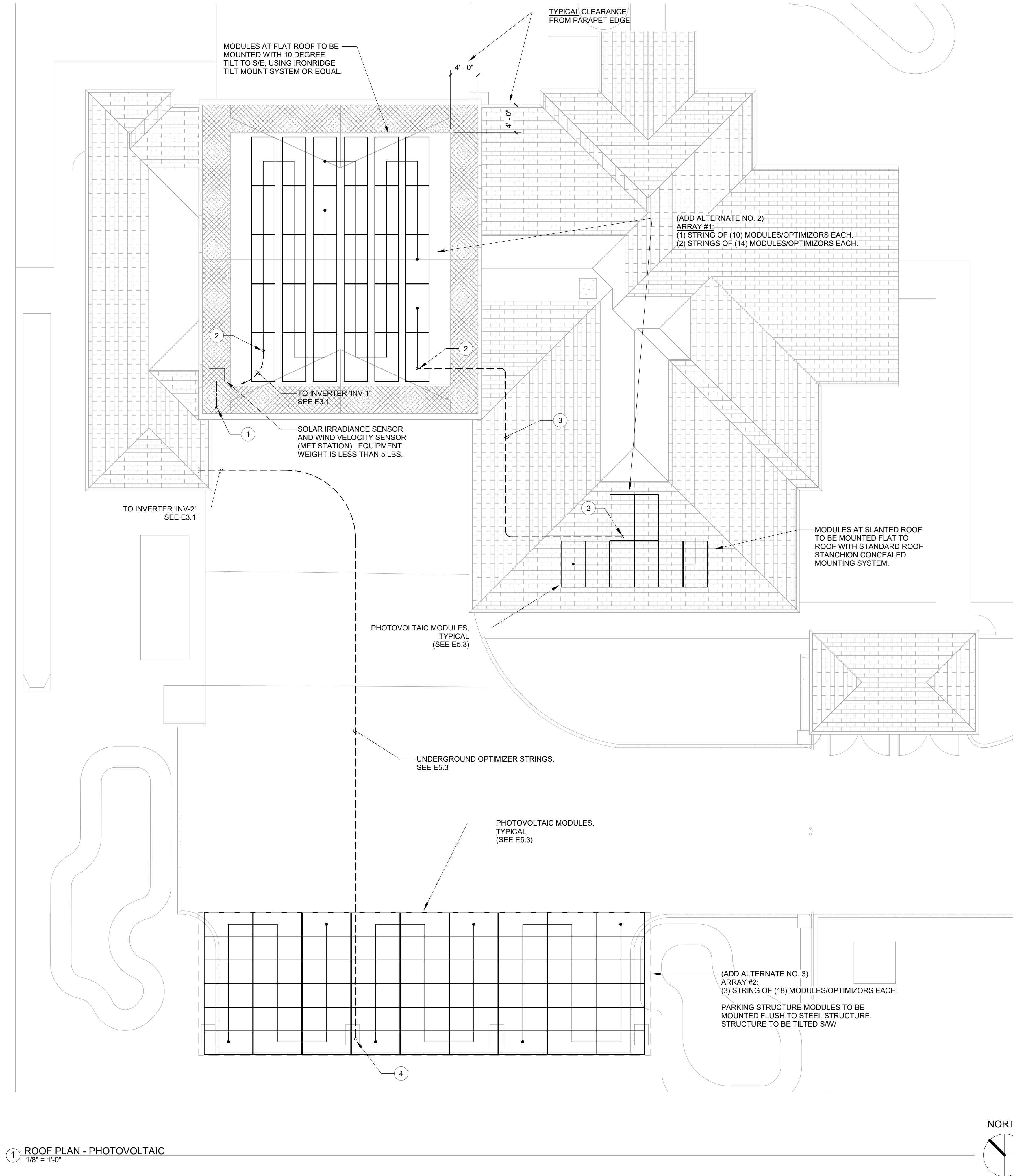
PROJECT:
CITY OF MORGAN HILL
BUTTERFIELD FIRE STATION
17285 Butterfield Blvd.
Morgan Hill, CA 95037



DESCRIPTION: 95% CONSTR. 01/16/23
DRAWINGS
100% CONSTR. 03/23/23
DRAWINGS

NOT FOR CONSTRUCTION
PROJECT NUMBER:
211101
SHEET TITLE:
FLOOR PLAN - MECHANICAL POWER

SHEET NUMBER:
E3.2



SHEET NUMBERED NOTES

- 1 CONDUIT PENETRATION TO APP BAY BELOW. CONTINUE TO ELEC ROOM 127.
- 2 PROVIDE WEATHER-TIGHT PENETRATION TO ATTIC BELOW.
- 3 STRING ROUTED CONCEALED UNDER ROOF, IN ATTIC, IN 3/4"C. LABELED "SOLAR CIRCUIT".
- 4 RUN UP INSIDE STRUCTURAL COLUMN AND PIER.

PV SYSTEM SUMMARY

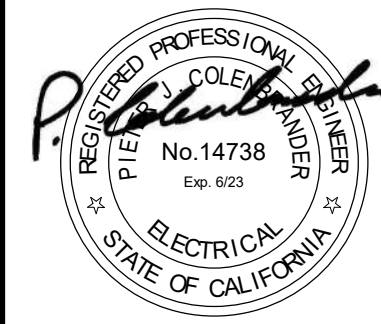
PV MODULE:	CANADIAN SOLAR BIHIKU CS3W-445
WATTS/MODULE:	445 W
QUANTITY:	92
PV INVERTER:	SOLAR EDGE SE17.3KUS EACH
INVERTER KW:	17.3 EACH
CEC EFFECENCY:	97.5%
QUANTITY:	(2)
TOTAL DC RATING:	40940 W
CEC RATING:	39000 W

MOUNTING & STRUCTURE:
MODULES SHALL BE MOUNTED IN CONFIGURATIONS SHOWN ON ROOF PLAN. ATTACH MODULES TO STRUCTURE AS DEPICTED ON ARCHITECTURAL AND STRUCTURAL DRAWINGS.

CALCULATIONS:
PROVIDE ALL STRUCTURAL AND WIND LOADING CALCULATIONS REQUIRED.

GENERAL NOTES

- A. ALL STRING/OPTIMIZER WIRING SHALL BE NEATLY ARRANGED ALONG UNDERSIDE OF MODULES UTILIZING APPROPRIATE STRING MANAGEMENT HARDWARE, MATCHING THE RACKING SYSTEM.
- B. ALL ROOF MOUNT CONDUIT SHALL BE SET ON DURA-BLOC (OR EQUAL) ROOFTOP SUPPORTS WITH MINIMUM 3'-1/2" DISTANCE FROM BOTTOM OF CONDUIT TO TOP OF ROOF SURFACE. WHERE STUBS ARE SHOWN; STUB OUT BENEATH MODULES WITH A MINIMUM 5" GOOSENECK IN DOWNWARD DIRECTION TO MITIGATE WATER ENTRY.
- C. CONTRACTOR SHALL DEVELOP STRINGING PLAN WITH STRINGS SEPARATED IN QUANTITIES SHOWN ON DIAGRAM. PROVIDE ALL CONDUIT JUMPERS REQUIRED TO ACCOMMODATE CONSTRUCTED STRING PLAN.
- D. ALL SOLAR CIRCUITS FOR ROOF ARRAY TO BE RUN CONCEALED. NO VISIBLE EMT OR RGS ON ROOF, EXCEPT AS REQUIRED AND NOT VISIBLE FROM BELOW.



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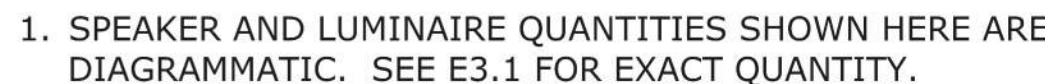
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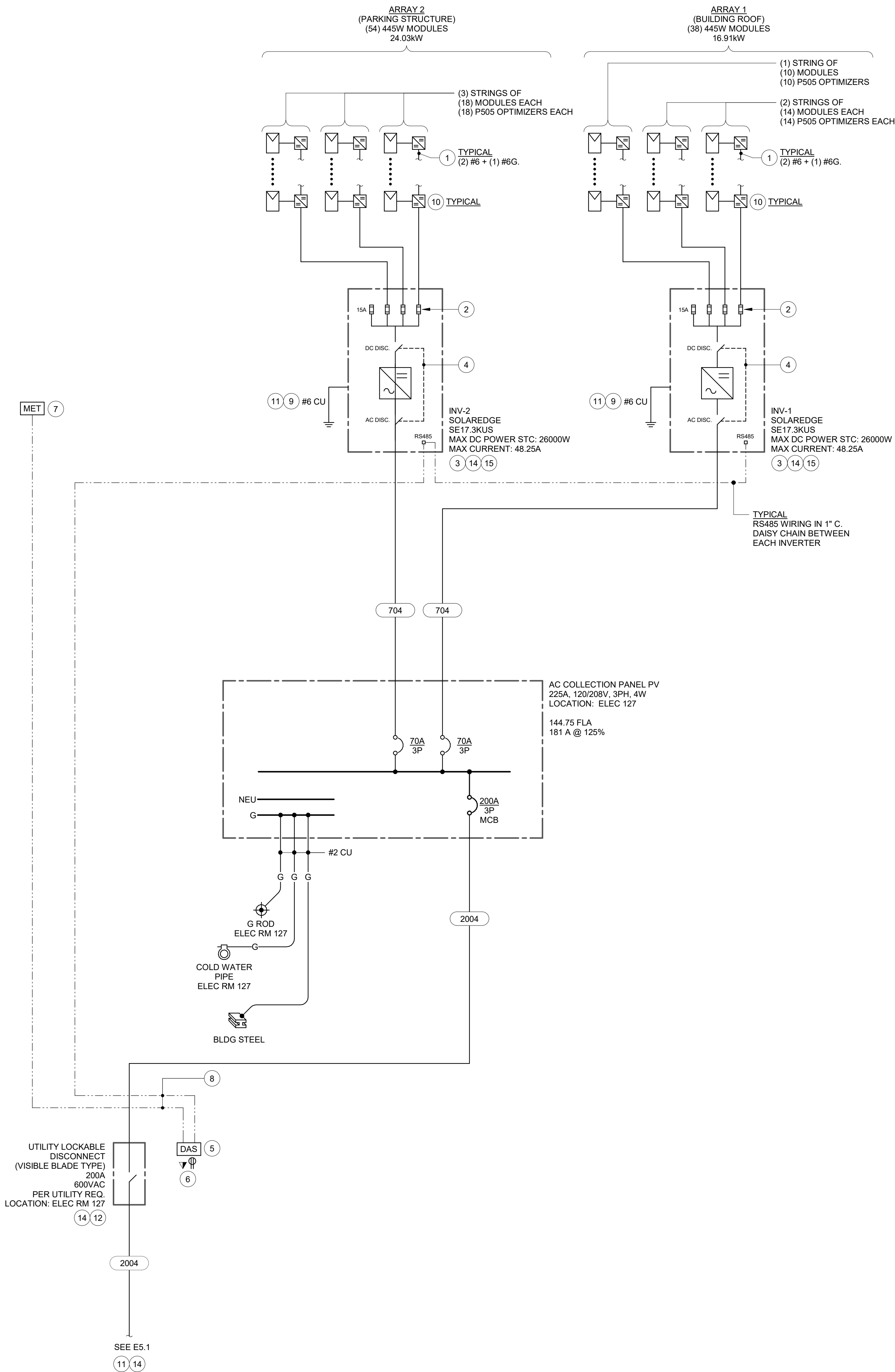
SHEET TITLE:
ROOF PLAN -
PHOTOVOLTAIC

SHEET NUMBER:

E3.3

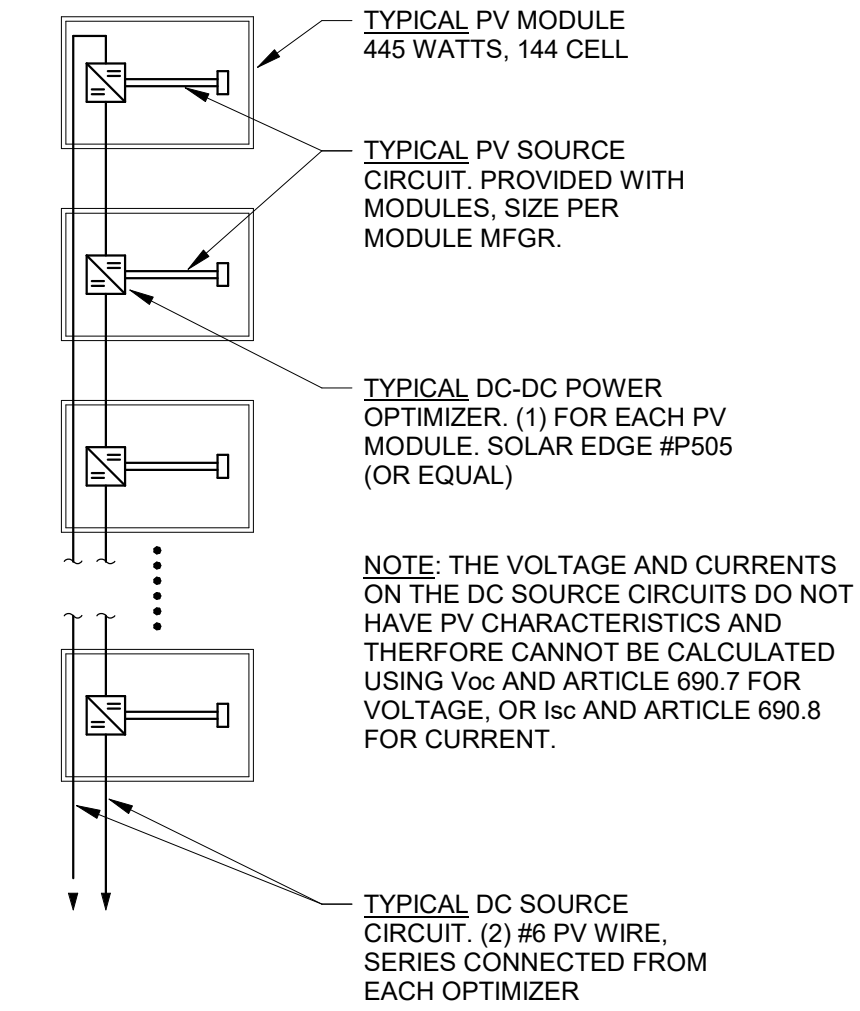
E5.1





1 SINGLE LINE DIAGRAM - PHOTOVOLTAIC
NO SCALE

PV SYSTEM FEEDER SCHEDULE		
ID	SETS	FEEDER DESCRIPTION
2004	(1) SET	(4) #3/0 & (1) #4 G IN (1) 2" C.
704	(1) SET	(4) #4 & (1) #6 G IN (1) 1 1/4" C.
FEEDER TAG KEY		
400 4 N		
INDICATES DOUBLE NEUTRAL		
WIRE QUANTITY		
FEEDER AMPACITY		



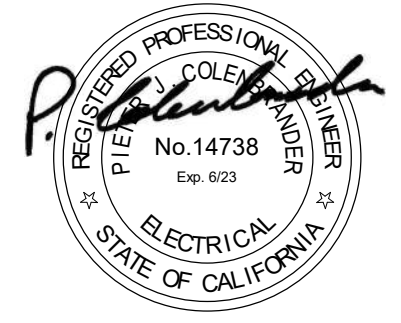
2 DC-DC OPTIMIZER DETAIL
NO SCALE

SHEET NUMBERED NOTES

- DC OUTPUT CIRCUIT FROM DC-DC OPTIMIZER TO INVERTER DC INPUT CONNECTIONS. SEE DETAIL 2, THIS SHEET. TYPICAL (2) #6 + (1) #6G. ALSO SEE GENERAL NOTES ON SHEET E9.01 FOR WIRING MANAGEMENT AND REQUIREMENTS.
- DC INTERFACE - 1000V DC, NEMA 3R, UL 1741. PROVIDE WITH 15A FUSES IN DIN RAIL MOUNTED FUSE HOLDERS. FUSE BOX SHALL BE LOCATED IN ELEC 127, ADJACENT TO PV INVERTER, FOR DC STRING INPUT ONTO INVERTER DC BUS.
- INVERTERS TO BE CERTIFIED WITH THE CALIFORNIA ENERGY COMMISSION AND UL1741 LISTED. INVERTER ASSEMBLY TO INCLUDE INTEGRAL AC AND DC DISCONNECTS (NOT SHOWN).
- INVERTERS SHALL BE PROVIDED WITH A COMBINED DC AND AC DISCONNECTION MEANS. ASSEMBLY SHALL INCLUDE DC FUSES AND FUSE HOLDERS IN QUANTITIES REQUIRED.
- PROVIDE A COMPLETE PHOTOVOLTAIC DATA ACQUISITION SYSTEM (DAS) TO ALLOW OWNER TO REMOTELY VIEW REAL-TIME VISUAL METERS OF SYSTEM PERFORMANCE, HISTORICAL GRAPHS, LOCAL WEATHER DATA AND CUSTOM PROJECT DETAILS TO CLIENT SATISFACTION. THIS INCLUDES ALL HEAD END EQUIPMENT, SOFTWARE AND LICENSING FEES. THIS SHALL ALSO INCLUDE INSTALLATION OF ANY METEOROLOGICAL EQUIPMENT REQUIRED AT THE ROOF FOR WEATHER MONITORING. UTILIZE DECK MONITORING SOFTWARE OR EQUAL.
- PROVIDE AND INSTALL FULLY TERMINATED CAT. 6 NETWORK JACK/CABLE AT DAS EQUIPMENT & 120V DUPLEX RECEPTACLE. DAS EQUIPMENT LOCATION BY SOLAR INTEGRATOR. CONFIRM EXACT LOCATION PRIOR TO ROUGH-IN.
- PROVIDE COMPACT METEOROLOGICAL WEATHER STATION (MET) AT ROOF. COLUMBIA WEATHER SYSTEMS OR EQUAL. MET STATION SHALL BE FULLY COMPATIBLE WITH DAS EQUIPMENT, CONFIRM COMPATIBILITY PRIOR TO ROUGH-IN.
- PROVIDE 1" RACEWAY WITH RS-485 WIRING FOR MONITORING EQUIPMENT COMMUNICATIONS. CONFIRM EXACT WIRING TYPE AND QUANTITY WITH DAS AND MET SYSTEMS SUPPLIERS PRIOR TO ROUGH-IN.
- PROVIDE GROUNDING CONNECTION FROM INVERTER ASSEMBLY TO MAIN SWITCHBOARD GROUND BUS PER CEC 690.47 AND MANUFACTURERS INSTRUCTIONS.
- PV SYSTEM SINGLE MODULE OPTIMIZER. OPTIMIZERS SHALL BE PROVIDED WITH INTERNAL CURRENT LIMITING FUNCTIONS AS DESCRIBED IN 690.8(B)(2) EXCEPTION 1.
- ALL PV SYSTEM INTERCONNECTIONS WITH THE DISTRIBUTION SYSTEM SHALL COMPLY WITH CEC ARTICLE 690. PROVIDE AND INSTALL ALL GROUNDING AS REQUIRED BY NEC ARTICLE 250, 690, 705 AND SYSTEM MANUFACTURER.
- KNIFE TYPE DISCONNECT TO BE CLEARLY LABELED "SOLAR SYSTEM DISCONNECT" AND LOCATED WITHIN 10 FT. OF THE MAIN SERVICE METER PER UTILITY COMPANY REGULATIONS. EXACT DISCONNECT TYPE SHALL BE A UTILITY CO. APPROVED PHOTOVOLTAIC DISCONNECT. COORDINATE EXACT TYPE WITH PG&E PRIOR TO ROUGH-IN.
- PROVIDE PV SYSTEM PLACARDS AS REQUIRED BY LOCAL UTILITY COMPANY.
- PHOTOVOLTAIC SYSTEM INVERTER, RAPID SHUT-DOWN COMPLIANT, WITH MODULE LEVEL OPTIMIZED STRING CONFIGURATIONS.

PV SYSTEM GENERAL NOTES

- UNLESS OTHERWISE NOTED, ALL WORK SHOWN ON DRAWINGS IS NEW AND TO BE PROVIDED AND INSTALLED UNDER THIS CONTRACT.
- ALL EXTERIOR EQUIPMENT / BOXES / COMBINERS SHALL BE NEMA 3R, RATED FOR OUTDOOR USE.
- ALL BARE METAL RACEWAYS SHALL BE RGS; U.O.N.
- ALL CONDUCTORS TO BE SIZED PER NEC T310.15(B)(16) WITH APPROPRIATE AMBIENT TEMPERATURE CORRECTION FACTORS PER NEC T310.15(B)(3)(c) AND/OR NEC 310.15(B)(2). CONDUCTORS SHALL ALSO BE DERATED PER NEC TABLE 310.15(B)(3)(a) FOR MORE THAN (3) CURRENT CARRYING CONDUCTORS IN A COMMON RACEWAY.
- ALL CONDUIT CONNECTORS TO OUTLET OR JUNCTION BOXES SHALL HAVE INSULATED THROATS (MANUFACTURED AS AN INTEGRAL PART OF THE CONNECTOR). AFTER-MARKET INSERTABLE THROATS ARE NOT ACCEPTABLE.
- COORDINATE ALL UTILITY SYSTEM RELATED ITEMS WITH UTILITY CO, INCLUDING APPLICATION FOR UTILITY INTERCONNECT AGREEMENT.
- ALL SYSTEM COMPONENTS AND EQUIPMENT / WIRING TO COMPLY WITH NEC ARTICLE 690 AND OTHER APPLICABLE CODES, REGULATIONS AND LOCAL ORDINANCES.
- ALL INVERTERS TO BE CERTIFIED WITH THE CALIFORNIA ENERGY COMMISSION AND UL1741 LISTED, UTILITY INTERACTIVE, SIZE AND RATINGS AS NOTED ON PLANS.
- ALL AC FEEDERS SIZED TO NEC T310.15(B)(16), 75 DEGREE COLUMN WHEN USED WITH 75 DEGREE TERMINATIONS AND EQUIPMENT. USE 90 DEGREE COLUMN WHEN USED WITH PV EQUIPMENT RATED WITH 90 DEGREE TERMINATIONS. CONDUCTOR SIZES TO BE BASED ON FULL EQUIPMENT NAMEPLATE LOADS.
- ALL CONDUIT FILL (AC AND DC) TO BE 40% MAXIMUM.
- PROVIDE AN EQUIPMENT GROUNDING CONDUCTOR IN EACH CONDUIT (BOTH AC AND DC), SIZED PER NEC 250 AND 690.
- ALL DC CIRCUITS TO BE 2-WIRE + GROUND, (90 DEGREE COLUMN), RATED FOR MIN. 156% OF I_{sc}.
- ALL AC AND DC WIRING IN CONDUIT TO BE RHW-2, THWN-2, OR XHHW-2 (90 DEGREE) WET RATED FOR USE WITH 90 DEGREE LISTED TERMINALS ON PV EQUIPMENT.
- ALL DC WIRING SHALL BE MINIMUM 1000V RATED.
- ALL DC WIRING TO BE LISTED "PV WIRE" (90 DEGREE) WET RATED AND SUNLIGHT RESISTANT. LOOSELY BUNDLE AND SECURE ALL INDIVIDUAL STRING WIRING NEATLY ALONG UNDERSIDE OF MODULES.
- ALL GROUNDING PER NEC 250 AND 690-45. BOND ALL METAL SURFACES / FRAMES / EQUIPMENT WITH MIN. #2 GROUND STRAP AS NOTED ON DETAIL 3/EPV5.2. WEEB MAY BE UTILIZED AT MODULE FRAME GROUNDING.
- ALL PV MODULES TO BE UL 1703 LISTED, WITH TYPE 2 FIRE RATING.
- ALL UNDERGROUND RACEWAYS TO BE SCHEDULE 40 PVC, UNLESS OTHERWISE NOTED.
- ALL IN-GROUND PULL BOXES TO BE FLUSH STYLE WITH CONCRETE LIDS.
- INVERTER(S) SHALL BE PROVIDED WITH DC AFCI PROTECTION PER CEC 690.11. INVERTERS SHOWN ON THIS SET OF DOCUMENTS ARE PROVIDED WITH DC AFCI PROTECTION, ANY SUBSTITUTIONS SHALL BE SUBMITTED TO THE ELECTRICAL ENGINEER OF RECORD FOR REVIEW.
- INVERTER(S) SHALL BE PROVIDED WITH INTEGRAL GROUND-FAULT PROTECTION PER CEC 690.35. INVERTERS SHOWN ON THIS SET OF DOCUMENTS ARE PROVIDED WITH INTEGRAL GROUND FAULT PROTECTION, ANY SUBSTITUTIONS SHALL BE SUBMITTED TO THE ELECTRICAL ENGINEER OF RECORD FOR REVIEW.
- ALL PV MODULE DC CABLE CONNECTORS SHALL BE MC4 TYPE (OR EQUAL), REQUIRING A TOOL FOR OPENING OF PV CONNECTIONS PER CEC 690.33. PV MODULES SHOWN ON THIS SET OF DOCUMENTS ARE PROVIDED WITH MC4 CONNECTORS, ANY SUBSTITUTIONS SHALL BE SUBMITTED TO THE ELECTRICAL ENGINEER OF RECORD FOR REVIEW.



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PROJECT:

CITY OF MORGAN HILL

BUTTERFIELD FIRE
STATION

17285 Butterfield Blvd.
Morgan Hill, CA 95037



DESCRIPTION: DATE:

95% CONSTR. 01/16/23
DRAWINGS

100% CONSTR. 03/23/23
DRAWINGS

NOT FOR
CONSTRUCTION

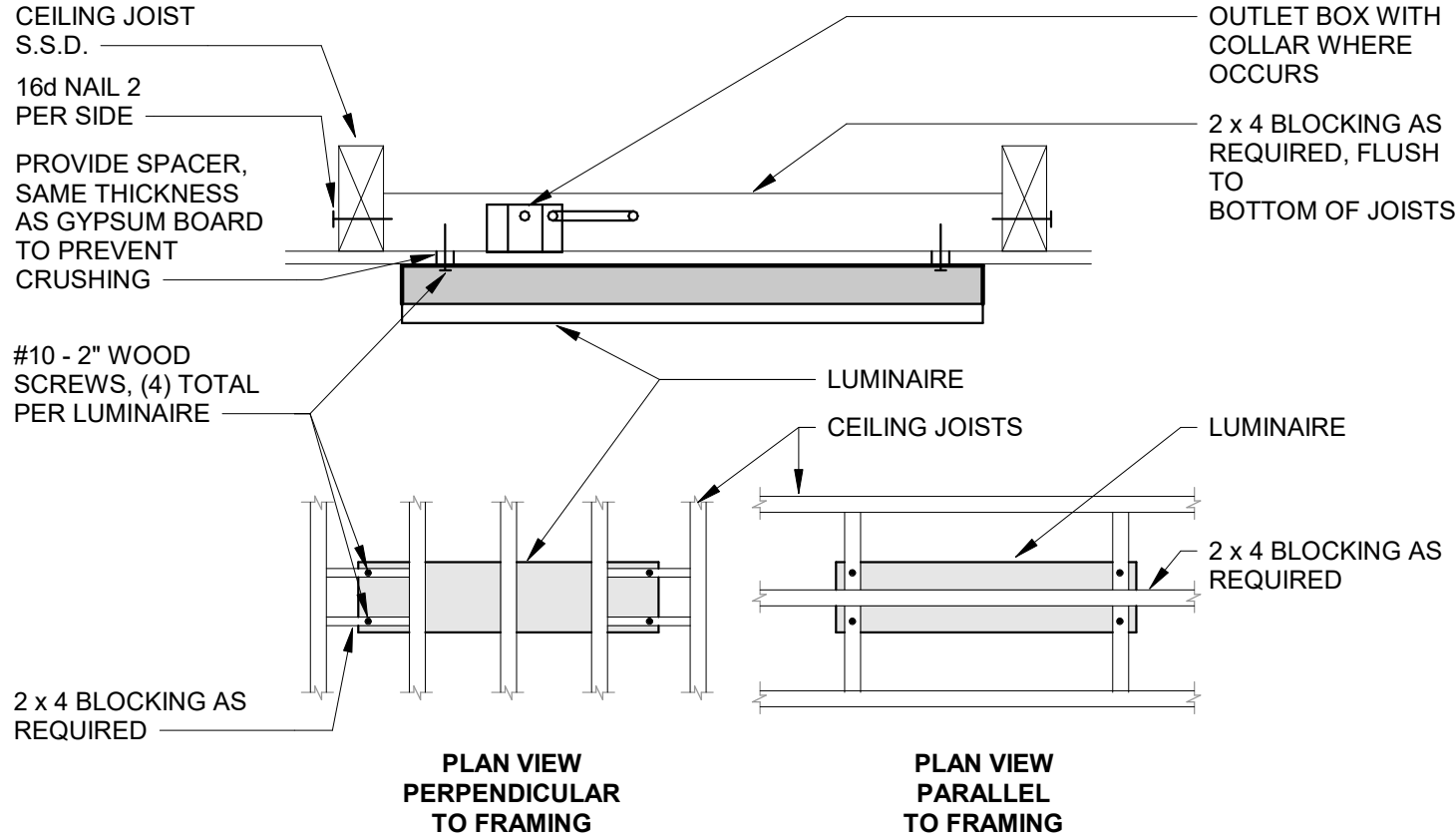
PROJECT NUMBER:
211101

SHEET TITLE:
SINGLE LINE
DIAGRAM -
PHOTOVOLTAIC

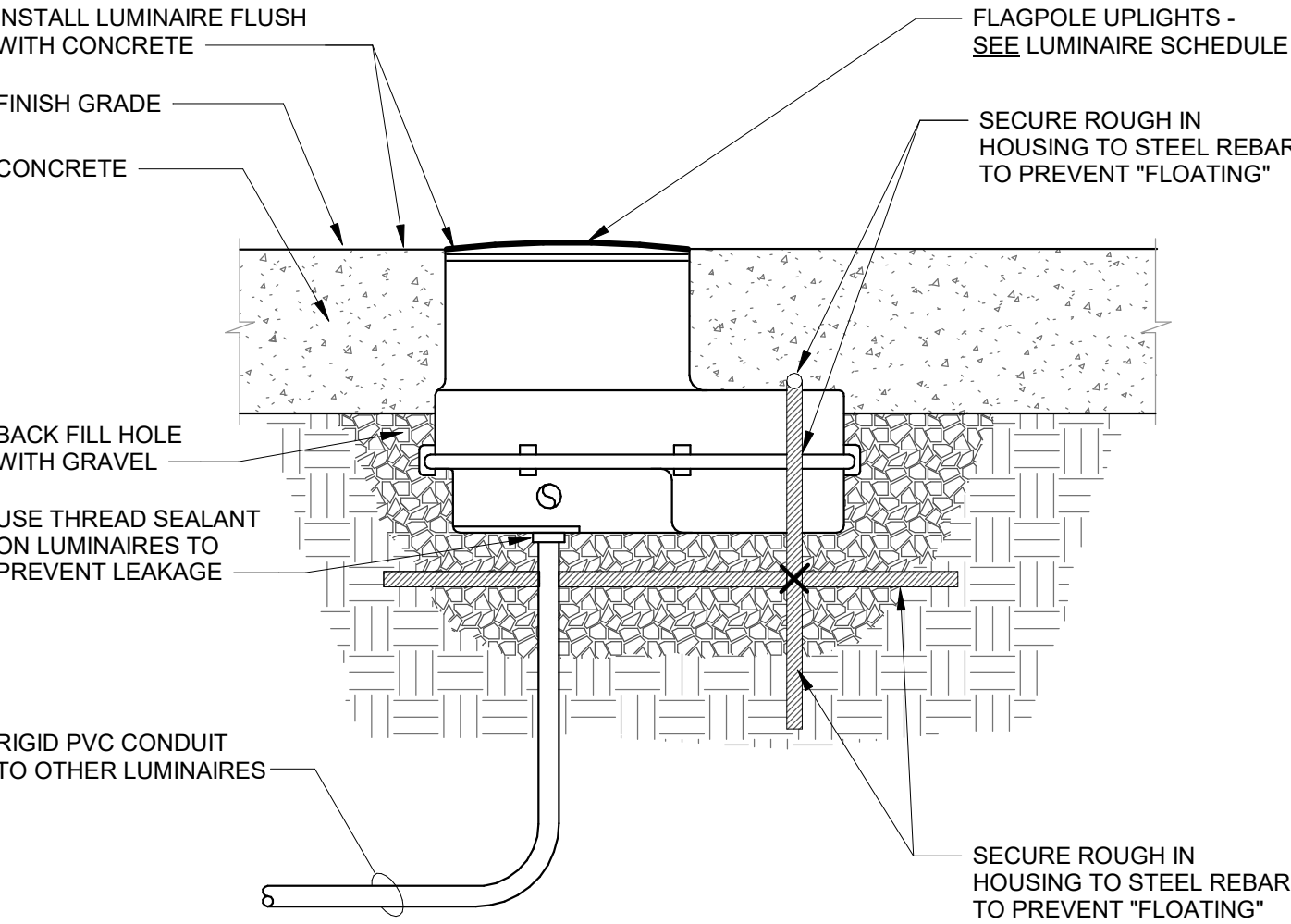
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E5.3

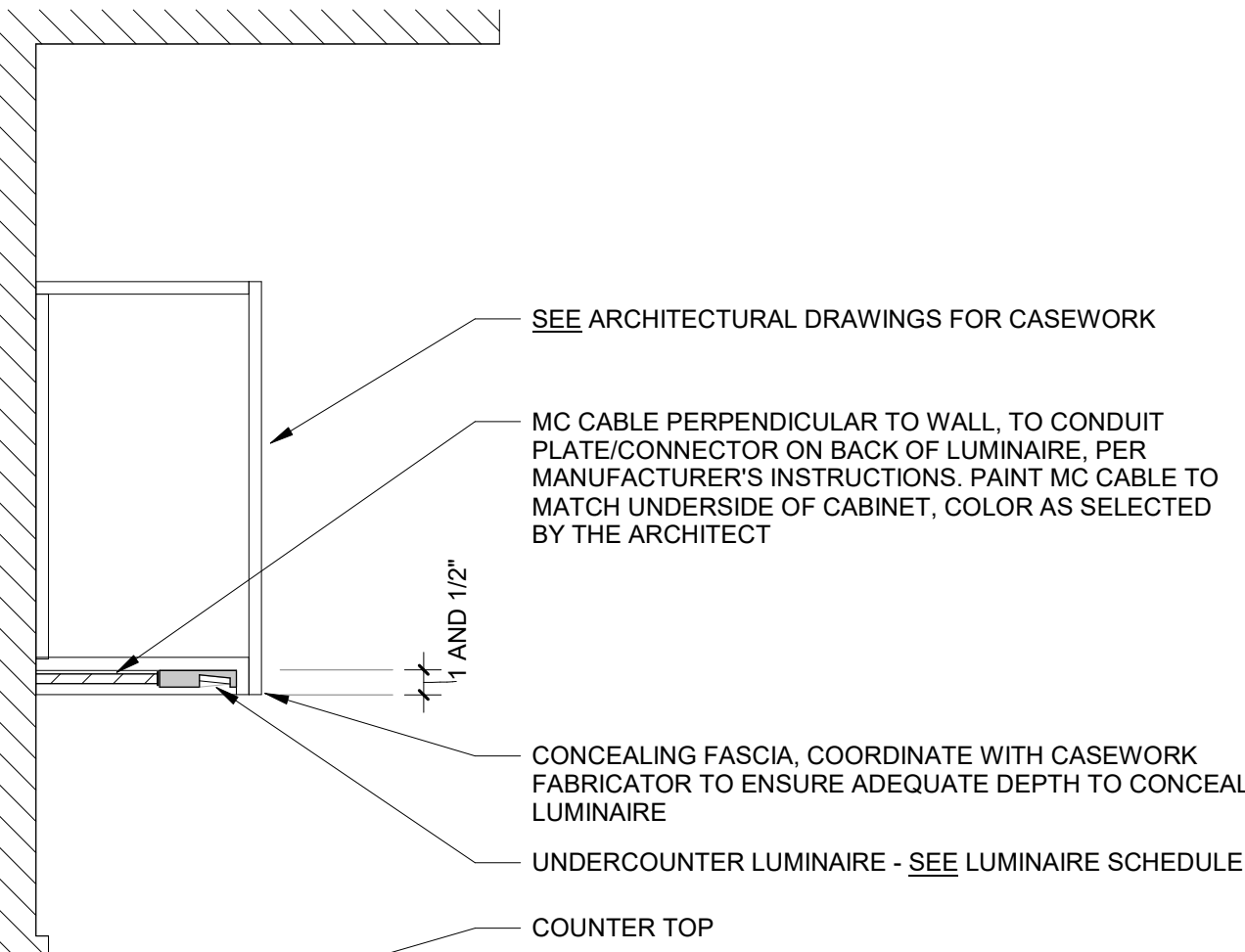
"SEE PLAN VIEW BELOW" TO "2 x 4 BLOCKING AS REQUIRED, FLUSH TO BOTTOM OF JOISTS



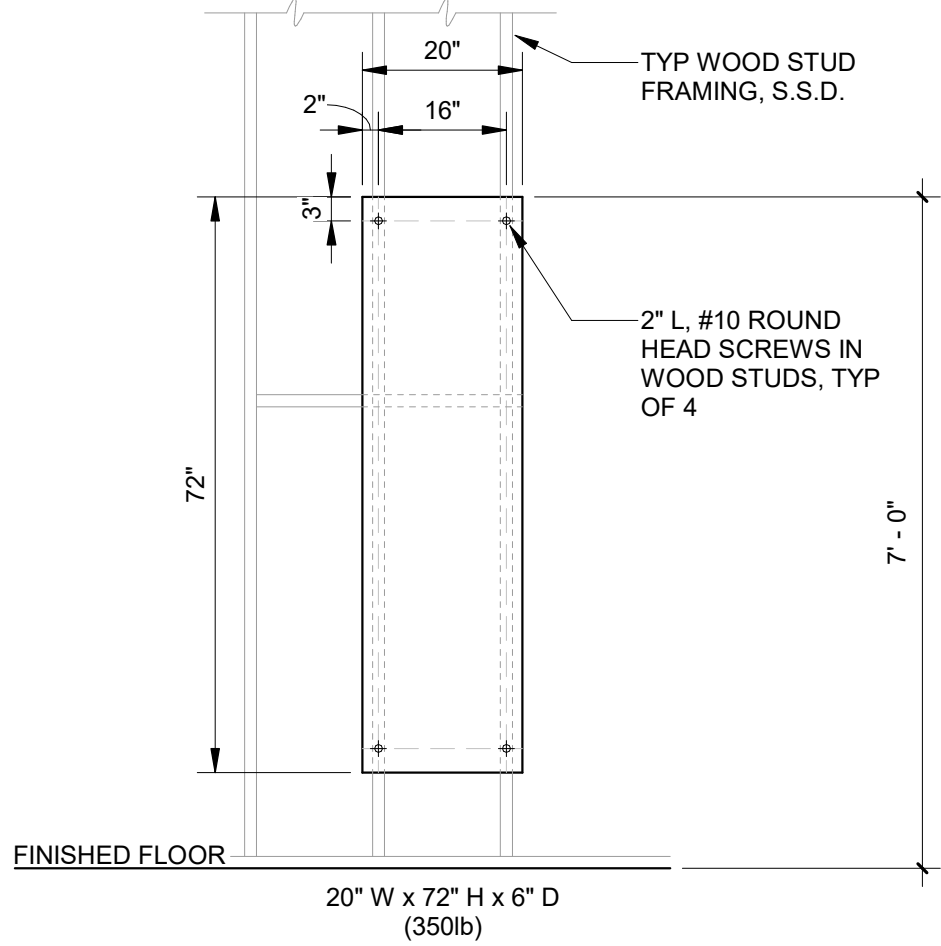
8 TYPICAL SURFACE CEILING MOUNTED LUMINAIRE
NO SCALE



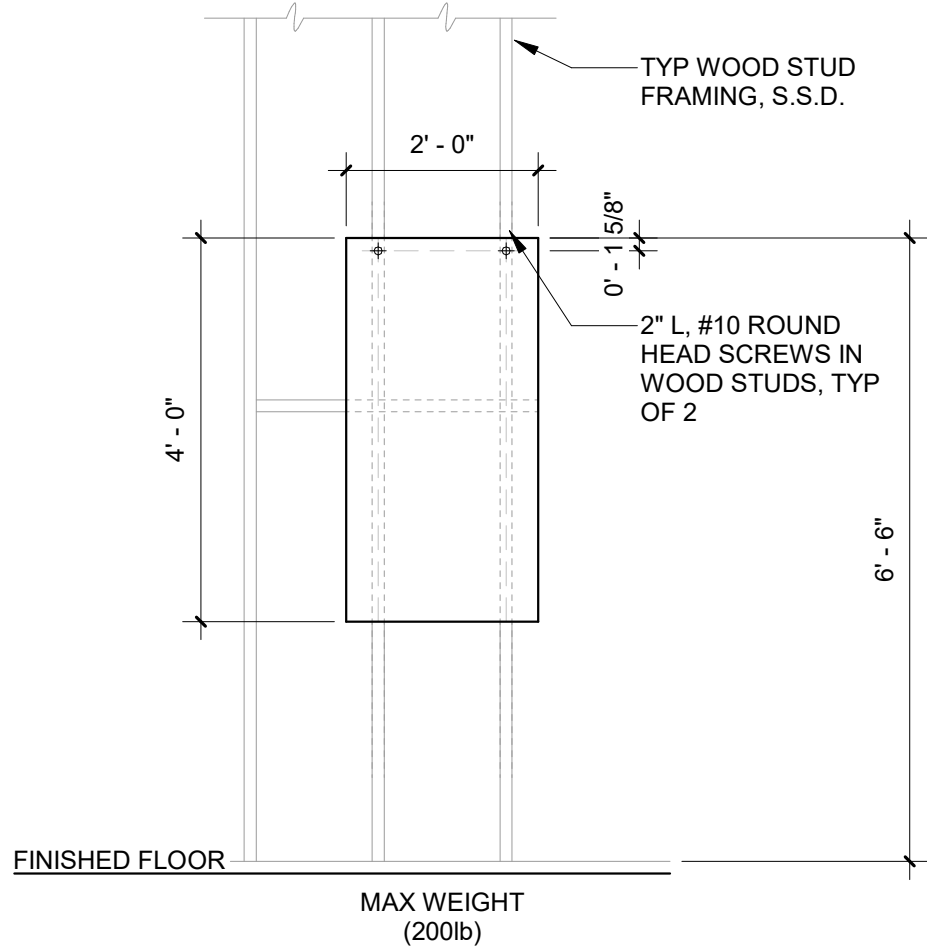
9 IN GRADE UPLIGHT
NO SCALE



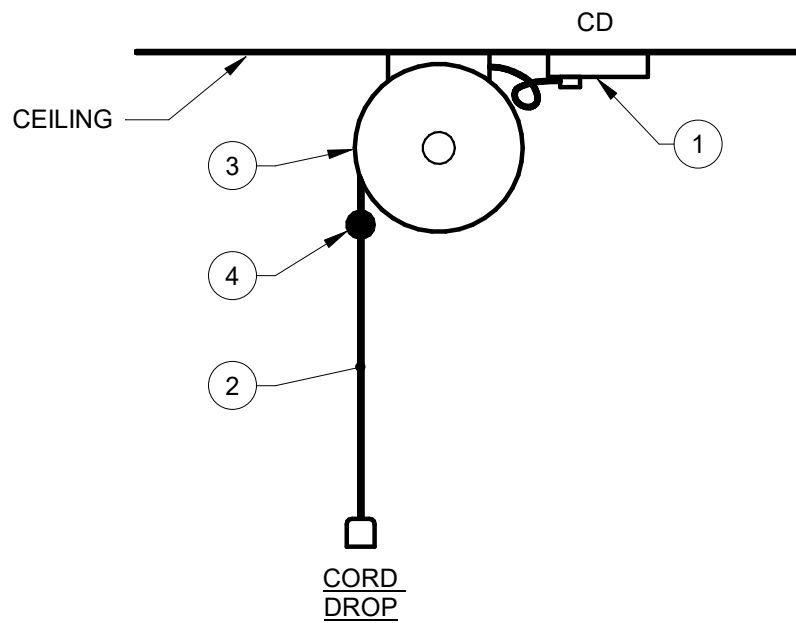
10 UNDERCABINET LUMINAIRE
NO SCALE



5 PANELBOARD MOUNTING - WOOD STUDS
NOT TO SCALE



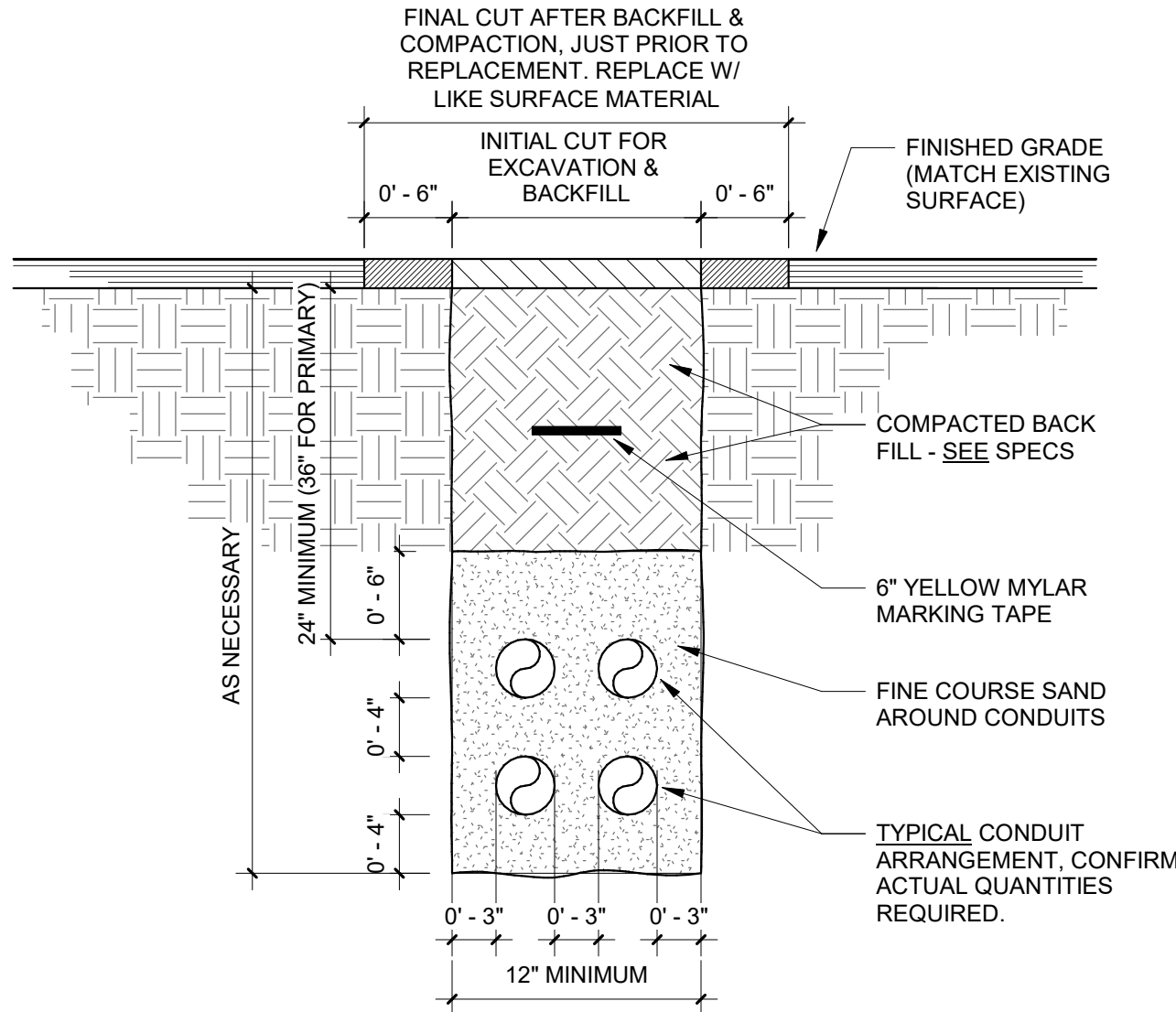
6 FIRE ALARM PANEL MOUNTING - WOOD STUDS
NOT TO SCALE



NUMBERED DETAIL NOTES

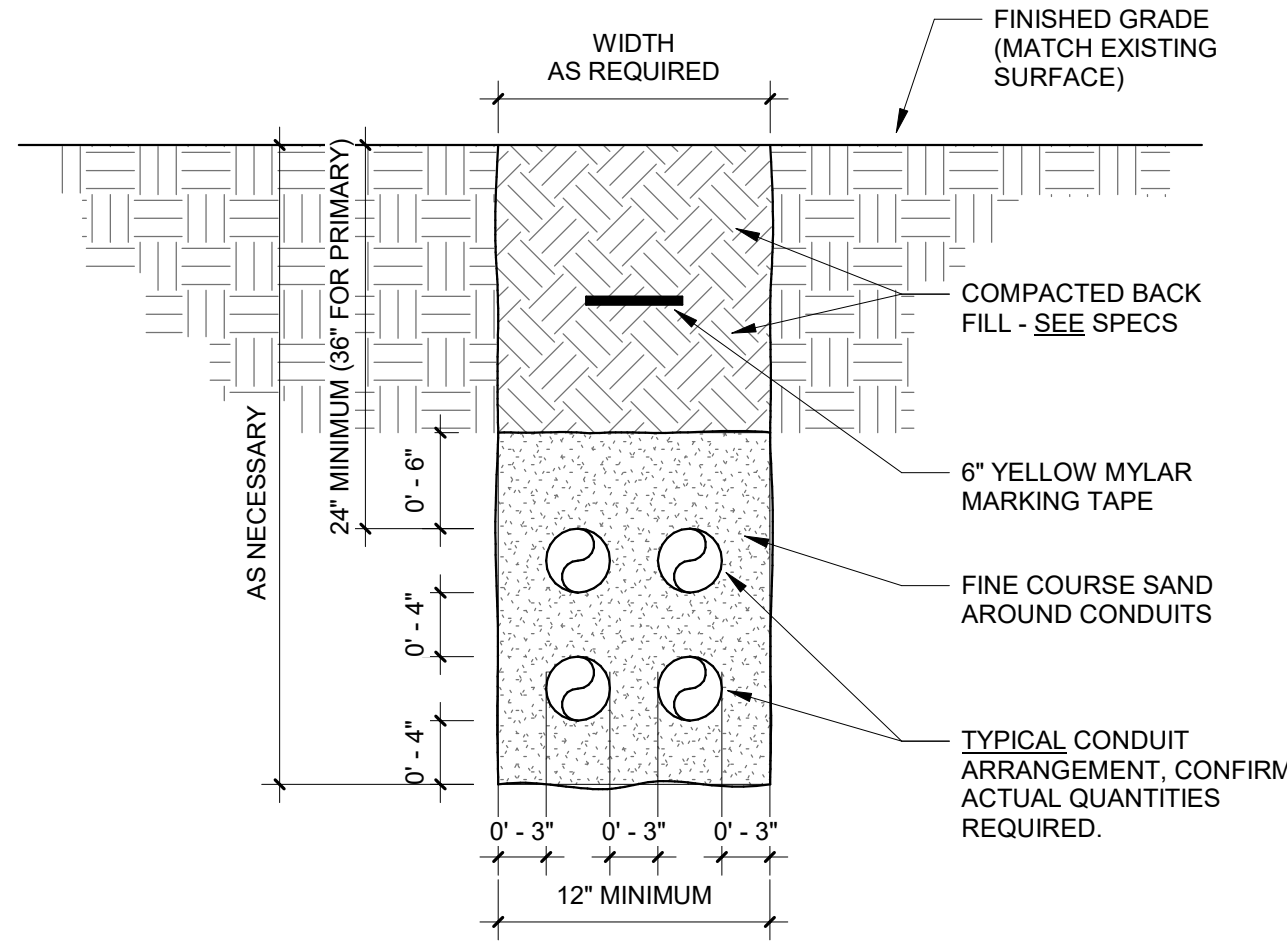
1. TWIST-LOCK RATED RECEPTACLE AS NOTED ON ELECTRICAL PLAN.
2. PROVIDE 40FT MIN. CORD LENGTH.
3. CEILING MOUNTED INDUSTRIAL CORD REEL. ATTACH USING (3) 5/16" TEKS 5 SELF DRILLING SCREWS.
4. SET HEIGHT OF CORD AS DIRECTED BY THE OWNER VIA BALL STOP.

7 TYPICAL CORD DROP REEL
NOT TO SCALE

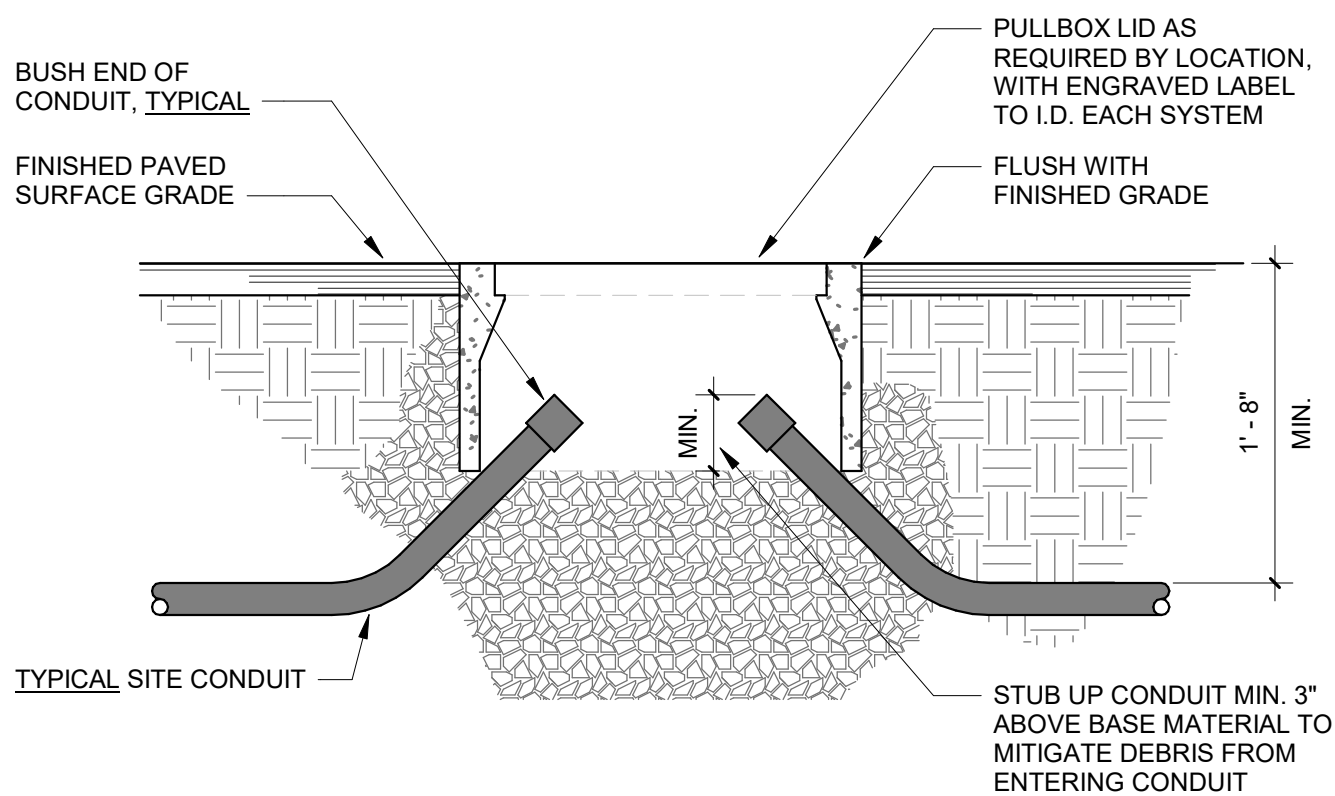


PAVED AREAS

1 TYPICAL TRENCH SECTIONS
NO SCALE

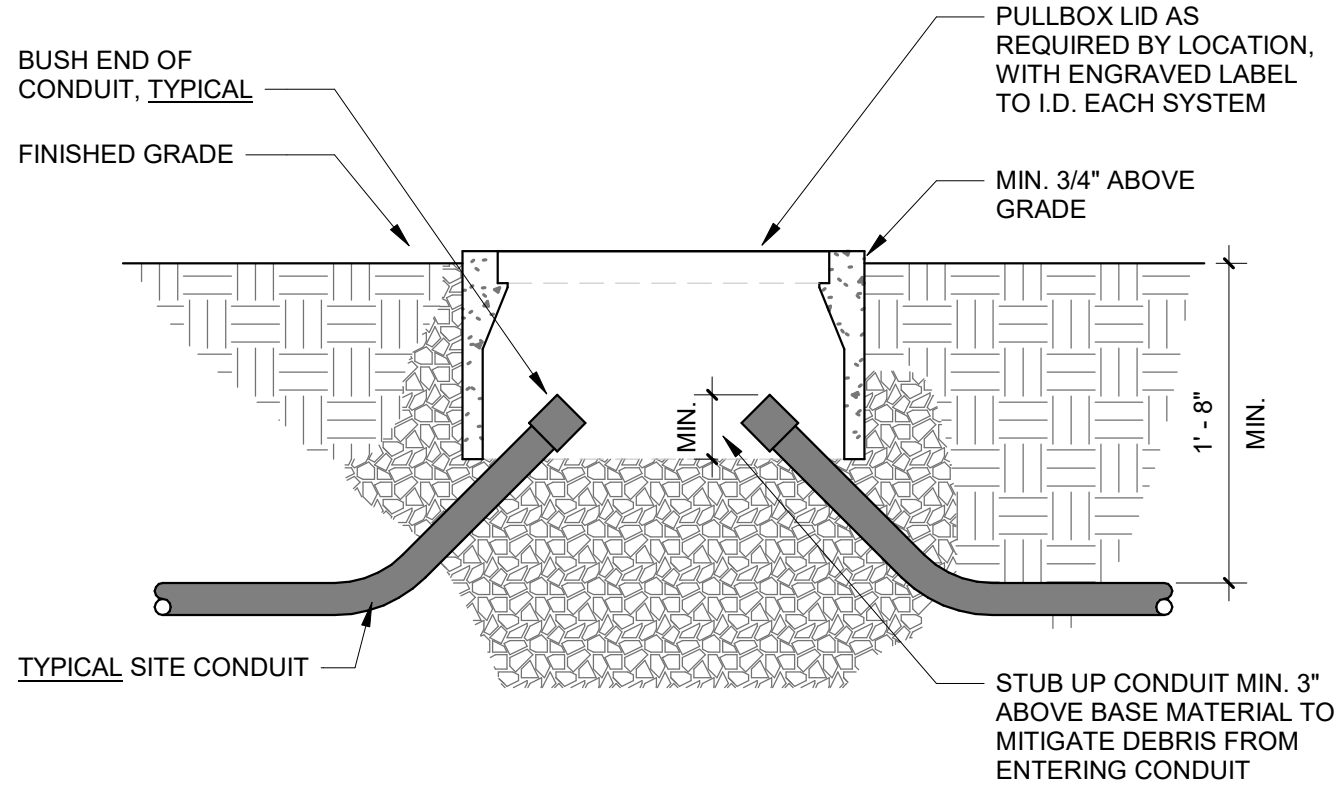


NON-PAVED AREAS

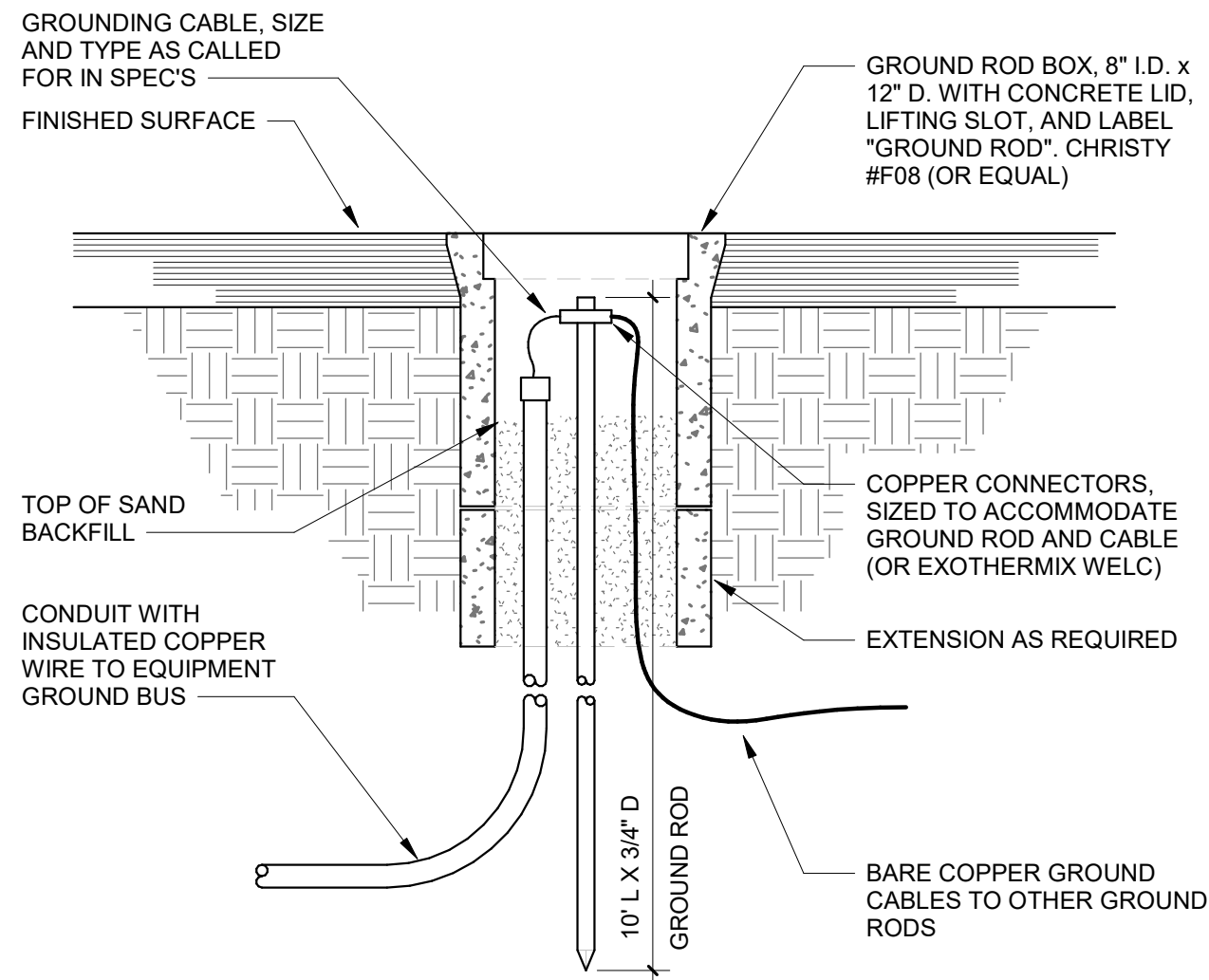


PAVED AREAS

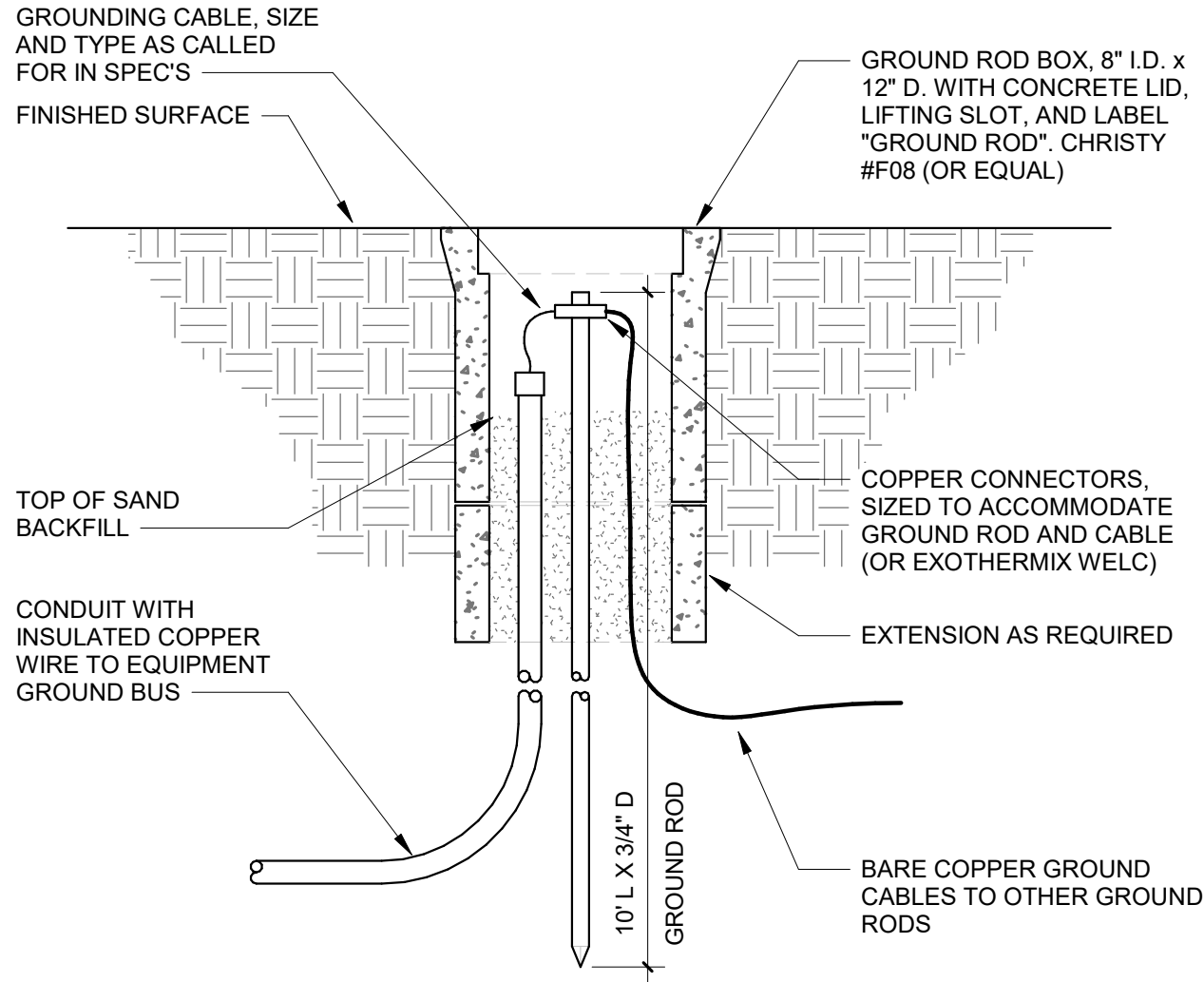
2 TYPICAL PULLBOX SECTION
NO SCALE



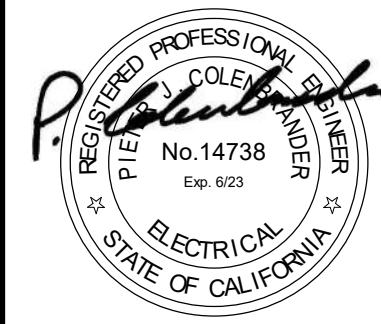
NON-PAVED AREAS



4 GROUND ROD - FINISHED SURFACE
NO SCALE



3 GROUND ROD - EARTH
NO SCALE



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ELECTRICAL ENGINEERING & CONSULTING SERVICES
4340 REDWOOD HWY., SUITE 245
SAN RAFAEL, CALIFORNIA 94903
(415) 402-0420/FAX (415) 429-9602

PROJECT:

CITY OF MORGAN HILL

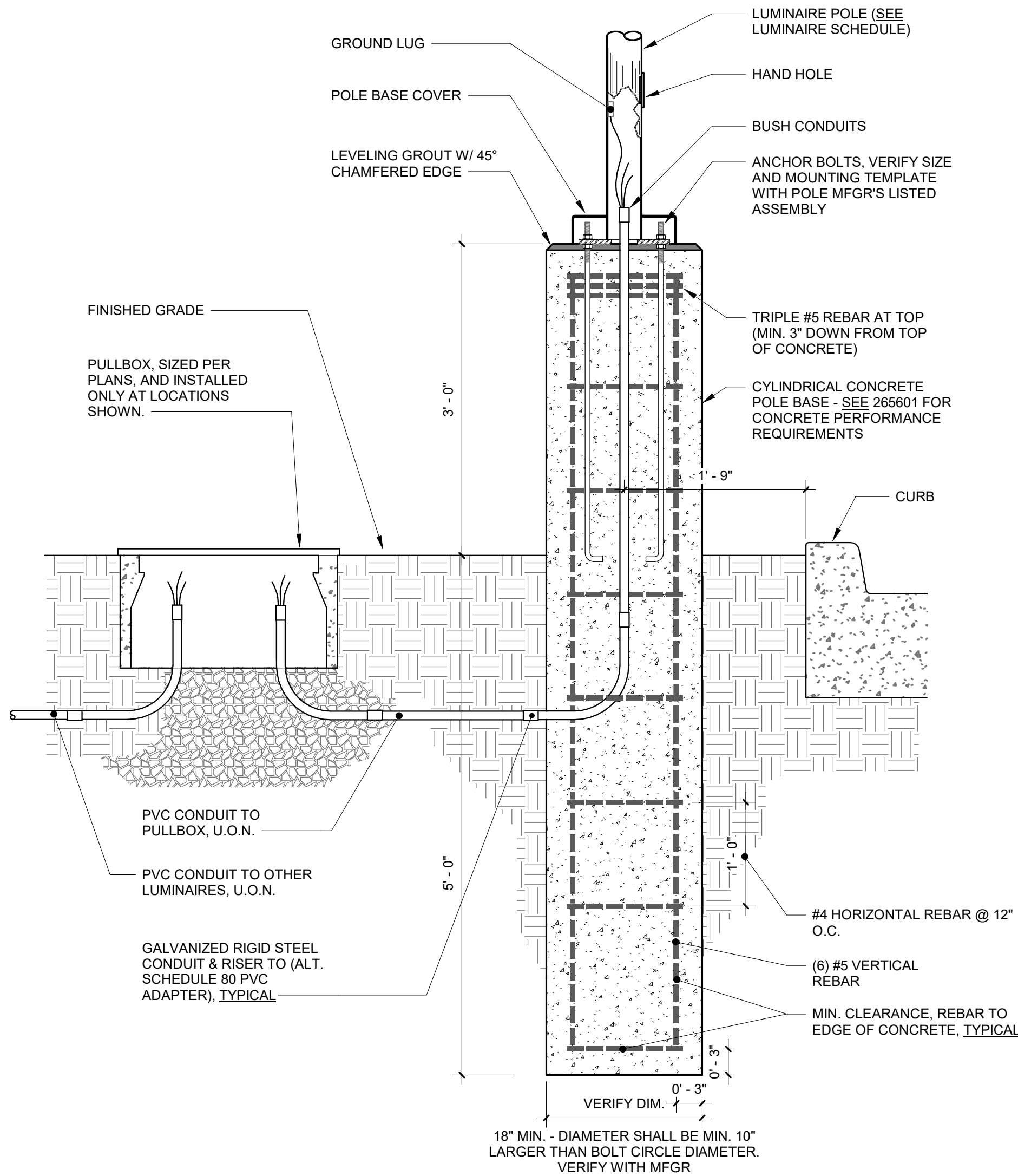
BUTTERFIELD FIRE STATION

17285 Butterfield Blvd.
Morgan Hill, CA 95037

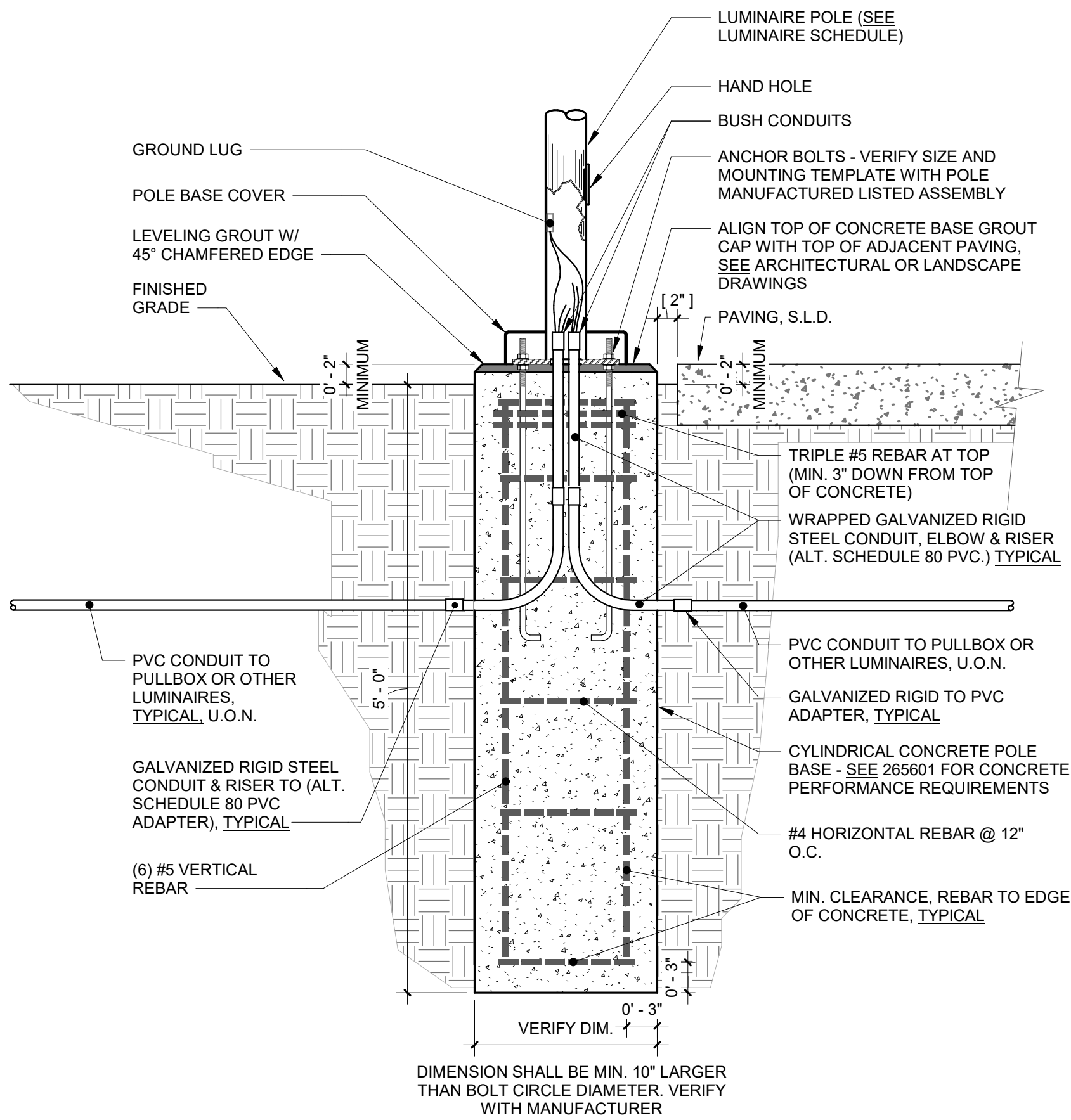


DESCRIPTION: DATE:
95% CONSTR. 01/16/23
DRAWINGS
100% CONSTR. 03/23/23
DRAWINGS

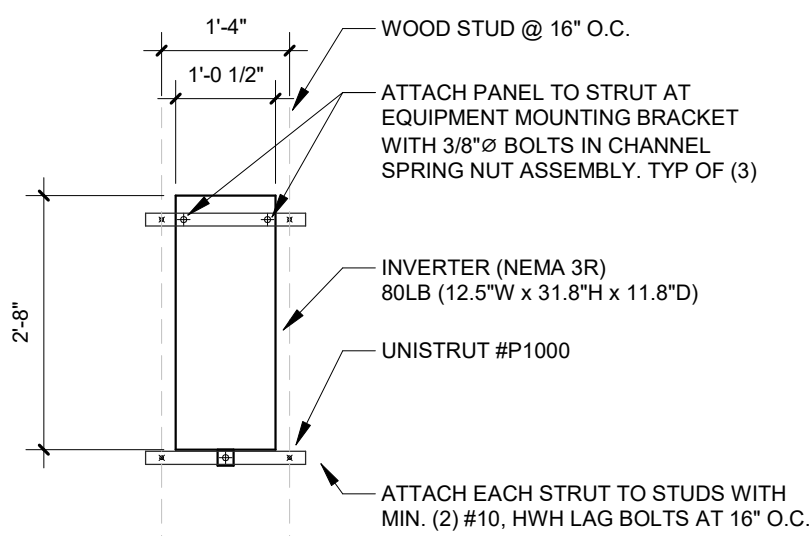
NOT FOR CONSTRUCTION
PROJECT NUMBER:
211101
SHEET TITLE:
DETAILS
SHEET NUMBER:
E7.1



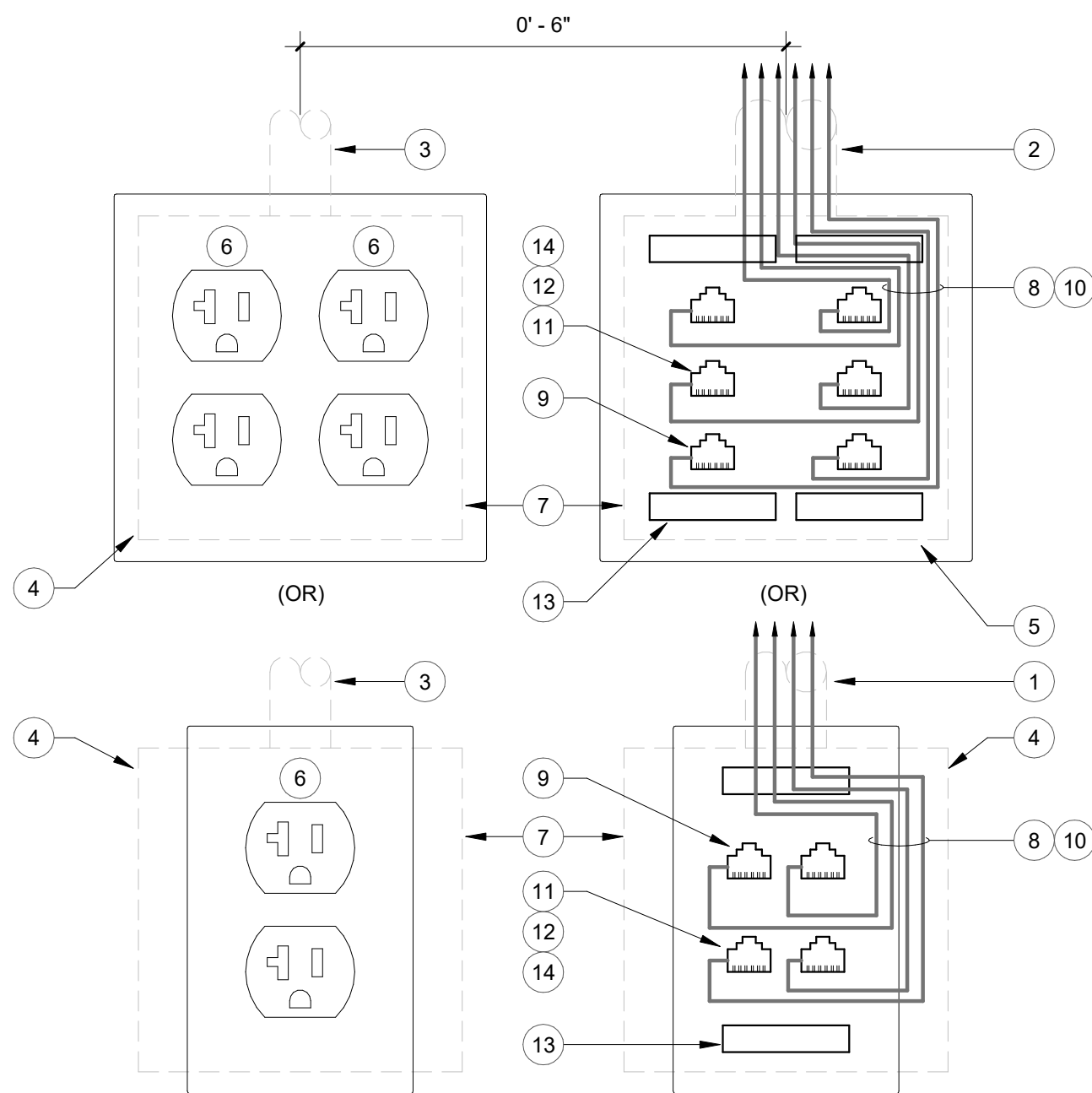
6 RAISED POLE BASE RAISED WITH PULLBOX
NO SCALE



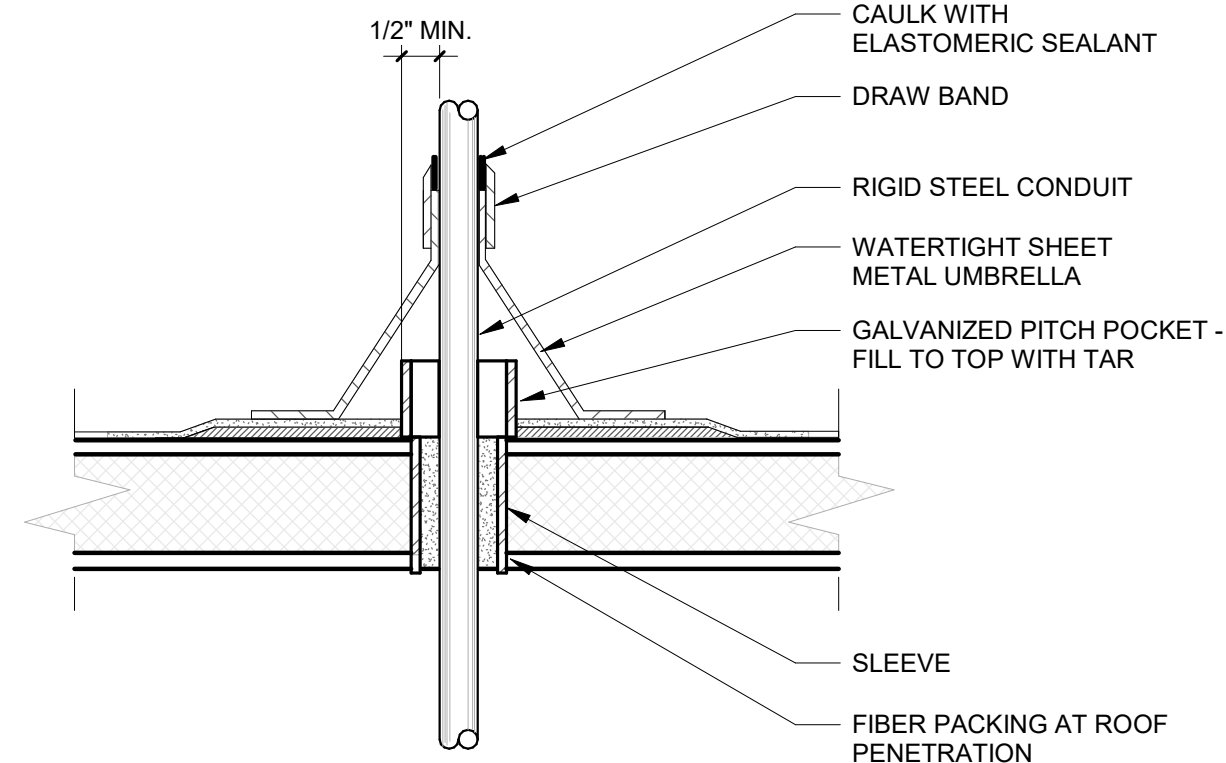
7 POLE BASE IN PLANTING
NO SCALE



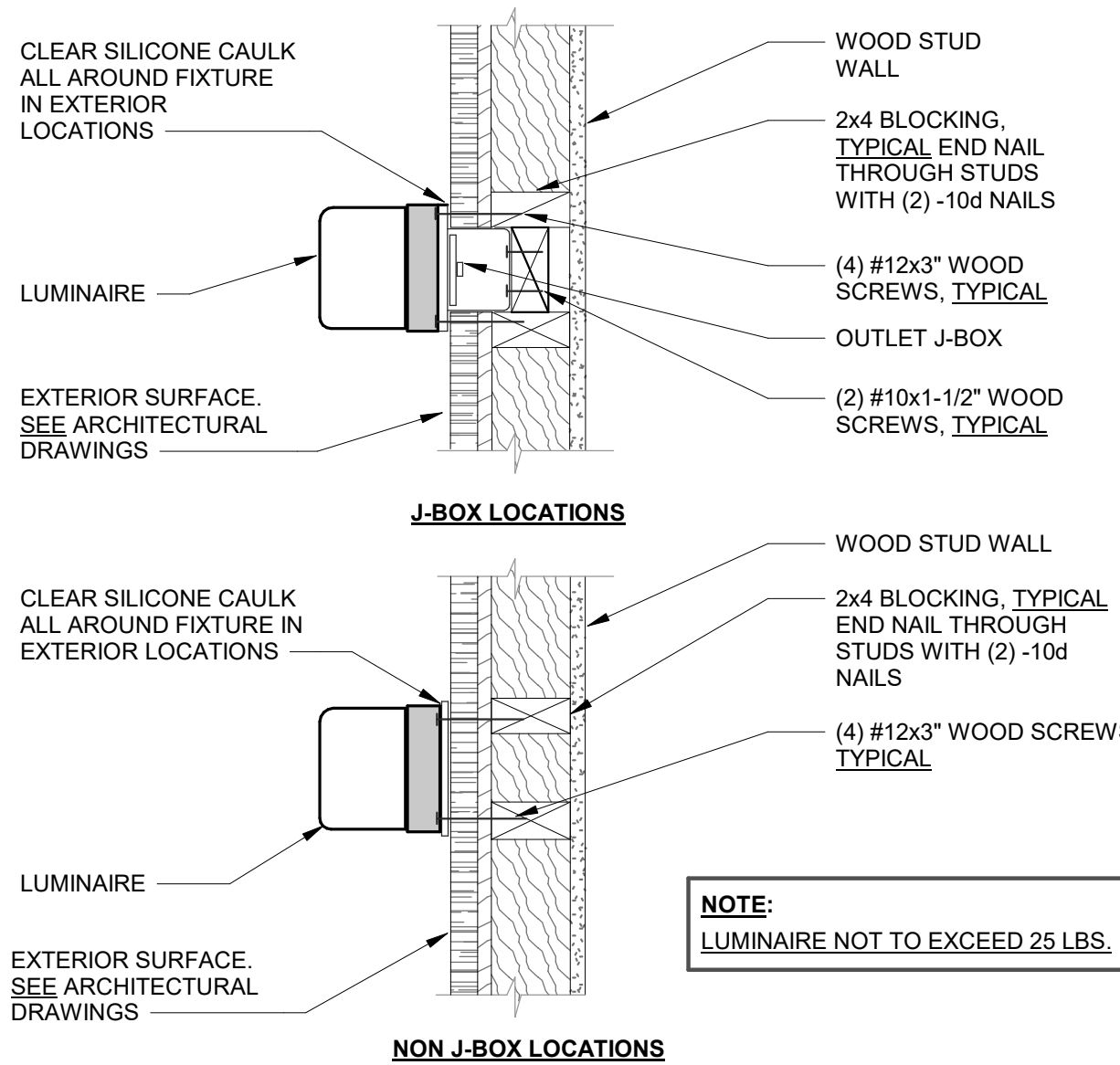
8 INTERIOR INVERTER MOUNTING
NOT TO SCALE



1 COMBINATION POWER & DATA OUTLETS
NO SCALE



2 TYPICAL STATION CABLE LABELING
NO SCALE



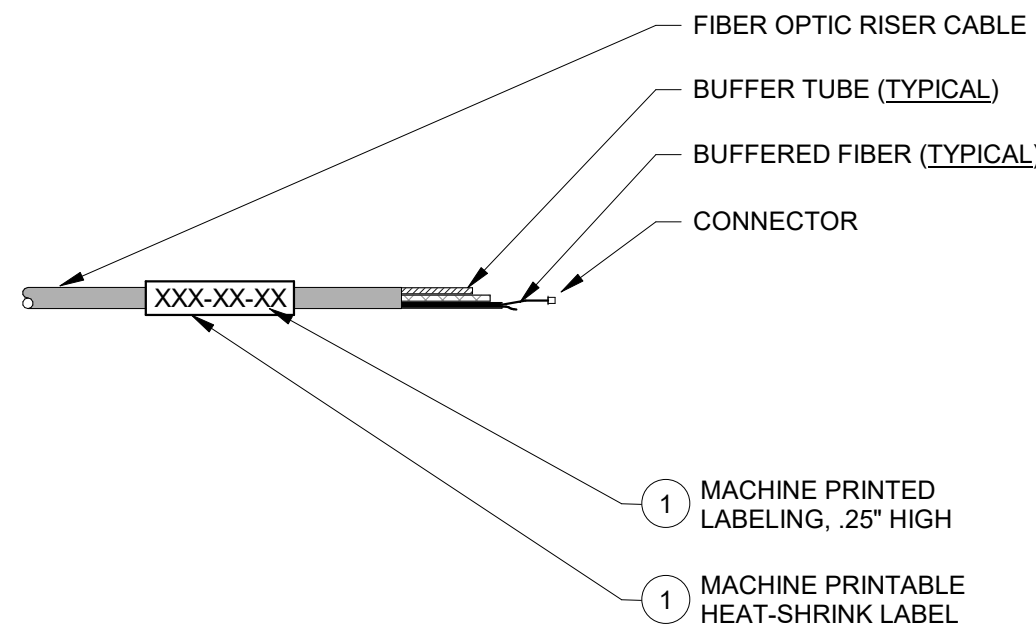
3 TYPICAL FIBER OPTIC CABLE LABELING
NO SCALE

DETAIL NUMBERED NOTES

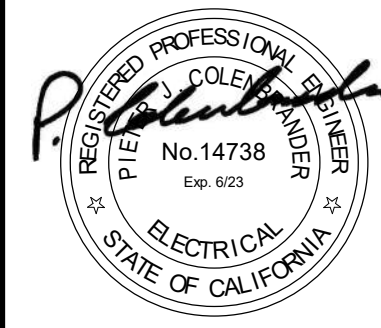
- 1 PROVIDE 1" CONDUIT STUBBED UP TO NEAREST ACCESSIBLE CONCEALED CEILING, U.O.N. - BUSH CONDUIT END(S)
- 2 PROVIDE 1-1/4" CONDUIT STUBBED UP TO NEAREST ACCESSIBLE CONCEALED CEILING, U.O.N. - BUSH CONDUIT END(S)
- 3 POWER CONDUIT & WIRES - SEE FLOOR PLANS
- 4 PROVIDE 4" SQUARE BOX, 2-1/8" DEEP, WITH TWO-GANG PLASTER RING
- 5 PROVIDE 4" SQUARE BOX, 3-1/2" DEEP, WITH TWO-GANG PLASTER RING
- 6 PROVIDE 20AMP, 120V, 3PG DOUBLE DUPLEX RECEPTACLE
- 7 LOCATE BOXES ADJACENT TO EACH OTHER & ALIGN COVER PLATES
- 8 TERMINATE CABLES AT EACH JACK. PROVIDE QUANTITY OF JACKS AS NOTED ON PLANS
- 9 WHERE NO DATA JACK IS REQUIRED, PROVIDE BLANK SPACER.
- 10 PROVIDE HOMERUN DATA CABLES TO RESPECTIVE PATCH PANEL.
- 11 MODULAR FACEPLATE WITH ANGLED OR FLAT EXIT (VERIFY WITH I.T. STANDARDS).
- 12 TYPICAL 8 PIN 8 POS, T568B DATA JACK, COLORED BLUE (VERIFY WITH I.T. STANDARDS).
- 13 PANDUIT, EFPL-1, 1.80 x 0.375 INCH, WHITE, POLYESTER LABEL. LABELS SHALL HAVE CAPACITY FOR 2 COLUMNS FOR LASER PRINTERS AND SHALL BE LABELED AS NOTED IN SPECIFICATIONS (VERIFY WITH I.T. STANDARDS).
- 14 SEE FLOOR PLANS FOR QUANTITY & TYPE OF JACKS IN EACH LOCATION.
- 15 TYPICAL SINGLE OR DOUBLE GANG FACEPLATE, FINISH AS DIRECTED BY ARCHITECT.

DETAIL NOTES

1. LABEL LOCATION AT FIBER OPTIC RISER CABLE. LABEL TYPE PANDUIT MMP350-C-75" x 3.5" 316 STAINLESS, MACHINE .25" H PRINTING.



4 TYPICAL SURFACE WALL MOUNTED LUMINAIRE
NO SCALE



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PROJECT:

CITY OF MORGAN HILL

BUTTERFIELD FIRE STATION

17285 Butterfield Blvd.
Morgan Hill, CA 95037



DESCRIPTION: DATE:

95% CONSTR. 01/16/23
DRAWINGS

100% CONSTR. 03/23/23
DRAWINGS

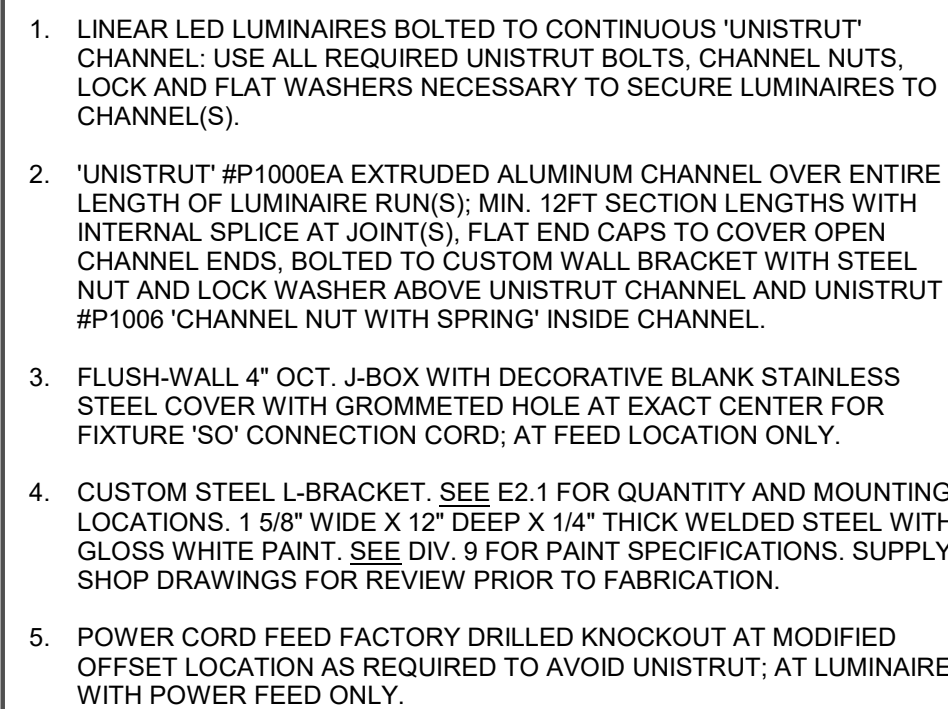
NOT FOR
CONSTRUCTION

PROJECT NUMBER:
211101

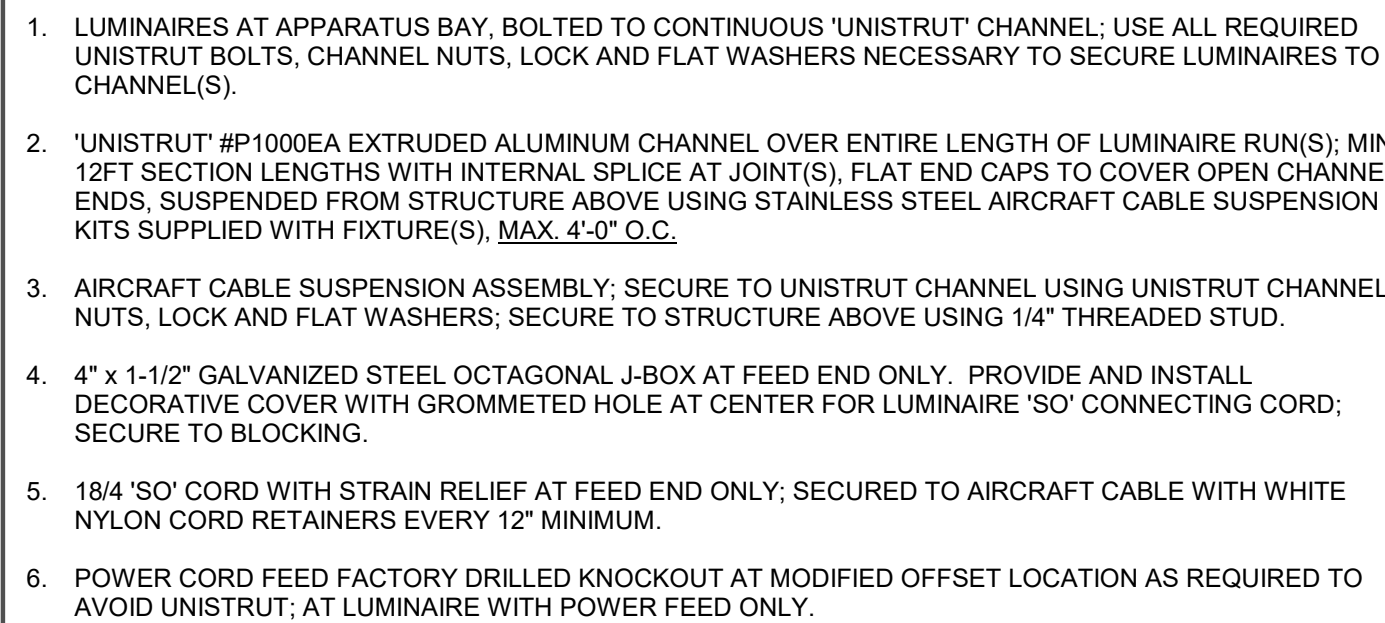
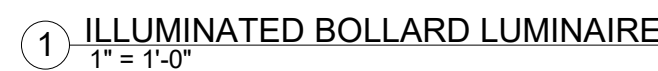
SHEET TITLE:
DETAILS

SHEET NUMBER:

E7.2



3 BRACKET MOUNTED LUMINAIRE IN APPARATUS BAY
12" = 1'-0"



2 PENDANT MOUNTED LUMINAIRE IN APPARATUS BAY
NO SCALE



STATE OF CALIFORNIA		CALIFORNIA ENERGY COMMISSION	
Outdoor Lighting			
CERTIFICATE OF COMPLIANCE			
Project Name: Morgan Hill Butterfield Fire Station		Report Page: (Page 7 of 9)	NRCC-LTO-E
Project Address: 17285 Butterfield Blvd		Date Prepared: 2/10/2023	

M. LIGHTING ALLOWANCE: PER SPECIFIC AREA

This table includes areas using the wattage allowance per specific area from Table 140.7-B / Table 170.2-S. More than one specific area allowance may be taken in a single project, if applicable. However, multiple specific area allowances may not be taken for the exact same area on the site.

01	02	03	04	05	06	07	08	09	10
Area Description	Specific Area Type per Table 140.7-B	CALCULATED ALLOWANCE (Watts)			DESIGN WATTS				Additional Allowance (Watts)
		Specific Area (ft²) ¹	Allowed Density (W/ft²)	Extra Allowance (Watts)	Luminaire Name or Item Tag	Watts per Luminaire	# of Luminaires	Design Watts	
Canopy	NonSalesCanopy	536	0.3	144.7	BA1	30	4	120	144.7
					BD1	20	4	80	
Total Design Watts for this Area:								200	
Total Allowance (Watts) All Areas:									144.7

¹ FOOTNOTES: See Table 140.7-B / Table 170.2-S for rules for calculating the specific area (ft²) for these additional lighting allowances.

² For luminaires indicated in Table F as linear, wattage in column 07 is W/ft instead of Watts/luminaire. Total linear feet should be indicated in column 08 instead of number of luminaires.

N. EXISTING CONDITIONS POWER ALLOWANCE (alterations only)

This section does not apply to this project.

O. DECLARATION OF REQUIRED CERTIFICATES OF INSTALLATION

Form/Title
NRCI-LTO-E - Must be submitted for all buildings

Registration Number:	Generated Date/Time:	Documentation Software: EnergyPro
CA Building Energy Efficiency Standards - 2022 Nonresidential Compliance	Report Version: 2022.0.000 Schema Version: rev 20220101	Compliance ID: EnergyPro-8069-0223-0023 Report Generated: 2023-02-10 13:50:11

STATE OF CALIFORNIA		CALIFORNIA ENERGY COMMISSION	
Outdoor Lighting			
CERTIFICATE OF COMPLIANCE			
Project Name: Morgan Hill Butterfield Fire Station		Report Page: (Page 8 of 9)	NRCC-LTO-E
Project Address: 17285 Butterfield Blvd		Date Prepared: 2/10/2023	

P. DECLARATION OF REQUIRED CERTIFICATES OF ACCEPTANCE

Form/Title	Systems/Spaces To Be Field Verified
NRCA-LTO-02-A - Must be submitted for all outdoor lighting controls except for alterations where controls are added to <= 20 luminaires.	Canopy; Automotive; Pedestrian;

Registration Number:	Generated Date/Time:	Documentation Software: EnergyPro
CA Building Energy Efficiency Standards - 2022 Nonresidential Compliance	Report Version: 2022.0.000 Schema Version: rev 20220101	Compliance ID: EnergyPro-8069-0223-0023 Report Generated: 2023-02-10 13:50:11

STATE OF CALIFORNIA		CALIFORNIA ENERGY COMMISSION	
Outdoor Lighting			
CERTIFICATE OF COMPLIANCE			
Project Name: Morgan Hill Butterfield Fire Station		Report Page: (Page 9 of 9)	NRCC-LTO-E
Project Address: 17285 Butterfield Blvd		Date Prepared: 2/10/2023	

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

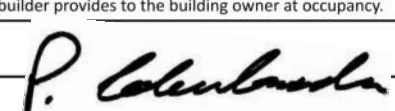
I certify that this Certificate of Compliance documentation is accurate and complete.

Documentation Author Name: Pieter Colenbrander	Documentation Author Signature: 
Company: O'Mahony & Myer	Signature Date: 2023-02-10
Address: 4340 Redwood Hwy, Ste 245	CEA/HERS Certification Identification (if applicable): E14738
City/State/Zip: San Rafael CA 94903	Phone: 415-492-0420

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

- The information provided on this Certificate of Compliance is true and correct.
- I am eligible under Division 3 of the Business and Professions Code to accept responsibility for the building design or system design identified on this Certificate of Compliance (responsible designer).
- The energy features and performance specifications, materials, components, and manufactured devices for the building design or system design identified on this Certificate of Compliance conform to the requirements of Title 24, Part 1 and Part 6 of the California Code of Regulations.
- The building design features or system design features identified on this Certificate of Compliance are consistent with the information provided on other applicable compliance documents, worksheets, calculations, plans and specifications submitted to the enforcement agency for approval with this building permit application.
- I will ensure that a completed signed copy of this Certificate of Compliance shall be made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections. I understand that a completed signed copy of this Certificate of Compliance is required to be included with the documentation the builder provides to the building owner at occupancy.

Responsible Designer Name: Pieter Colenbrander	Responsible Designer Signature: 
Company: O'Mahony & Myer	Date Signed: 2023-02-10
Address: 4340 Redwood Hwy Suite 245	License: E14738
City/State/Zip: San Rafael CA 94903	Phone: 415-492-0420

Registration Number:	Generated Date/Time:	Documentation Software: EnergyPro
CA Building Energy Efficiency Standards - 2022 Nonresidential Compliance	Report Version: 2022.0.000 Schema Version: rev 20220101	Compliance ID: EnergyPro-8069-0223-0023 Report Generated: 2023-02-10 13:50:11

STATE OF CALIFORNIA		CALIFORNIA ENERGY COMMISSION	
Outdoor Lighting			
CERTIFICATE OF COMPLIANCE			
Project Name: Morgan Hill Butterfield Fire Station		Report Page: (Page 4 of 9)	NRCC-LTO-E
Project Address: 17285 Butterfield Blvd		Date Prepared: 2/10/2023	

F. OUTDOOR LIGHTING FIXTURE SCHEDULE

For new or altered lighting systems demonstrating compliance with 140.7 / 170.2(e)6 all new luminaires being installed and any existing luminaires remaining or being moved within the spaces covered by the permit application are included in the Table below. For altered lighting systems using the Existing Power method per 141.0(b)2L only new luminaires being installed and replacement luminaires being installed as part of the project scope are included (i.e. existing luminaires remaining or existing luminaires being moved are not included). Outdoor lighting attached to multifamily buildings and controlled from the inside of a dwelling unit are included in Table H, and are not included here. All other multifamily outdoor lighting is included here.

Designed Wattage:

01	02	03	04	05	06	07	08	09	10
BG1	BG1-IN-GRADE	☐ Linear	19	Mfr. Spec	3	New	☒	NA: < 6200 lumens	☐ ☐
BH1	BH1-WALLWASHER	☐ Linear	14	Mfr. Spec	1	New	☒	NA: < 6200 lumens	☐ ☐
Total Design Watts:								1104	

* NOTES: Selections with a * require a note in the space below explaining how compliance is achieved.

EX: Luminaire is lighting a statue; EXCEPTION 2 to 130.2(b)

¹ FOOTNOTES: Authority Having Jurisdiction may ask for Luminaire cut sheets to confirm wattage used for compliance per 130.0(c) / 160.5(b)

² For linear luminaires, wattage should be indicated as W/ft instead of Watts/luminaire. Total linear feet should be indicated in column 05 instead of number of luminaires.

³ Select "New" for new luminaires in a new outdoor lighting project, or for added luminaires in an alteration. Select "Altered" for replacement luminaires in an alteration. Select "Existing to Remain" for existing luminaires within the project scope that are not being altered and are remaining. Select "Existing Reinstalled" for existing luminaires which are being removed and reinstalled as part of the project scope.

* Compliance with mandatory shielding requirements is required for luminaires with initial lumen output >= 6,200 unless exempted by 130.2(b) / 160.5(c)

G. SHIELDING REQUIREMENTS (BUG)

This section does not apply to this project.

Registration Number:	Generated Date/Time:	Documentation Software: EnergyPro
CA Building Energy Efficiency Standards - 2022 Nonresidential Compliance	Report Version: 2022.0.000 Schema Version: rev 20220101	Compliance ID: EnergyPro-8069-0223-0023 Report Generated: 2023-02-10 13:50:11

STATE OF CALIFORNIA		CALIFORNIA ENERGY COMMISSION	
Outdoor Lighting			
CERTIFICATE OF COMPLIANCE			
Project Name: Morgan Hill Butterfield Fire Station		Report Page: (Page 5 of 9)	NRCC-LTO-E
Project Address: 17285 Butterfield Blvd		Date Prepared: 2/10/2023	

H. OUTDOOR LIGHTING CONTROLS

This table demonstrates compliance with controls requirements for all new or altered luminaires installed as part of the permit application. For alteration projects, luminaires which are existing to remain (ie untouched) and luminaires which are removed and reinstalled (wiring only) do not need to be included in this table even if they are within the spaces covered by the permit application.

Outdoor lighting for nonresidential buildings, parking garages and common service areas in multifamily buildings must be documented separately from outdoor lighting attached to multifamily buildings and controlled from the inside of a dwelling unit

Mandatory Controls for Nonresidential Occupancies, Parking Garages & Common Areas in Multifamily Buildings

01	02	03	04	05
Area Description	Shut-Off 130.2(c)1 / 160.5(c)	Auto-Schedule 130.2(c)2 / 160.5(c)	Motion Sensor 130.2(c)3 / 160.5(c)	Field Inspector
Canopy	Astronomical Timer	Provided	NA: Not permitted by H&LS	☐ ☐
Automotive	Astronomical Timer	Provided	NA: Not permitted by H&LS	☐ ☐
Pedestrian	Astronomical Timer	Provided	NA: Not permitted by H&LS	☐ ☐

¹ FOOTNOTE: Text has been abbreviated, please refer to Table 160.5-A to confirm compliance with the specific light source technologies listed.

² Authority having jurisdiction may ask for cut sheets or other documentation to confirm compliance of light source.

³ Recessed luminaires marked for use in fire-rated installations, and recessed luminaires installed in non-insulated ceilings are excepted from i and iii.

Registration Number:	Generated Date/Time:	Documentation Software: EnergyPro
CA Building Energy Efficiency Standards - 2022 Nonresidential Compliance	Report Version: 2022.0.000 Schema Version: rev 20220101	Compliance ID: EnergyPro-8069-0223-0023 Report Generated: 2023-02-10 13:50:11

STATE OF CALIFORNIA		CALIFORNIA ENERGY COMMISSION	
Outdoor Lighting			
CERTIFICATE OF COMPLIANCE			
Project Name: Morgan Hill Butterfield Fire Station		Report Page: (Page 6 of 9)	NRCC-LTO-E
Project Address: 17285 Butterfield Blvd		Date Prepared: 2/10/2023	

I. LIGHTING POWER ALLOWANCE (per 140.7 / 170.2(e))

01									
This table includes areas using allowance calculations per 140.7 / 170.2(e). General Hardscape Allowance is per Table 140.7-A/ Table 170.2-R while "Use it or lose it" Allowances are per Table 140.7-B/ Table 170.2-S. Indicate which allowances are being used to expand sections for user input. Luminaires that qualify for one of the "Use it or lose it" allowances shall not qualify for another "Use it or lose it" allowance. Outdoor lighting attached to multifamily buildings and controlled from the inside of a dwelling unit are included in Table H, and are not included here. All other multifamily outdoor lighting is included here.									
☒ General Hardscape Allowance Table I (below)		☐ "Use it or lose it" Allowance (select all that apply) (select all that apply)							
		☐ Per Application Table J	☐ Sales Frontage Table K	☐ Ornamental Table L	☒ Per Specific Area Table M				

Calculated General Hardscape Lighting Power Allowance per Table 140.7-A for Nonresidential & Hotel/Motel

02	03	04	05	06	07	08	09
Area Description	Area Wattage Allowance (AWA) Illuminated Area (ft ²)	Allowed Density (W/ft ²)	Area Allowance (Watts)	Perimeter Length (lf)	Allowed Density (W/lf)	Linear Allowance (Watts)	Total General AWA + LWA (Watts)
Automotive	7888	0.021	165.6	852	0.2	170.4	336
Pedestrian	8677	0.021	182.2	977	0.2	195.4	378
Initial Wattage Allowance for Entire Site (Watts):							250
Instances of Initial Wattage Allowance (L2 0 only) ¹							
Total General Hardscape Allowance (Watts):							964

J. LIGHTING ALLOWANCE: PER APPLICATION

This section does not apply to this project.

K. LIGHTING ALLOWANCE: SALES FRONTAGE

This section does not apply to this project.

L. LIGHTING ALLOWANCE: ORNAMENTAL

This section does not apply to this project.

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Outdoor Lighting			
CERTIFICATE OF COMPLIANCE			
Project Name: Morgan Hill Butterfield Fire Station		Report Page: (Page 1 of 9)	NRCC-LTO-E
Project Address: 17285 Butterfield Blvd		Date Prepared: 2/10/2023	

A. GENERAL INFORMATION

01 Project Location (city)	Morgan Hill	04 Total Illuminated Hardscape Area (ft ²)	16565
02 Climate Zone	4		
03 Outdoor Lighting Zone per Title 24 Part 10.114 or as designated by Authority Having Jurisdiction (AHJ):			
☐ L2-0: Very Low - Undeveloped Parkland	☐ L2-2: Moderate - Urban Clusters	☐ L2-4: High - Must be reviewed by CA Energy Commission for Approval	
☐ L2-1: Low - Rural Areas	☒ L2-3: Moderately High - Urban Areas		
05 Occupancy Types within Project			
• Gymnasium • Office • Support Areas • Warehouse			

B. PROJECT SCOPE

This table includes outdoor lighting systems that are within the scope of the permit application and are demonstrating compliance using the prescriptive path outlined in 140.7 / 170.2(e)6 or 141.0(b)2L / 180.2(b)4Bv for alterations.

My Project Consists of:

01		02		
☒	New Lighting System	Must Comply with Allowances from 140.7 / 170.2(e)6		
☐	Altered Lighting System	Is your alteration increasing the connected lighting load (Watts)? ☐ Yes ☐ No		
03		04		05
% of Existing Luminaires Being Altered ¹		Sum Total of Luminaires Being Added or Altered		Calculation Method
☐ < 10%	☐ >= 10% and < 50%	☐ >= 50%		

Please proceed to Table F. Outdoor Lighting Fixture Schedule to define the project's luminaires.

¹ FOOTNOTES: % of Existing Luminaires Being Altered = (Sum Total of Luminaires Being Added or Altered / Existing Luminaires within the Scope of the Permit Application) x 100.

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Project Name: Morgan Hill Butterfield Fire Station		Report Page: (Page 2 of 9)	NRCC-LTO-E
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C. COMPLIANCE RESULTS

Results in this table are automatically calculated from data input and calculations in Tables F through N. Note: If any cell on this table says "COMPLIES with Exceptional Conditions" refer to Table O. Exceptional Conditions for guidance or see applicable Table referenced below.

Calculations of Total Allowed Lighting Power (Watts) 140.7 / 170.2(e)6 or 141.0(b)2L / 180.2(b)4Bv										Compliance Results			
01	02	03	04	05	06	07	08	09					
General Hardscape Allowance 140.7(d)1 / 170.2(e)6 (See Table J)	+	Per Application 140.7(d)2 / 170.2(e)6 (See Table K)	+	Sales Frontage 140.7(d)2 (See Table L)	+	OR	Existing Power Allowance 141.0(b)2L / 180.2(b)4Bv (See Table N)	=	Total Allowed (Watts)	≥	Total Actual (Watts)	07 must be >= 08	
964	+	---	+	---	+	144.7	OR	---	=	1,108	≥	1,104	COMPLIES
Shielding Compliance (See Table G for Details)										N/A			
Controls Compliance (See Table H for Details)										COMPLIES		N/A	

D. EXCEPTIONAL CONDITIONS

This table is auto-filled with uneditable comments because of selections made or data entered in tables throughout the form.

E. ADDITIONAL REMARKS

This table includes remarks made by the permit applicant to the Authority Having Jurisdiction.

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STATE OF CALIFORNIA		CALIFORNIA ENERGY COMMISSION	
Outdoor Lighting			
CERTIFICATE OF COMPLIANCE			
Project Name: Morgan Hill Butterfield Fire Station		Report Page: (Page 3 of 9)	NRCC-LTO-E
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F. OUTDOOR LIGHTING FIXTURE SCHEDULE

For new or altered lighting systems demonstrating compliance with 140.7 / 170.2(e)6 all new luminaires being installed and any existing luminaires remaining or being moved within the spaces covered by the permit application are included in the Table below. For altered lighting systems using the Existing Power method per 141.0(b)2L only new luminaires being installed and replacement luminaires being installed as part of the project scope are included (i.e. existing luminaires remaining or existing luminaires being moved are not included). Outdoor lighting attached to multifamily buildings and controlled from the inside of a dwelling unit are included in Table H, and are not included here. All other multifamily outdoor lighting is included here.

Designed Wattage:

01	02		03	04	05	06	07	08	09	10	
Name or Item Tag	Complete Luminaire Description		Watts per luminaire ^{1,2}	How is Wattage determined	Total Number Luminaires ²	Luminaire Status ³	Excluded per 140.7(a) / 170.2(e)6A	Design Watts	Cutoff Req. > 6,200 initial lumen output 130.2(b) / 160.5(c)1 ⁴	Field Inspector	
										Pass	Fail
BA1	BA1-CANOPY DOWN	☐ Linear	30	Mfr. Spec	4	New	☐	120	NA: < 6200 lumens	☐	☐
BB1	BB1-SCONCE	☐ Linear	11	Mfr. Spec	3	New	☐	33	NA: < 6200 lumens	☐	☐
BC1	BC1-SCONCE	☐ Linear	28	Mfr. Spec	6	New	☐	168	NA: < 6200 lumens	☐	☐
BC2	BC2-SCONCE	☐ Linear	14	Mfr. Spec	15	New	☐	210	NA: < 6200 lumens	☐	☐
BD1	BD1-WALL UPLIGHT	☐ Linear	20	Mfr. Spec	4	New	☐	80	NA: < 6200 lumens	☐	☐
BE1	BE1-AREA POLE	☐ Linear	39	Mfr. Spec	9	New	☐	351	NA: < 6200 lumens	☐	☐
BE2	BE2-AREA POLE	☐ Linear	39	Mfr. Spec	2	New	☐	78	NA: < 6200 lumens	☐	☐
BE3	BE3-AREA POLE	☐ Linear	20	Mfr. Spec	2	New	☐	40	NA: < 6200 lumens	☐	☐
BF1	BF1-BOLLARD	☐ Linear	8	Mfr. Spec	3	New	☐	24	NA: < 6200 lumens	☐	☐

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Project Name: Morgan Hill Butterfield Fire Station Report Page: (Page 7 of 9)

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K. TAILORED METHOD GENERAL LIGHTING POWER ALLOWANCE

This section does not apply to this project.

L. ADDITIONAL LIGHTING ALLOWANCE: TAILORED WALL DISPLAY

This section does not apply to this project.

M. ADDITIONAL LIGHTING ALLOWANCE: TAILORED FLOOR AND TASK LIGHTING

This section does not apply to this project.

N. ADDITIONAL LIGHTING ALLOWANCE: TAILORED DECORATIVE /SPECIAL EFFECTS

This section does not apply to this project.

O. ADDITIONAL LIGHTING ALLOWANCE: TAILORED VERY VALUABLE MERCHANDISE

This section does not apply to this project.

P. POWER ADJUSTMENT: LIGHTING CONTROL CREDIT (POWER ADJUSTMENT FACTOR (PAF))

This section does not apply to this project.

Q. RATED POWER REDUCTION COMPLIANCE FOR ONE-FOR-ONE ALTERATIONS

This section does not apply to this project.

R. 80% LIGHTING POWER FOR ALL ALTERATIONS - CONTROLS EXCEPTIONS

This section does not apply to this project.

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S. DAYLIGHT DESIGN POWER ADJUSTMENT FACTOR (PAF)

This section does not apply to this project.

T. DWELLING UNIT LIGHTING

This section does not apply to this project.

U. DECLARATION OF REQUIRED CERTIFICATES OF INSTALLATION

Form/Title

NRCC-LTI-E - Must be submitted for all buildings

V. DECLARATION OF REQUIRED CERTIFICATES OF ACCEPTANCE

Form/Title Systems/Spaces To Be Field Verified

NRCA-LTI-02-A - Must be submitted for occupancy sensors and automatic time switch controls. Whole Building Time Switch; Office; Corridor; Restroom; Kitchen; Storage; Fitness; Laundry;

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Project Name: Morgan Hill Butterfield Fire Station Report Page: (Page 9 of 9)

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DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

I certify that this Certificate of Compliance documentation is accurate and complete.

Documentation Author Name: Pieter Colenbrander

Documentation Author Signature: *P. Colenbrander*

Company: O'Mahony & Myer

Address: 4340 Redwood Hwy, Ste 245

City/State/Zip: San Rafael CA 94903

Signature Date: 2023-02-10

CEA/HERS Certification Identification (if applicable): E14738

Phone: 415-492-0420

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

- The information provided on this Certificate of Compliance is true and correct.
- I am eligible under Division 3 of the Business and Professions Code to accept responsibility for the building design or system design identified on this Certificate of Compliance (responsible designer).
- The energy features and performance specifications, materials, components, and manufactured devices for the building design or system design identified on this Certificate of Compliance conform to the requirements of Title 24, Part 1 and Part 6 of the California Code of Regulations.
- The building design features or system design features identified on this Certificate of Compliance are consistent with the information provided on other applicable compliance documents, worksheets, calculations, plans and specifications submitted to the enforcement agency for approval with this building permit application.
- I will ensure that a completed signed copy of this Certificate of Compliance shall be made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections. I understand that a completed signed copy of this Certificate of Compliance is required to be included with the documentation the builder provides to the building owner at occupancy.

Responsible Designer Name: Pieter Colenbrander

Responsible Designer Signature: *P. Colenbrander*

Company: O'Mahony & Myer

Address: 4340 Redwood Hwy Suite 245

City/State/Zip: San Rafael CA 94903

Date Signed: 2023-02-10

License: E14738

Phone: 415-492-0420

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Project Name: Morgan Hill Butterfield Fire Station Report Page: (Page 4 of 9)

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F. INDOOR LIGHTING FIXTURE SCHEDULE									
AN1	AN1-2X2	No	NA	29	Mfr. Spec	5	No	145	☐
AN2	AN2-1X4	No	NA	29	Mfr. Spec	2	No	58	☐
Total Designed Watts: CONDITIONED SPACES								2,903	
01	02	03	04	05	06	07	08	09	10
Name or Item Tag	Complete Luminaire Description	Modular (Track) Fixture	Small Aperture & Color Change ¹	Watts per luminaire ²	How is Wattage determined	Total Number of Luminaires	Excluded per 140.6(a)3 / 170.2(e)2C	Design Watts	Field Inspector
AJ1	AJ1-SURFACE LINEAR 4	No	NA	25	Mfr. Spec	3	No	75	☐
Total Designed Watts: UNCONDITIONED SPACES								75	

¹FOOTNOTE: Design Watts for small aperture and color changing luminaires which qualify per 140.6(a)4B / 170.2(e)2D is adjusted to be 75%/80% of their rated wattage. Table F automatically makes this adjustment, the permit applicant should enter full rated wattage in column 05.

²Authority Having Jurisdiction may ask for Luminaire cut sheets to confirm wattage used for compliance per 130.0(c) / 160.5(b). Wattage used must be the maximum rated for the luminaire, not the lamp.

G. MODULAR LIGHTING SYSTEMS

This section does not apply to this project.

H. INDOOR LIGHTING CONTROLS (Not including PAFs)

This table includes lighting controls for conditioned and unconditioned spaces.

Building Level Controls

01	02	03
Mandatory Demand Response 110.12(c)	Shut-off controls 130.1(c) / 160.5(b)4C	Field Inspector
		Pass
		Fail
NA < 4,000W subject to multilevel	Whole Building Auto Time Switch	☐

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Project Name: Morgan Hill Butterfield Fire Station Report Page: (Page 5 of 9)

Project Address: 17285 Butterfield Blvd Date Prepared: 2/10/2023

H. INDOOR LIGHTING CONTROLS (Not including PAFs)									
Area Level Controls									
04	05	06	07	08	09	10	11	12	
Area Description	Complete Building or Area Category Primary Function Area	Manual Area Controls 130.1(a) / 160.5(b)4A	Multi-Level Controls 130.1(b) / 160.5(b)4B	Shut-Off Controls 130.1(c) // 160.5(b)4C	Primary/Sky lit Daylighting 130.1(d) / 160.5(b)4D	Secondary Daylighting 130.1(d) / 160.5(b)4D	Interlocked Systems 140.6(a)1/ 170.2(e)2A	Field Inspector	
					NA: Rm < 24sf Glazing	NA: Rm < 24sf Glazing	No	Pass	Fail
Office	Office (<=250 square feet)	Readily Accessible	Dimmer	Occupancy Sensor	NA: Rm < 24sf Glazing	NA: Rm < 24sf Glazing	No	☐	☐
Corridor	Corridor	Readily Accessible	Dimmer	Occupancy Sensor	NA: Rm < 24sf Glazing	NA: Rm < 24sf Glazing	No	☐	☐
Restroom	Restroom	Readily Accessible	NA: Enclosed area <100SF	Occupancy Sensor	NA: Rm < 24sf Glazing	NA: Rm < 24sf Glazing	No	☐	☐
Kitchen	Kitchen/ Food Preparation	Readily Accessible	Dimmer	Occupancy Sensor	NA: Rm < 24sf Glazing	NA: Rm < 24sf Glazing	No	☐	☐
Bunks	All Other Space Types	Readily Accessible	Dimmer	NA: Elec. equip. rm	NA: Rm < 24sf Glazing	NA: Rm < 24sf Glazing	No	☐	☐
Storage	All Other Space Types	Readily Accessible	NA: Enclosed area <100SF	Occupancy Sensor	NA: Rm < 24sf Glazing	NA: Rm < 24sf Glazing	No	☐	☐
Electrical	Electrical Mechanical Telephone Room	Readily Accessible	NA: Enclosed area <100SF	NA: Elec. equip. rm	NA: Rm < 24sf Glazing	NA: Rm < 24sf Glazing	No	☐	☐
Garage	Auto Repair	Readily Accessible	Dimmer	See Building Level	NA: Garage < 36sf opening	NA: Garage < 36sf opening	No	☐	☐
Fitness	Exercise Center Gymnasium	Readily Accessible	Dimmer	Occupancy Sensor	NA: Rm < 24sf Glazing	NA: Rm < 24sf Glazing	No	☐	☐
Laundry	Laundry	Readily Accessible	Dimmer	Occupancy Sensor	NA: Rm < 24sf Glazing	NA: Rm < 24sf Glazing	No	☐	☐
13									
Plan Sheet Showing Daylit Zones:									

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Project Name: Morgan Hill Butterfield Fire Station Report Page: (Page 6 of 9)

Project Address: 17285 Butterfield Blvd Date Prepared: 2/10/2023

I. LIGHTING POWER ALLOWANCE: COMPLETE BUILDING OR AREA CATEGORY METHODS					
Each area complying using the Complete Building or Area Category Methods per 140.6(b) are included in this table. Column 06 indicates if additional lighting power allowances per 140.6(c) or adjustments per 140.6(a) are being used.					
Conditioned Spaces					
01	02	03	04	05	06
Area Description	Complete Building or Area Category Primary Function Area	Allowed Density (W/ft ²)	Area (ft ²)	Allowed Wattage (Watts)	Additional Allowance / Adjustment Area Category PAF
Office	Office (<=250 square feet)	0.65	731	475.2	No
Restroom	Restroom	0.65	240	156	No
Storage	All Other Space Types	0.4	278	111.2	No
Laundry	Laundry	0.45	84	37.8	No
Kitchen / Dining	Kitchen/ Food Preparation	0.95	784	744.8	No
Fitness	Exercise Center Gymnasium	0.5	350	175	No
Corridor	Corridor	0.4	498	199.2	No
Garage	Auto Repair	0.55	1,770	973.5	No
Workshop	All Other Space Types	0.6	84	50.4	No
Electrical	Electrical Mechanical Telephone Room	0.4	178	71.2	No
TOTALS:				4,997	See Tables I, or P for detail
Unconditioned Spaces					
01	02	03	04	05	06
Area Description	Complete Building or Area Category Primary Function Area	Allowed Density (W/ft ²)	Area (ft ²)	Allowed Wattage (Watts)	Additional Allowance / Adjustment Area Category PAF
Trash Enclosure	All Other Space Types	0.4	234	93.6	No
TOTALS:				234	See Tables I, or P for detail

J. ADDITIONAL ALLOWANCE: AREA CATEGORY METHOD QUALIFYING LIGHTING SYSTEM

This section does not apply to this project.

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This document is used to demonstrate compliance with requirements in 110.9, 110.12(c), 130.0, 130.1, 140.6 and 141.0(b)2 for indoor lighting scopes using the prescriptive path for nonresidential and hotel/motel occupancies. It is also used to document compliance with requirements in 160.5, 170.2(e) and 180.2(b)4 for indoor lighting scopes using the prescriptive path for multifamily occupancies. Multifamily includes dormitory and senior living facilities.

Project Name: Morgan Hill Butterfield Fire Station Report Page: (Page 1 of 9)

Project Address: 17285 Butterfield Blvd Date Prepared: 2/10/2023

A. GENERAL INFORMATION

01 Project Location (city)	Morgan Hill	04 Total Conditioned Floor Area (ft ²)	4,997
02 Climate Zone	4	05 Total Unconditioned Floor Area (ft ²)	234
03 Occupancy Types Within Project (select all that apply):	06 # of Stories (Habitable Above Grade)		
• Gymnasium • Office • Support Areas • All Other Occupancies	1		

B. PROJECT SCOPE

This table includes any lighting systems that are within the scope of the permit application and are demonstrating compliance using the prescriptive path outlined in 140.6 / 170.2(e) or 141.0(b)2 / 180.2(b)4 for alterations.

Scope of Work	Conditioned Spaces	Unconditioned Spaces		
01	02	03	04	05
My Project Consists of (check all that apply):	Calculation Method	Area (ft ²)	Calculation Method	Area (ft ²)
☒ New Lighting System	Area Category Method	4997	Area Category Method	234
☐ New Lighting System - Parking Garage				
Total Area of Work (ft ²)	4997		234	

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C. COMPLIANCE RESULTS									
If any cell on this table says "DOES NOT COMPLY" or "COMPLIES with Exceptional Conditions" refer to Table D, for guidance.									
Lighting in conditioned and unconditioned spaces must not be combined for compliance per 140.6(b)1 / 170.2(e)	Allowed Lighting Power per 140.6(b) / 170.2(e) (Watts)					Adjusted Lighting Power per 140.6(a) / 170.2(e) (Watts)			Compliance Results
	01	02	03	04	05	06	07	08	09
	Complete Building 140.6(c)1	Area Category 140.6(c)2 / 170.2(e)4	Area Category Additional 140.6(c)2E / 170.2(e)4Av (+)	Tailored 140.6(c)3 / 170.2(e)4B (-)	Total Allowed (Watts)	Total Designed (Watts)	PAF Lighting Control Credits 140.6(a)2 / 170.2(e)1B (-)	Total Adjusted (Watts) *Includes Adjustments	05 must be >= 08 140.6 / 170.2(e)
	(See Table I)	(See Table I)	(See Table J)	(See Table K)	=	(See Table F)	=	=	
Conditioned	2,994.3	0			=	2,994	0	=	2903
Unconditioned		93.6	0		=	94	0	=	75
Controls Compliance (See Table H for Details)									
Rate Power Reduction Compliance (See Table Q for Details)									

D. EXCEPTIONAL CONDITIONS

This table is auto-filled with uneditable comments because of selections made or data entered in tables throughout the form.

E. ADDITIONAL REMARKS

This table includes remarks made by the permit applicant to the Authority Having Jurisdiction.

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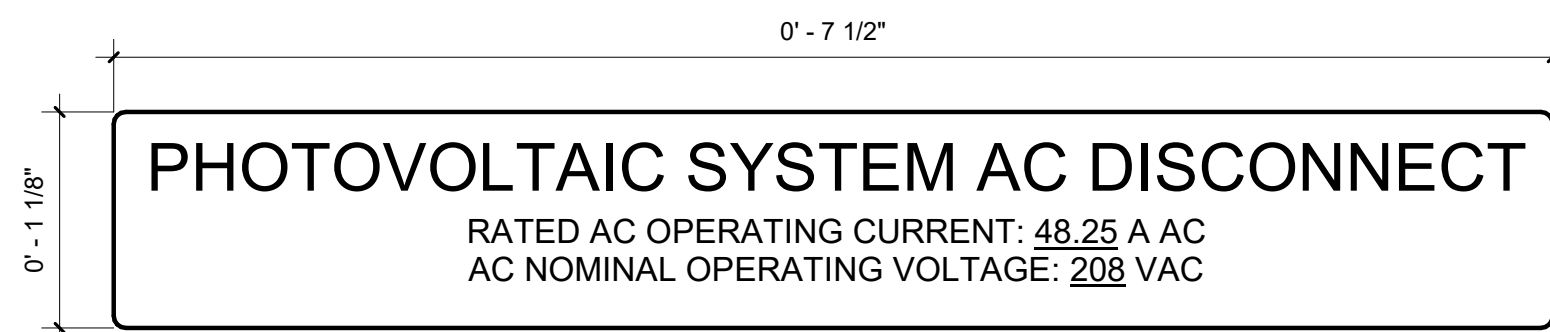
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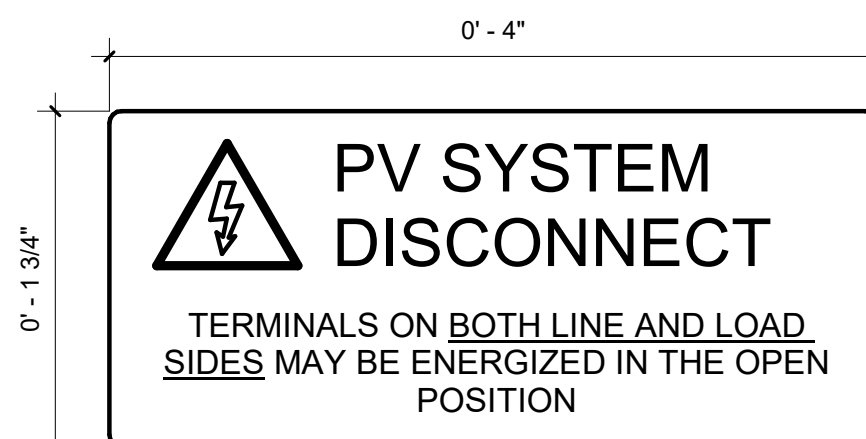
F. INDOOR LIGHTING FIXTURE SCHEDULE										
This table includes all planned permanent and portable lighting other than dwelling unit/ hotel/ motel room lighting. Multifamily dwelling unit and hotel/motel room lighting is documented in Table T. If using Table T to document lighting in multifamily common use areas providing shared provisions for living, eating, cooking or sanitation, those luminaires are not included here.										
Designed Wattage: Conditioned Spaces										
01	02	03	04	05	06	07	08	09	10	
Name or Item Tag	Complete Luminaire Description	Modular (Track) Fixture	Small Aperture & Color Change ¹	Watts per luminaire ²	How is Wattage determined	Total Number of Luminaires	Excluded per 140.6(a)3 / 170.2(e)2C	Design Watts	Field Inspector	
									Pass	Fail
AA1	AA1-LINEAR 4	No	NA	26	Mfr. Spec	5	No	130	☐	☐
AA2	AA2-LINEAR 6	No	NA	39	Mfr. Spec	1	No	39	☐	☐
AB1	AB3-STRIP	No	NA	33	Mfr. Spec	3	No	99	☐	☐
AB2	AB2-STRIP	No	NA	11	Mfr. Spec	3	No	33	☐	☐
AB3	AB3-STRIP	No	NA	40	Mfr. Spec	2	No	80	☐	☐
AB4	AB4-STRIP	No	NA	33	Mfr. Spec	1	No	33	☐	☐
AC1	AC1-LINEAR 8	No	NA	70	Mfr. Spec	1	No	70	☐	☐
AD1	AD1-SURFACE DOWN	No	NA	13	Mfr. Spec	8	No	104	☐	☐
AE1	AE1-UNDERCABINET	No	NA	11	Mfr. Spec	17	No	187	☐	☐
AG1	AG1-DRUM 36"	No	NA	35	Mfr. Spec	4	No	140	☐	☐
AH1	AH1-SURFACE SQUARE	No	NA	35	Mfr. Spec	4	No	140	☐	☐
AH1E	AH1E-SURFACE SQUARE	No	NA	35	Mfr. Spec	4	No	140	☐	☐
AJ1	AJ1-SURFACE LINEAR 4	No	NA	25	Mfr. Spec	3	No	75	☐	☐
AK1	AJ2-SURFACE LINEAR 8	No	NA	50	Mfr. Spec	2	No	100	☐	☐
AK1	AK1-SURFACE STRIP 8	No	NA	66	Mfr. Spec	3	No	660	☐	☐
AL2	AK2-PENDANT STRIP 8	No	NA	66	Mfr. Spec	5	No	330	☐	☐
AL1	AL1-RECESSED ROUND	No	NA	19	Mfr. Spec	3	No	57	☐	☐
AL2	AL2-RECESSED ROUND	No	NA	14	Mfr. Spec	12	No	168	☐	☐
AL3	AL3-RECESSED ROUND	No	NA	19	Mfr. Spec	3	No	57	☐	☐
AM1	AM1-DECORATIVE	No	NA	29	Mfr. Spec	2	No	58	☐	☐



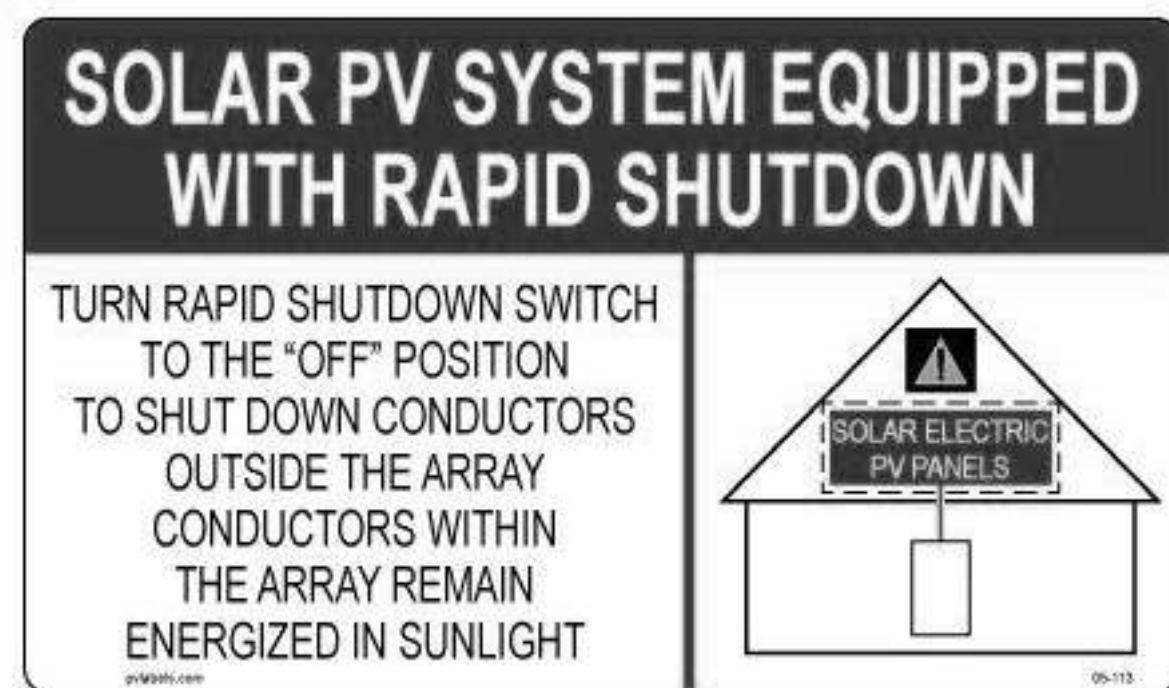
- NOTES:**
1. LOCATE AT PHOTOVOLTAIC INTERCONNECTION DISCONNECT SYSTEM CONDUIT RUNS.
 2. RED BACKGROUND, REFLECTIVE WHITE LETTERING (MIN. 1/4" TEXT).
 3. LABELS SHALL BE UV SUN RESISTANT, WITH ADHESIVE SUITABLE FOR OUTDOOR ENVIRONMENT.



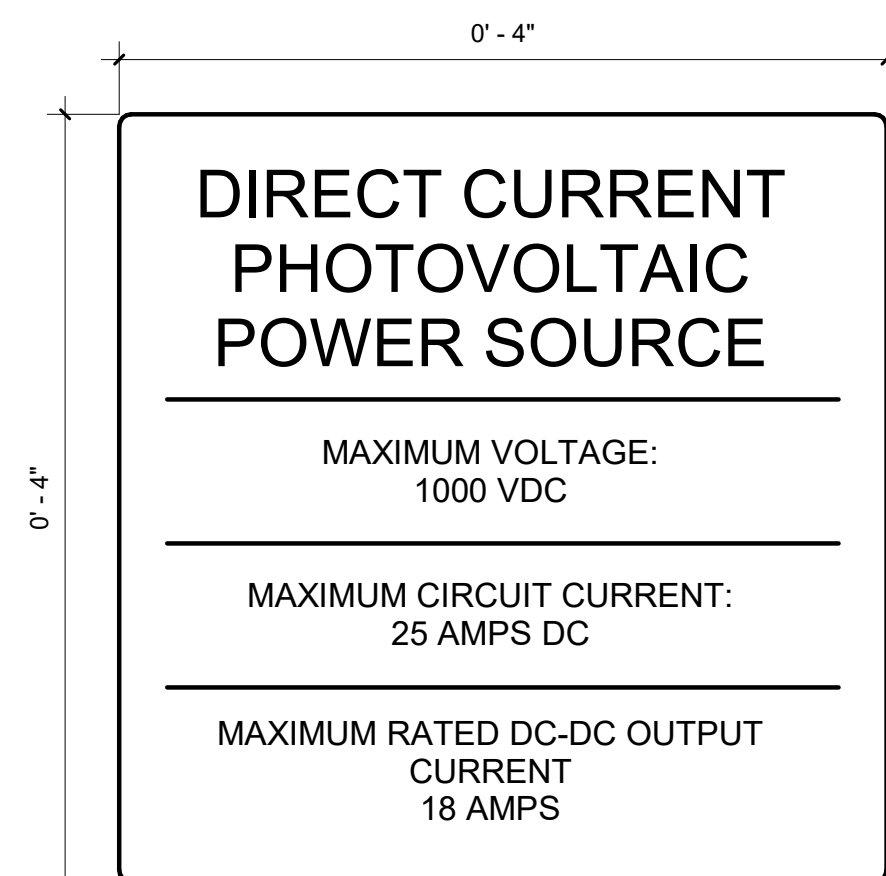
- NOTES:**
1. LOCATE AT 10' INCREMENTS ALONG ALL EXPOSED PHOTOVOLTAIC SYSTEM CONDUIT RUNS.
 2. LOCATE AT ALL COVERS, ENCLOSURES, J-BOXES AND CONDUIT BODIES.
 3. RED BACKGROUND, REFLECTIVE WHITE LETTERING (MIN. 3/8" TEXT).
 4. LABELS SHALL BE UV SUN RESISTANT, WITH PERMANENT ADHESIVE BOND SUITABLE FOR SURFACE MATERIAL APPLICATION.
 5. REFERENCE: CEC 690.31(G)



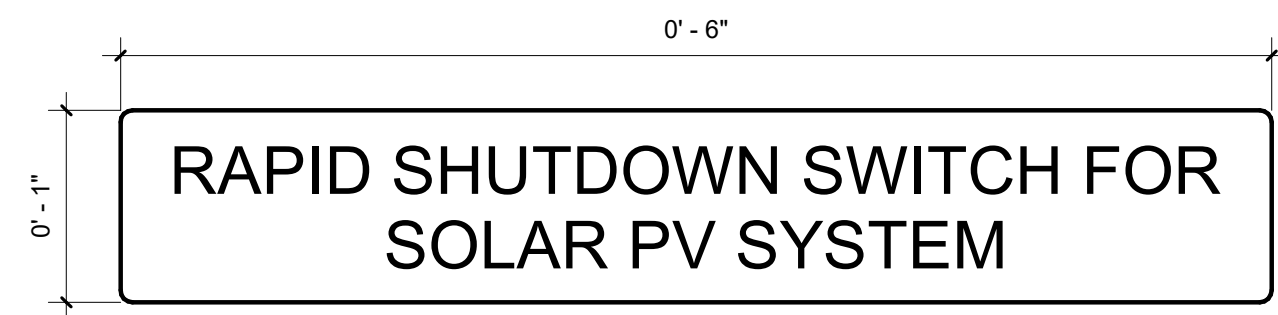
- NOTES:**
1. LOCATE AT ALL INVERTER AC/DC DISCONNECT(S).
 2. RED BACKGROUND, REFLECTIVE WHITE LETTERING (MIN. 1/4" TEXT).
 3. LABELS SHALL BE UV SUN RESISTANT, WITH PERMANENT ADHESIVE BOND SUITABLE FOR SURFACE MATERIAL APPLICATION.
 4. REFERENCE: CEC 690.15(D)



- NOTES:**
1. LOCATE NOT MORE THAN 3' FROM THE SERVICE DISCONNECTING MEANS.
 2. RED BACKGROUND, REFLECTIVE WHITE LETTERING (MIN. 3/8" & 3/16" TEXT).
 3. LABELS SHALL BE UV SUN RESISTANT, WITH PERMANENT ADHESIVE BOND SUITABLE FOR SURFACE MATERIAL APPLICATION.
 4. REFERENCE: CEC 690.56(C)(1)



- NOTES:**
1. LOCATE AT DC INPUT DEVICE.
 2. RED BACKGROUND, REFLECTIVE WHITE LETTERING (MIN. 1/4" & 1/8" TEXT).
 3. LABELS SHALL BE UV SUN RESISTANT, WITH ADHESIVE SUITABLE FOR OUTDOOR ENVIRONMENT.
 4. REFERENCE: CEC 690.53



- NOTES:**
1. LOCATE WITHIN 3' OF RAPID SHUTDOWN SWITCH.
 2. RED BACKGROUND, REFLECTIVE WHITE LETTERING (MIN. 1/4" TEXT).
 3. LABELS SHALL BE UV SUN RESISTANT, WITH ADHESIVE SUITABLE FOR OUTDOOR ENVIRONMENT.
 4. REFERENCE: CEC 690.56(C)(3)



PV power optimization at the module-level

- Specifically designed to work with SolarEdge inverters
- Up to 25% more energy
- Superior efficiency (99.5%)
- Mitigates all types of module mismatch losses, from manufacturing tolerance to partial shading
- Flexible system design for maximum space utilization
- Fast installation with a single bolt
- Next generation maintenance with module-level monitoring
- Meets NEC requirements for arc fault protection (AFCI) and Photovoltaic Rapid Shutdown System (PVRSS)
- Module-level voltage shutdown for installer and firefighter safety

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/ Power Optimizer for North America P370 / P400 / P401 / P485 / P505

Optimizer model (typical module compatibility)	P370 (for higher power 60 and 72 cell modules)	P400 (for 72 & 96 cell modules)	P401 (for high power 60 and 72 cell modules)	P485 (for high voltage modules)	P505 (for higher current modules)
INPUT					
Rated Input DC Power ⁽¹⁾	370	400	401	485	505
Absolute Maximum Input Voltage (V _{oc} at 25°C)	60	80	60	125 ⁽²⁾	83 ⁽²⁾
MPPT Operating Range	8 - 60	8 - 80	8-40	12.5 - 100	12.5 - 83
Maximum Short Circuit Current (I _{sc})	11	10.1	11.5	11	10
Maximum Efficiency	99.5	99.5	99.5	99.5	99.5
Voltage Efficiency	99.8	99.8	99.8	99.8	99.8
Conversion Loss ⁽³⁾	0	0	0	0	0
OUTPUT DURING OPERATION (POWER OPTIMIZER CONNECTED TO OPERATING SOLAREGE INVERTER)					
Maximum Output Current	60	51	60	80	80
Maximum Output Voltage	60	80	60	125	83
OUTPUT DURING STANDBY (POWER OPTIMIZER DISCONNECTED FROM SOLAREGE INVERTER OR SOLAREGE INVERTER OFF)					
Standby Output Current (typical)	0	0	0	0	0
Standby Output Voltage (typical)	0	0	0	0	0
STANDARD COMPLIANCE					
Photovoltaic Rapid Shutdown System	NEC 2014, 2017 & 2020	NEC 2014, 2017 & 2020	NEC 2014, 2017 & 2020	NEC 2014, 2017 & 2020	NEC 2014, 2017 & 2020
UL1741	Yes	Yes	Yes	Yes	Yes
UL1741 SA	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class B	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class C	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class D	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class E	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class F	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class G	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class H	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class I	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class J	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class K	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class L	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class M	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class N	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class O	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class P	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class Q	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class R	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class S	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class T	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class U	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class V	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class W	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class X	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class Y	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class Z	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AA	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AB	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AC	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AD	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AE	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AF	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AG	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AH	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AI	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AJ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AK	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AL	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AM	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AN	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AO	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AP	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AQ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AR	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AS	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AT	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AU	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AV	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AW	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AX	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AY	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class AZ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BA	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BB	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BC	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BD	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BE	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BF	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BG	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BH	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BI	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BJ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BK	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BL	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BM	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BN	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BO	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BP	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BQ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BR	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BS	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BT	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BU	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BV	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BW	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BX	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BY	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class BZ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CA	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CB	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CC	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CD	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CE	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CF	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CG	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CH	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CI	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CJ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CK	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CL	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CM	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CN	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CO	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CP	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CQ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CR	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CS	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CT	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CU	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CV	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CW	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CX	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CY	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class CZ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DA	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DB	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DC	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DD	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DE	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DF	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DG	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DH	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DI	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DJ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DK	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DL	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DM	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DN	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DO	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DP	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DQ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DR	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DS	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DT	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DU	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DV	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DW	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DX	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DY	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class DZ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EA	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EB	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EC	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class ED	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EE	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EF	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EG	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EH	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EI	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EJ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EK	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EL	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EM	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EN	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EO	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EP	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EQ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class ER	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class ES	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class ET	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EU	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EV	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EW	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EX	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EY	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class EZ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FA	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FB	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FC	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FD	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FE	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FF	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FG	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FH	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FI	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FJ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FK	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FL	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FM	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FN	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FO	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FP	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FQ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FR	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FS	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FT	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FU	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FV	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FW	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FX	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FY	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class FZ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GA	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GB	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GC	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GD	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GE	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GF	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GG	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GH	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GI	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GJ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GK	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GL	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GM	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GN	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GO	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GP	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GQ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GR	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GS	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GT	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GU	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GV	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GW	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GX	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GY	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class GZ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HA	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HB	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HC	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HD	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HE	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HF	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HG	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HH	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HI	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HJ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HK	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HL	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HM	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HN	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HO	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HP	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HQ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HR	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HS	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HT	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HU	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HV	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HW	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HX	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HY	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class HZ	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class IA	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class IB	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class IC	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class ID	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class IE	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class IF	Yes	Yes	Yes	Yes	Yes
UL1741 SA Class IG	Yes	Yes	Yes	Yes	Yes
UL1					